

Identifying modifiable factors and their joint effect on dementia risk in the UK Biobank

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Supplementary Note

Data cleaning and processing

All exposure variables surviving data processing were summarized in Supplementary Table1-4. From the initial 7214 available exposure variables, we excluded those with missing value >20% of the full sample and retained data collected at baseline. We then manually removed obviously unmodifiable exposure variables, which mainly came from the “Genomics”, “Cognitive function” and “Biological samples” categories. Multiple-choice categorical variables (e.g., types of physical activity attended) were dummy coded into dichotomous variables (i.e., whether participated in one type of activity, assigning NA to those missing on the top-level array variable). Variables with only one level were removed. We further limited the lowest requirement for number of cases ($N > 500$) in analysis. For diseases, we adopted data from “First occurrences” category, which was coded with 3-character ICD 10 and recorded the first date of diagnosis. We set abnormal date as missing, and excluded diagnosis from the follow up period (i.e., after 2010). Diseases with a prevalence less than 2% were also excluded, except for some well-established factors like stroke and COPD. For single-choice categorical variables, we reset the order of levels (following logical ordinal progression) and removed variables with only one level. For continuous variables, we divided them into tertiles after imputation. The majority of them used the lowest third as reference, except for time of using computer, which used the median part as reference¹. Substantive results were largely unchanged after excluding outliers (retained mean-3SD ~ mean+3SD, results not shown here). As outliers are not necessarily errors, we thus left data unchanged with full records². All meaningless negative values were set as missing (NA). Some negative values were recoded (e.g., changed -10, which means less than 1, to 0.5, details are shown in Supplementary Table2-4).

Some variables were rearranged. For some dietary habits, we determined the recommendation intake level for fruit, fish, wholegrain, and et al. based on previous research³. A general healthy diet level score was also generated³. A set of diseases were generated from a combination of several 3-character ICD10 codes⁴. Lung function was evaluated as the mean value of z-scores of FVC, FEV1 and PEF⁵. As the time of using computer has a U-shaped relationship with dementia, we set the middle category of tertiles as reference¹. The accurate evaluation of ever smoke⁶ and grams of alcohol intake⁷ was based on previous research. Moderate alcohol intake is considered to be 5-15 grams per day for women and 5-30 grams per day for men³.

For factors survived the EWAS analysis, collinearity was checked and we excluded one of two factors from a highly-associated pair ($r^2 > 0.9$) and retained the one that were less correlated with other factors in the dataset by using the R program: findCorrelation (from the caret package⁸). We eventually excluded one highly-correlated variables (FieldID: X24506, Natural environment percentage, buffer 1000m).

Sensitivity analysis using hierarchical clustering

As the previous analysis only excluded variables with extremely high collinearity ($r^2 > 0.9$), a remaining correlation of variables and confounding might exist. Therefore, we used a machine learning method to check collinearity in the sensitivity analysis. To alleviate the issue of multicollinearity, we calculated Spearman rank-order correlations⁹ (Supplementary Fig. 8 left panel) to the 62 pre-selected

variables and then converted the correlation matrix to a distance matrix defined by

$$dist_{matrix} = 1 - \frac{|corr_{matrix} + transpose(corr_{matrix})|}{2}$$

We then performed hierarchical clustering of the distance matrix to group variables based on Ward's linkage, predictors were clustered together if they were highly correlated¹⁰ (Supplementary Fig. 8 right panel). We then used 0.5 as a threshold to cut the dendrogram and chose the best representative variable within each cluster. Among the 62 pre-selected variables, eight of them were removed (Supplementary Table7). The results based on variables selected by machine learning were presented in Supplementary Table 12-13 and Supplementary Fig. 8, Extended Data Fig. 3. Analyses were performed on python 3.9 with scikit-learn v1.2.1.

Procedure used to select predictors for imputation

We used the *quickpred* function from the mice package in R to select predictors for each variable. The function creates a predictor matrix as described by Van Buuren et al¹¹. We modified the original function to ensure a maximum of 11 predictors (sex was forced to be included). This precaution allow working of this function in large sample sizes¹².

AD-PRS Generation and adoption in PAF calculation

Genotype data were available for all 502,493 participants in the UKB. Detailed genotyping and quality control procedures by the UKB are available in a previous publication¹³. We excluded single nucleotide polymorphisms with call rates <95%, minor allele frequency <0.1%, and deviation from the Hardy-Weinberg equilibrium with $p < 1 \times 10^{-10}$. We selected subjects who were estimated to have recent British ancestry based on self-report information and principal component analyses of the genotypes and had no more than 10 putative third-degree relatives in the kinship table. After the quality control procedures, we obtained a total of 591,050 single nucleotide polymorphisms and 337,199 participants.

We calculated the PRS with the summary statistics from a meta-analysis of GWASs from 4 large AD consortia¹⁴, which included a total of 7,055,881 single nucleotide polymorphisms and 54,062 individuals (17,008 cases and 37,054 control subjects).

PRSice was the software (<http://www.PRSice.info>) used for the calculation of AD-PRS. We used p value-informed clumping with a cutoff of $r^2 = 0.1$ in a 250-kb window in the analysis¹⁵. P thresholds for scoring were set at $p < 0.0005$, $p < 0.001$, $p < 0.005$, $p < 0.01$, $p < 0.05$, $p < 0.1$, $p < 0.5$, and $p < 1$ ¹⁶.

We have incorporated the polygenic risk score in the calculation of PAF. The polygenic risk score was calculated across all SNPs associated with Alzheimer disease with a P value less than 0.50 as a threshold for inclusion¹⁷. This threshold was chosen as previous work has shown it contains the optimal set of variants for predicting AD¹⁸. The distribution of polygenic-risk scores was divided into tertiles, representing high/intermediate/low genetic risk¹⁹.

Traits selection for Mendelian randomization

Similar UKB traits were used for replacement when the exact trait was missing (e.g., we used "Number

of days/week of moderate/vigorous physical activity 10+ minutes” to replace “MET minutes per week for vigorous activity”). For some variables, such as hand grip strength, left and right-hand data are integrated in correlation analysis, but GWAS summary statistics are still separate, so MR analysis is performed separately (Supplementary Table8 records the GWAS ID for each trait).

Sample characteristics

A flow diagram of eligible study participants selection is shown in Fig. 1. In the full sample, participants were 54.2% female and had a mean baseline age of 56.0 (standard deviation, SD = 8.04). 43.2% reported higher level of education (College/University degree and/or Other professional qualifications like nursing, teaching), and 55.1% reported average pre-tax household income of 31,000 pounds or higher. Participants had a mean BMI of 27.4 (SD = 4.82). Overall, 4654 (1.35%) met the diagnosis of dementia during follow-up, and the mean age at diagnoses was 72.8 years (SD = 5.9). For further details, see Supplementary Table5.

Supplementary Discussion 1: the results of B vitamins

The finding that self-reported use of folic acid-containing B vitamin supplements increased the risk of dementia, at least in those under 60 at recruitment, is puzzling in view of evidence that folate status is related to better cognition²⁰⁻²². A possible explanation is that these supplements contained folic acid rather than natural folates. Folic acid use has been found to impair cognition in the elderly under certain circumstances^{20,23}.

Supplementary Discussion 2: the results of MR using GWAS for all-cause dementia

Less significant results were found when replace the GAWS for AD in the outcome with GWAS for all-cause dementia. The differences might be attributed to the reduced power of FinnGen’s GWAS, as the patients were half of the number of that in AD’s GWAS. Moreover, the heterogeneity in the outcome will reduce the precision of our causal estimates, making it harder to identify a true causal effect should one exist²⁴.

Supplementary Discussion 3: the results of PAF for PRS and modifiable factors

In our PAF calculations incorporating both genetic risk (tertiles of PRS) and modifiable risk factors, the PAF for PRS (8.8%) in Model2 was similar to that of *APOEε4* allele (7.11%) reported in previous literature²⁵. Full results are shown in Supplementary Table 18-19. Interestingly, after excluding the overlap between variables, the overall PAF didn’t change much (overall PAF changed from 47.0% to 46.8% in model1, and changed from 72.6% to 72.2% in model2), possibly due to the highest communality of genetic risk, which means most variance in genetic factor was accounted for by other six domains²⁶, this may be related to the shared genetic risk between exposure and dementia²⁷, as well as

the interaction of genes and environment²⁸.

References

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Supplementary Table 1 Original data for modification

FieldID	Field	Coding				coding notes
Lifestyles						
1289	Cooked vegetable intake	100373	-1	Do not know	tablespoons/day	set as NA
			-10	Less than one		set as 0.5
			-3	Prefer not to answer		set as NA
1299	Salad / raw vegetable intake	100373	same as above		tablespoons/day	
1309	Fresh fruit intake	100373	same as above		pieces/day	
1319	Dried fruit intake	100373	same as above		pieces/day	
1329	Oily fish intake	100377	-1	Do not know		set as NA
			-3	Prefer not to answer		set as NA
			0	Never		0
			1	Less than once a week		0
			2	Once a week		1
			3	2-4 times a week		1
			4	5-6 times a week		1
			5	Once or more daily		1
1339	Non-oily fish intake	100377	same as above			
1438	Bread intake	100373	same as above		slices/week	
1448	Bread type	100391	-1	Do not know		set as NA
			-3	Prefer not to answer		set as NA
			1	White		
			2	Brown		
			3	Wholemeal or wholegrain		
			4	Other type of bread		
1458	Cereal intake	100373	same as above		bowls/week	
1468	Cereal type	100393	-1	Do not know		set as NA
			-3	Prefer not to answer		set as NA
			1	Bran cereal (e.g. All Bran, Branflakes)		1
			2	Biscuit cereal (e.g. Weetabix)		0

			3	Oat cereal (e.g. Ready Brek, porridge)		1
			4	Muesli		0
			5	Other (e.g. Cornflakes, Frosties)		0
1369	Beef intake	100377	0	Never		1
			1	Less than once a week		1
			2	Once a week		0
			3	2-4 times a week		0
			4	5-6 times a week		0
			5	Once or more daily		0
1379	Lamb/mutton intake	100377	same as above			
1389	Pork intake	100377	same as above			
1349	Processed meat intake	100377	same as above			
2644	Light smokers, at least 100 smokes in lifetime	100349	-1	Do not know		set as NA
			-3	Prefer not to answer		set as NA
			0	No		
			1	Yes		
1249	Past tobacco smoking	100348	-3	Prefer not to answer		set as NA
			1	Smoked on most or all days		
			2	Smoked occasionally		
			3	Just tried once or twice		
			4	I have never smoked		
1239	Current tobacco smoking	100347	-3	Prefer not to answer		set as NA
			0	No		
			1	Yes, on most or all days		
			2	Only occasionally		
1160	Sleep duration	100291			hours/day	set -1 -3 as NA Improper sleep duration <7/>9 hours
1568	Average weekly red wine intake	100291			glasses	set -1 -3 as NA
4407	Average monthly red wine intake	100291			glasses	set -1 -3 as NA
1578	Average weekly champagne plus white wine intake	100291			glasses	set -1 -3 as NA

4418	Average monthly champagne plus white wine intake	100291			glasses	set -1 -3 as NA
1588	Average weekly beer plus cider intake	100291			pints	set -1 -3 as NA
4429	Average monthly beer plus cider intake	100291			pints	set -1 -3 as NA
1598	Average weekly spirits intake	100291			measures	set -1 -3 as NA
4440	Average monthly spirits intake	100291			measures	set -1 -3 as NA
1608	Average weekly fortified wine intake	100291			glasses	set -1 -3 as NA
4451	Average monthly fortified wine intake	100291			glasses	set -1 -3 as NA
5364	Average weekly intake of other alcoholic drinks	100291			glasses	set -1 -3 as NA
4462	Average monthly intake of other alcoholic drinks	100291			glasses	set -1 -3 as NA
1558	Alcohol intake frequency.	100402	-3	Prefer not to answer		set as NA
			1	Daily or almost daily		
			2	Three or four times a week		
			3	Once or twice a week		
			4	One to three times a month		
			5	Special occasions only		
			6	Never		

Supplementary Table 2 Multiple-choice categorical variables and processing

FieldID	Field	Coding							coding notes
SES									
6142	Current employment status	100295	-3	Prefer not to answer				set as NA	
			-7	None of the above				set as NA	
			1	In paid employment or self-employed				set as 0	Turn data into binary classification : 1 : unemployed; 0: employed
			2	Retired				set as 0	
			3	Looking after home and/or family				set as 1	
			4	Unable to work because of sickness or disability				set as 1	
			5	Unemployed				set as 1	
			6	Doing unpaid or voluntary work				set as 0	
			7	Full or part-time student				set as 0	
6138	Qualifications	100305	1	College or University degree				set as 0	
			2	A levels/AS levels or equivalent				set as 1	Turn data into binary classification: 1: lower education; 0: higher education
			3	O levels/GCSEs or equivalent				set as 1	
			4	CSEs or equivalent				set as 1	
			5	NVQ or HND or HNC or equivalent				set as 1	
			6	Other professional qualifications eg: nursing, teaching				set as 0	
			-1	None of the above				set as 1	Regarded as below high school diploma
			-3	Prefer not to answer				set as NA	
Medical history									
6146	Attendance/disability/mobility allowance	100510	-1	Do not know				set as NA	Turn data into binary classification : 1 : disability; 0: no disability
			-3	Prefer not to answer				set as NA	
			-7	None of the above				set as 0	
			1	Attendance allowance (means severely disabled)				set as 1	
			2	Disability living allowance				set as 1	
			3	Blue badge (Disabilities or health conditions cause immobility)				set as 1	
6149	Mouth/teeth dental problems	100538	-3	Prefer not to answer				set as NA	
			-7	None of the above					

			1	Mouth ulcers					
			2	Painful gums					
			3	Bleeding gums					
			4	Loose teeth					
			5	Toothache					
			6	Dentures					
Lifestyles			1						
6160	Leisure/social activities	100328	-3	Prefer not to answer				set as NA	
			-7	None of the above					
			1	Sports club or gym					
			2	Pub or social club					
			3	Religious group					
			4	Adult education class					
			5	Other group activity					
6155	Vitamin and mineral supplements	100629	-3	Prefer not to answer				set as NA	
			-7	None of the above				set as 0	
			1	Vitamin A					
			2	Vitamin B					
			3	Vitamin C					
			4	Vitamin D					
			5	Vitamin E					
			6	Folic acid or Folate (Vit B9)					
			7	Multivitamins +/- minerals					
6179	Mineral and other dietary supplements	100630	-3	Prefer not to answer				set as NA	
			-7	None of the above					
			1	Fish oil (including cod liver oil)					
			2	Glucosamine					
			3	Calcium					
			4	Zinc					
			5	Iron					
			6	Selenium					

SES: socioeconomic status

Supplementary Table 3 Single-choice categorical variables and processing

FieldID	Field	Coding			coding notes
SES					
738	Average total household income before tax	100294	-1	Do not know	set as NA
			-3	Prefer not to answer	set as NA
			1	Less than 18,000	
			2	18,000 to 30,999	
			3	31,000 to 51,999	
			4	52,000 to 100,000	
			5	Greater than 100,000	
Lifestyles					
1210	Snoring	100345	-1	Do not know	set as NA
			-3	Prefer not to answer	set as NA
			2	Yes	
			1	No	
Derived	Ever smoke		2	Yes	Original data are listed in Supplementary Table1
			1	No	
Derived	Moderate drinking		2	No	
			1	Yes	
2159	Ever had same-sex intercourse	100352	-3	Prefer not to answer	set as NA
			0	No	
			1	Yes	
22036	Above moderate/vigorous/walking recommendation	7	2	No	
			1	Yes	
Derived	Moderate sleep		2	No	
			1	Yes	
1190	Nap during day	100343	-3	Prefer not to answer	set as NA
			1	Never/rarely	

			2	Sometimes	
			3	Usually	
1200	Sleeplessness / insomnia	100343	-3	Prefer not to answer	set as NA
			1	Never/rarely	
			2	Sometimes	
			3	Usually	
1220	Daytime dozing / sleeping (narcolepsy)	100346	-1	Do not know	set as NA
			-3	Prefer not to answer	set as NA
			0	Never/rarely	
			1	Sometimes	
			2	Often	
			3	All of the time	
1269	Exposure to tobacco smoke at home	100291	set -1 -3 as NA		Turn data into binary classification: 1: exposure to tobacco smoke at home; 0: no exposure
1279	Exposure to tobacco smoke outside home	100291	set -1 -3 as NA		Turn data into binary classification: 1: exposure to tobacco smoke outside; 0: no exposure
1120	Weekly usage of mobile phone in last 3 months	100336	-1	Do not know	set as NA
			-3	Prefer not to answer	set as NA
			0	Less than 5mins	
			1	5-29 mins	
			2	30-59 mins	
			3	1-3 hours	
			4	4-6 hours	
			5	More than 6 hours	
1130	Hands-free device/speakerphone use with mobile phone in last 3 month	100337	-1	Do not know	set as NA
			-3	Prefer not to answer	set as NA
			0	Never or almost never	
			1	Less than half the time	
			2	About half the time	
			3	More than half the time	

			4	Always or almost always	
Derived	Fruit eaten above recommendation				
Derived	Fish eaten above recommendation				
Derived	White bread/refined cereals intake reduced				
Derived	Wholegrain eaten above recommendation				
Derived	Red meat intake reduced				
Derived	Processed meat intake reduced				
Medical history					
130176	Date B01 first reported (varicella [chickenpox])				
130178	Date B02 first reported (zoster [herpes zoster])				
130188	Date B07 first reported (viral warts)				
130226	Date B35 first reported (dermatophytosis)				
130622	Date D50 first reported (iron deficiency anaemia)				
130648	Date D64 first reported (other anaemias)				
130696	Date E03 first reported (other hypothyroidism)				
130792	Date E66 first reported (obesity)				
130814	Date E78 first reported (disorders of lipoprotein metabolism and other lipidaemias)				
130868	Date F17 first reported (mental and behavioural disorders due to use of tobacco)				
130906	Date F41 first reported (other anxiety disorders)				
130910	Date F43 first reported (reaction to severe stress, and adjustment disorders)				
131052	Date G43 first reported (migraine)				
131060	Date G47 first reported (sleep disorders)				
131074	Date G56 first reported (mononeuropathies of upper limb)				
131128	Date H00 first reported (hordeolum and chalazion)				
131142	Date H10 first reported (conjunctivitis)				
131166	Date H26 first reported (other cataract)				
131222	Date H60 first reported (otitis externa)				
131224	Date H61 first reported (other disorders of external ear)				
131262	Date H92 first reported (otalgia and effusion of ear)				
131264	Date H93 first reported (other disorders of ear, not elsewhere classified)				
131350	Date I48 first reported (atrial fibrillation and flutter)				
131396	Date I80 first reported (phlebitis and thrombophlebitis)				

131402	Date I83 first reported (varicose veins of lower extremities)			
131404	Date I84 first reported (haemorrhoids)			
131426	Date J01 first reported (acute sinusitis)			
131428	Date J02 first reported (acute pharyngitis)			
131430	Date J03 first reported (acute tonsillitis)			
131436	Date J06 first reported (acute upper respiratory infections of multiple and unspecified sites)			
131442	Date J11 first reported (influenza, virus not identified)			
131456	Date J18 first reported (pneumonia, organism unspecified)			
131462	Date J22 first reported (unspecified acute lower respiratory infection)			
131464	Date J30 first reported (vasomotor and allergic rhinitis)			
131468	Date J32 first reported (chronic sinusitis)			
131472	Date J34 first reported (other disorders of nose and nasal sinuses)			
131494	Date J45 first reported (asthma)			
131584	Date K21 first reported (gastro-oesophageal reflux Medical history)			
131598	Date K29 first reported (gastritis and duodenitis)			
131600	Date K30 first reported (dyspepsia)			
131612	Date K40 first reported (inguinal hernia)			
131620	Date K44 first reported (diaphragmatic hernia)			
131630	Date K52 first reported (other non-infective gastro-enteritis and colitis)			
131636	Date K57 first reported (diverticular Medical history of intestine)			
13163	Date K58 first reported (irritable bowel syndrome)			
131640	Date K59 first reported (other functional intestinal disorders)			
131646	Date K62 first reported (other Medical historys of anus and rectum)			
131650	Date K64 first reported (haemorrhoids and perianal venous thrombosis)			
131674	Date K80 first reported (cholelithiasis)			
131702	Date L03 first reported (cellulitis)			
131708	Date L08 first reported (other local infections of skin and subcutaneous tissue)			
131720	Date L20 first reported (atopic dermatitis)			
131740	Date L30 first reported (other dermatitis)			
131742	Date L40 first reported (psoriasis)			
131768	Date L57 first reported (skin changes due to chronic exposure to nonionising radiation)			
131794	Date L72 first reported (follicular cysts of skin and subcutaneous tissue)			

131806	Date L82 first reported (seborrhoeic keratosis)			
131836	Date L98 first reported (other disorders of skin and subcutaneous tissue, not elsewhere classified)			
131858	Date M10 first reported (gout)			
131864	Date M13 first reported (other arthritis)			
131872	Date M17 first reported (gonarthrosis [arthrosis of knee])			
131876	Date M19 first reported (other arthrosis)			
131878	Date M20 first reported (acquired deformities of fingers and toes)			
131884	Date M23 first reported (internal derangement of knee)			
131888	Date M25 first reported (other joint disorders, not elsewhere classified)			
131916	Date M47 first reported (spondylosis)			
131924	Date M51 first reported (other intervertebral disk disorders)			
131928	Date M54 first reported (dorsalgia)			
131938	Date M65 first reported (synovitis and tenosynovitis)			
131946	Date M70 first reported (soft tissue disorders related to use, overuse and pressure)			
131950	Date M72 first reported (fibroblastic disorders)			
131954	Date M75 first reported (shoulder lesions)			
131958	Date M77 first reported (other enthesopathies)			
131960	Date M79 first reported (other soft tissue disorders, not elsewhere classified)			
131964	Date M81 first reported (osteoporosis without pathological fracture)			
132054	Date N30 first reported (cystitis)			
132070	Date N39 first reported (other disorders of urinary system)			
132072	Date N40 first reported (hyperplasia of prostate)			
132096	Date N60 first reported (benign mammary dysplasia)			
132102	Date N63 first reported (unspecified lump in breast)			
132104	Date N64 first reported (other disorders of breast)			
132124	Date N81 first reported (female genital prolapse)			
132128	Date N83 first reported (noninflammatory disorders of ovary, fallopian tube and broad ligament)			
132130	Date N84 first reported (polyp of female genital tract)			
132140	Date N89 first reported (other noninflammatory disorders of vagina)			
132146	Date N92 first reported (excessive, frequent and irregular menstruation)			
132148	Date N93 first reported (other abnormal uterine and vaginal bleeding)			
132150	Date N94 first reported (pain and other conditions associated with female genital organs and menstrual cycle)			

132152	Date N95 first reported (menopausal and other perimenopausal disorders)				
132276	Date O80 first reported (single spontaneous delivery)				
Derived	TBI (traumatic brain injury)				
Derived	Diabetes				
Derived	CHD (coronary heart Medical history)				
Derived	stroke				
Derived	COPD				
Derived	Depression				
Derived	Arthritis				
Derived	Hypertension				
Derived	Hearing_loss				
Derived	Vision_impairment				
Early life					
1677	Breastfed as a baby	100349	-1	Do not know	set as NA
			-3	Prefer not to answer	set as NA
			0	No	
			1	Yes	
1687	Comparative body size at age 10	100428	-1	Do not know	set as NA
			-3	Prefer not to answer	set as NA
			1	Thinner	
			2	Plumper	
			3	About average	recode: "0"=3
1697	Comparative height size at age 10	100429	-1	Do not know	set as NA
			-3	Prefer not to answer	set as NA
			1	Shorter	recode: "0"=3
			2	Taller	
			3	About average	
1787	Maternal smoking around birth	100349	set -1 -3 as NA		
Psychosocial factors					
1920	Mood swings	100349	set -1 -3 as NA		
1930	Miserableness	100349	set -1 -3 as NA		

1940	Irritability	100349	set -1 -3 as NA		
1950	Sensitivity / hurt feelings	100349	set -1 -3 as NA		
1960	Fed-up feelings	100349	set -1 -3 as NA		
1970	Nervous feelings	100349	set -1 -3 as NA		
1980	Worrier / anxious feelings	100349	set -1 -3 as NA		
1990	Tense / 'highly strung'	100349	set -1 -3 as NA		
2000	Worry too long after embarrassment	100349	set -1 -3 as NA		
2010	Suffer from 'nerves'	100349	set -1 -3 as NA		
2020	Loneliness, isolation	100349	set -1 -3 as NA		
2030	Guilty feelings	100349	set -1 -3 as NA		
2110	Able to confide	100501	-1	Do not know	set as NA
			-3	Prefer not to answer	set as NA
			0	Never or almost never	set as 1
			1	Once every few months	set as 1
			2	About once a month	set as 1
			3	About once a week	set as 0
			4	2-4 times a week	set as 0
			5	Almost daily	set as 0

Supplementary Table 4 Continuous variables and processing

FieldID	Field	Coding			coding notes	
SES						
189	Townsend deprivation index at recruitment					
Physical measures						
23099	Body fat percentage					
23105	Basal metabolic rate					
23111	Leg fat percentage (right)				mean value of left and right	Leg_fat_percentage
23115	Leg fat percentage (left)					
23119	Arm fat percentage (right)				mean value of left and right	arm_fat_percentage
23123	Arm fat percentage (left)					
23127	Trunk fat percentage					
3062	Forced vital capacity (FVC)				mean value of 3 arrays	mean value of z-scores of 3 fields= Lung_function
3063	Forced expiratory volume in 1-second (FEV1)					
3064	Peak expiratory flow (PEF)					
4079	Diastolic blood pressure, automated reading				mean value of 2 arrays	
4080	Systolic blood pressure, automated reading					
46	Hand grip strength (left)				mean value of left and right	hand_grip_strength
47	Hand grip strength (right)					
Lifestyles						
1050	Time spend outdoors in summer	100329	-1	Do not know	set as NA	
			-10	Less than an hour a day	set as 0.5	
			-3	Prefer not to answer	set as NA	
1060	Time spent outdoors in winter	100329				
2139	Age first had sexual intercourse	100504	-1	Do not know	set as NA	
			-2	Never had sex	set as NA	
			-3	Prefer not to answer	set as NA	
2149	Lifetime number of sexual partners	100291			set -1 -3 as NA	
22037	MET minutes per week for walking					
22038	MET minutes per week for moderate activity					

22039	MET minutes per week for vigorous activity					
22040	Summed MET minutes per week for all activity					
1070	Time spent watching television (TV)	100329	-1	Do not know	set as NA	
			-10	Less than an hour a day	set as 0.5	
			-3	Prefer not to answer	set as NA	
1080	Time spent using computer	100329				
1488	Tea intake	100373	-1	Do not know	set as NA	
			-10	Less than one	set as 0.5	
			-3	Prefer not to answer	set as NA	
1498	Coffee intake	100373				
1528	Water intake	100373				
Derived	Healthy_diet_level					
Derived	Vegetable intake					
Local environment						
24003	Nitrogen dioxide air pollution; 2010				Mean values	
24004	Nitrogen oxides air pollution; 2010					
24016	Nitrogen dioxide air pollution; 2005					
24017	Nitrogen dioxide air pollution; 2006					
24018	Nitrogen dioxide air pollution; 2007					
24005	Particulate matter air pollution (pm10); 2010				Mean values	
24019	Particulate matter air pollution (pm10); 2007					
24006	Particulate matter air pollution (pm2.5); 2010					
24008	Particulate matter air pollution 2.5-10um; 2010					
24020	Average daytime sound level of noise pollution					
24021	Average evening sound level of noise pollution					
24022	Average night-time sound level of noise pollution					
24023	Average 16-hour sound level of noise pollution					
24024	Average 24-hour sound level of noise pollution					
24500	Greenspace percentage, buffer 1000m					
24501	Domestic garden percentage, buffer 1000m					
24502	Water percentage, buffer 1000m					

24503	Greenspace percentage, buffer 300m					
24504	Domestic garden percentage, buffer 300m					
24505	Water percentage, buffer 300m					
24506	Natural environment percentage, buffer 1000m					
24507	Natural environment percentage, buffer 300m					

Supplementary Table 5 Baseline characteristics of studied population

	labels	n (%) or mean (sd)
n	max	344341
sex (%)	female	186720 (54.23)
	male	157621 (45.77)
age (mean (SD))		56.01 (8.04)
deprivation index (mean (SD))		-1.52 (2.96)
income (%)	<18,000	64051 (18.60)
	18,000-30,999	90551 (26.30)
	31,000-51,999	97406 (28.29)
	52,000-100,000	74294 (21.58)
	100,000+	18039 (5.24)
higher education (College/University degree and/or Other professional qualifications eg: nursing, teaching) (%)	no	148643 (43.17)
	yes	195698 (56.83)
bmi (mean (SD))		27.35 (4.82)
follow-up dementia (%)	no	339687 (98.65)
	yes	4654 (1.35)
age at diagnosis (mean (SD))		72.8 (5.9)
follow-up time (mean (SD)), years		8.81 (2.85)
APOE ε4 (%)	0	246323 (71.53)
	1	89789 (26.08)
	2	8229 (2.39)

Supplementary Table 6 Baseline characteristics for each genetic *APOE*-risk category

	Low risk ($\epsilon 2\epsilon 2$ or $\epsilon 2\epsilon 3$)	Intermediate risk ($\epsilon 3\epsilon 3$)	High risk ($\epsilon 2\epsilon 4$, $\epsilon 3\epsilon 4$ or $\epsilon 4\epsilon 4$)	p for difference
n	36996	171761	76018	
Female (%)	20234 (54.692)	93933 (54.688)	41625 (54.757)	0.949
dementia diagnosis (%)	268 (0.724)	1590 (0.926)	1953 (2.569)	<0.001
Lifestyle (%)				0.949
Favorable	12362 (33.414)	57121 (33.256)	25266 (33.237)	
Intermediate	12178 (32.917)	56720 (33.023)	25034 (32.932)	
Unfavorable	12456 (33.669)	57920 (33.721)	25718 (33.831)	
Medical history (%)				0.402
Favorable	18942 (51.200)	88552 (51.555)	39172 (51.530)	
Intermediate	6079 (16.432)	27565 (16.048)	12315 (16.200)	
Unfavorable	11975 (32.368)	55644 (32.396)	24531 (32.270)	
Locla environment (%)				0.534
Favorable	12298 (33.241)	56930 (33.145)	25186 (33.132)	
Intermediate	12382 (33.468)	58051 (33.798)	25824 (33.971)	
Unfavorable	12316 (33.290)	56780 (33.058)	25008 (32.897)	
Psychosocial factors (%)				0.503
Favorable	12798 (34.593)	59753 (34.788)	26218 (34.489)	
Intermediate	11637 (31.455)	54275 (31.599)	24074 (31.669)	
Unfavorable	12561 (33.952)	57733 (33.612)	25726 (33.842)	
Physical measures (%)				0.057
Favorable	12974 (35.069)	59412 (34.590)	26519 (34.885)	
Intermediate	11875 (32.098)	54948 (31.991)	24473 (32.194)	
Unfavorable	12147 (32.833)	57401 (33.419)	25026 (32.921)	
Socioeconomic status (%)				0.947
Favorable	13868 (37.485)	64146 (37.346)	28480 (37.465)	
Intermediate	11974 (32.366)	55916 (32.555)	24693 (32.483)	
Unfavorable	11154 (30.149)	51699 (30.099)	22845 (30.052)	

N, number of individuals at risk. Data are presented as the frequency in the stratum (percentage) for categorical values. We compared baseline characteristics across *APOE* strata using chi-squared tests to assess statistical significance and establish two-sided P values.

Supplementary Table 7 Associations between 210 risk factors and dementia in EWAS

FieldID	HR	CI5	CI95	P	case	Full sample	P.value.for.Schoenfeld.residuals	Section	Exposures	Choices	Category
Arthritis	1.1099	1.036	1.1891	0.0030297	48034	344324	0.003540416		Arthritis		Medical history
CHD	1.5055	1.3878	1.6333	6.96E-23	19010	344324	4.80E-06		CHD		Medical history
COPD	1.3926	1.1613	1.67	0.0003532	3819	344324	0.053840777		COPD		Medical history
Depression	1.7731	1.626	1.9335	2.01E-38	31483	344324	0.000522106		Depression		Medical history
Diabetes	1.8559	1.7041	2.0214	9.53E-46	17868	344324	0.314517562		Diabetes		Medical history
Hearing_loss	1.2888	1.1042	1.5042	0.001296	7713	344324	0.396480392		Hearing_loss		Medical history
Hypertension	1.3827	1.2946	1.4768	5.00E-22	44915	344324	0.000152327		Hypertension		Medical history
stroke	2.5015	2.2209	2.8176	1.57E-51	5573	344324	0.033315362		stroke		Medical history
TBI	2.9403	2.4485	3.5309	7.46E-31	2859	344324	7.06E-09		Traumatic brain injury		Medical history
Vision_impairment	1.2543	1.1275	1.3953	3.07E-05	11869	344324	0.121877584		Vision_impairment		Medical history
X130176	0.4992	0.3654	0.6818	1.26E-05	9368	344324	0.859259053		varicella [chickenpox]		Medical history
X130178	1.1535	0.9677	1.375	0.1111014	7139	344324	0.694379392		zoster [herpes zoster]		Medical history
X130188	0.9906	0.8269	1.1868	0.9186292	10928	344324	0.478905702		viral warts		Medical history
X130226	0.9806	0.862	1.1155	0.7656002	19322	344324	0.650726327		dermatophytosis		Medical history
X130622	1.3832	1.1723	1.632	0.0001216	9308	344324	0.926781788		iron deficiency anaemia		Medical history
X130648	1.6175	1.3932	1.8779	2.71E-10	8311	344324	0.121990677		other anaemias		Medical history
X130696	0.992	0.8801	1.1181	0.895345	18831	344324	0.211284901		other hypothyroidism		Medical history
X130792	1.3284	1.1596	1.5216	4.19E-05	11548	344324	0.048785034		obesity		Medical history
X130814	1.1394	1.0676	1.2161	8.48E-05	52392	344324	0.112098068		disorders of lipoprotein metabolism and other lipidaemias		Medical history
X130868	1.1405	0.991	1.3124	0.0666439	16547	344324	0.88088083		mental and behavioural disorders due to use of tobacco		Medical history

X130906	1.6008	1.3727	1.8667	1.97E-09	9663	344324	0.098036919		other anxiety disorders	Medical history
X130910	1.4325	1.1963	1.7152	9.23E-05	9021	344324	0.600431086		reaction to severe stress, and adjustment disorders	Medical history
X131052	0.8953	0.7644	1.0486	0.1700917	16116	344324	0.455558274		migraine)	Medical history
X131060	1.2964	1.1099	1.5143	0.0010573	8246	344324	0.248281879		sleep disorders)	Medical history
X131074	1.1976	1.0354	1.3853	0.0151878	10879	344324	0.929056461		mononeuropathies of upper limb	Medical history
X131128	1.1512	0.9462	1.4006	0.1592586	6965	344324	0.348360644		hordeolum and chalazion	Medical history
X131142	1.0173	0.8798	1.1764	0.8168621	15036	344324	0.525007173		conjunctivitis	Medical history
X131166	1.2157	1.0838	1.3637	0.0008592	10522	344324	0.24367712		other cataract	Medical history
X131222	0.8875	0.7423	1.0611	0.190513	11177	344324	0.734113071		otitis externa	Medical history
X131224	0.9971	0.8696	1.1432	0.9666113	14729	344324	0.314937813		other disorders of external ear	Medical history
X131262	0.8317	0.6709	1.031	0.0926461	8934	344324	0.514416553		otalgia and effusion of ear	Medical history
X131264	1.0347	0.8527	1.2556	0.7294727	6967	344324	0.712948965		other disorders of ear, not elsewhere classified	Medical history
X131350	1.3909	1.2297	1.5733	1.53E-07	7434	344324	0.055866135		atrial fibrillation and flutter	Medical history
X131396	1.1874	1.0267	1.3733	0.0206128	8778	344324	0.020494168		phlebitis and thrombophlebitis	Medical history
X131402	0.8734	0.7487	1.0189	0.0851383	12011	344324	0.614859001		varicose veins of lower extremities	Medical history
X131404	0.9946	0.8644	1.1444	0.9395629	12845	344324	0.770336549		haemorrhoids	Medical history
X131426	0.9008	0.7701	1.0536	0.1913342	16048	344324	0.085271929		acute sinusitis	Medical history
X131428	1.0094	0.8626	1.1811	0.9073379	15395	344324	0.656292065		acute pharyngitis	Medical history
X131430	0.7356	0.5849	0.9252	0.0086702	12065	344324	0.030060337		acute tonsillitis	Medical history
X131436	1.0175	0.9083	1.1399	0.7643227	27967	344324	0.489074843		acute upper respiratory infections of multiple and unspecified sites	Medical history
X131442	0.8263	0.6507	1.0493	0.1175873	7147	344324	0.959774336		influenza, virus not identified	Medical history

X131456	1.1942	1.027	1.3886	0.0210905	8982	344324	0.004943858		pneumonia, organism unspecified	Medical history
X131462	1.1345	1.0242	1.2567	0.0155936	26795	344324	0.399067984		unspecified acute lower respiratory infection	Medical history
X131464	0.9571	0.8603	1.0649	0.4210321	33895	344324	0.018347208		vasomotor and allergic rhinitis	Medical history
X131468	0.8325	0.6732	1.0295	0.0907813	7427	344324	0.783936915		chronic sinusitis	Medical history
X131472	0.961	0.803	1.15	0.6639604	9533	344324	0.490098743		other disorders of nose and nasal sinuses	Medical history
X131494	1.0582	0.9703	1.1541	0.2006994	42554	344324	0.048141039		asthma	Medical history
X131584	1.0274	0.9333	1.1311	0.5812542	27339	344324	0.084511359		gastro-oesophageal reflux Medical history	Medical history
X131598	1.2441	1.1208	1.381	4.11E-05	17868	344324	0.000839129		gastritis and duodenitis	Medical history
X131600	1.1432	0.9657	1.3533	0.1200129	7541	344324	0.545132172		dyspepsia	Medical history
X131612	0.986	0.8698	1.1178	0.8261324	12685	344324	0.440632043		inguinal hernia	Medical history
X131620	1.1237	1.013	1.2465	0.0275297	17509	344324	6.72E-05		diaphragmatic hernia	Medical history
X131630	1.2609	1.1092	1.4334	0.0003931	12514	344324	0.067476581		other non-infective gastro-enteritis and colitis	Medical history
X131636	1.0776	0.967	1.2008	0.1760016	14392	344324	0.129716718		diverticular Medical history of intestine	Medical history
X131638	0.9563	0.8405	1.0879	0.4967252	20308	344324	0.594061813		irritable bowel syndrome	Medical history
X131640	1.534	1.3161	1.7879	4.40E-08	7467	344324	0.236853359		other functional intestinal disorders	Medical history
X131646	1.0279	0.9159	1.1535	0.6403114	18603	344324	0.125607069		other Medical histories of anus and rectum	Medical history
X131650	0.9526	0.8353	1.0863	0.4687656	18312	344324	0.036732598		haemorrhoids and perianal venous thrombosis	Medical history
X131674	1.1044	0.9683	1.2596	0.1388035	12541	344324	0.114375104		cholelithiasis	Medical history
X131702	1.1361	0.9851	1.3103	0.0795742	12238	344324	0.288814252		cellulitis	Medical history
X131708	1.1593	0.9564	1.4053	0.1322121	7111	344324	0.158494609		other local infections of skin and subcutaneous tissue	Medical history
X131720	1.0632	0.8785	1.2867	0.5289569	8935	344324	0.231144216		atopic dermatitis	Medical history

X131740	1.163	1.0483	1.2901	0.004347 3	27556	344324	0.787662568		other dermatitis		Medical history
X131742	1.1642	0.9802	1.3828	0.083305 8	8356	344324	0.135729424		psoriasis		Medical history
X131768	0.9656	0.8215	1.135	0.671202 8	7559	344324	0.192756394		skin changes due to chronic exposure to nonionising radiation		Medical history
X131794	1.0774	0.895	1.2969	0.430834	7881	344324	0.386803298		follicular cysts of skin and subcutaneous tissue		Medical history
X131806	0.8157	0.7074	0.9405	0.005042 8	13591	344324	0.104305762		seborrheic keratosis		Medical history
X131836	0.8245	0.73	0.9313	0.001897 2	22266	344324	0.156227502		other disorders of skin and subcutaneous tissue, not elsewhere classified		Medical history
X131858	1.0295	0.8827	1.2006	0.711255	8093	344324	0.560654526		gout		Medical history
X131864	1.1508	1.005	1.3176	0.042062 8	10383	344324	0.01078465		other arthritis		Medical history
X131872	1.2112	1.0815	1.3564	0.000912 3	12408	344324	0.120094098		gonarthrosis [arthrosis of knee]		Medical history
X131876	1.0785	0.9992	1.164	0.052517 1	36971	344324	0.010713813		other arthrosis		Medical history
X131878	1.1104	0.9541	1.2923	0.176040 2	9702	344324	0.101624619		acquired deformities of fingers and toes		Medical history
X131884	0.9526	0.8055	1.1266	0.570230 8	10056	344324	0.119788024		internal derangement of knee		Medical history
X131888	1.0586	0.9741	1.1505	0.179936 9	45936	344324	0.984996098		other joint disorders, not elsewhere classified		Medical history
X131916	1.37	1.2229	1.5348	5.61E-08	12595	344324	0.072118165		spondylosis		Medical history
X131924	1.1882	1.0367	1.3619	0.013247 6	12129	344324	0.021728402		other intervertebral disk disorders		Medical history
X131928	1.1038	1.0082	1.2084	0.032599 5	38179	344324	0.051597133		dorsalgia		Medical history
X131938	1.0259	0.8623	1.2205	0.773086 1	8302	344324	0.104528181		synovitis and tenosynovitis		Medical history
X131946	1.1146	0.9293	1.3369	0.242189 4	7689	344324	0.042634885		soft tissue disorders related to use, overuse and pressure		Medical history
X131950	0.9673	0.8286	1.1292	0.673761 3	10159	344324	0.782699798		fibroblastic disorders		Medical history
X131954	1.141	1.0055	1.2948	0.040912	14412	344324	0.013460048		shoulder lesions		Medical history
X131958	0.952	0.8334	1.0874	0.468160 9	18978	344324	0.794533858		other enthesopathies		Medical history

X131960	1.0994	1.0074	1.1998	0.033557 5	40614	344324	0.666023873		other soft tissue disorders, not elsewhere classified		Medical history
X131964	1.2651	1.0968	1.4592	0.001241 9	7387	344324	0.211127629		osteoporosis without pathological fracture		Medical history
X132054	0.9733	0.8136	1.1642	0.767002 6	9716	344324	0.000872808		cystitis		Medical history
X132070	1.3158	1.1875	1.458	1.58E-07	21594	344324	0.016469219		other disorders of urinary system		Medical history
X132072	1.025	0.9152	1.1479	0.669726 1	11447	344324	0.197387144		hyperplasia of prostate		Medical history
X132096	0.8696	0.7062	1.0708	0.188306 9	8220	344324	0.30335981		benign mammary dysplasia		Medical history
X132102	1.1087	0.9187	1.338	0.28188	11054	344324	0.773560759		unspecified lump in breast		Medical history
X132104	1.1535	0.9326	1.4268	0.187975 3	8059	344324	0.350217845		other disorders of breast		Medical history
X132124	1.2356	1.0516	1.4517	0.010112 1	7874	344324	0.583095261		female genital prolapse		Medical history
X132128	1.0448	0.8333	1.3099	0.704052 6	7527	344324	0.479878282		noninflammatory disorders of ovary, fallopian tube and broad ligament		Medical history
X132130	0.8817	0.7247	1.0729	0.20864	9771	344324	0.461252892		polyp of female genital tract		Medical history
X132140	0.885	0.6954	1.1263	0.320525 7	7619	344324	0.131245018		other noninflammatory disorders of vagina		Medical history
X132146	1.0315	0.8602	1.2369	0.737836 8	20684	344324	0.986766603		excessive, frequent and irregular menstruation		Medical history
X132148	0.9718	0.7296	1.2945	0.845174 5	7191	344324	0.011956418		other abnormal uterine and vaginal bleeding		Medical history
X132150	0.9183	0.7021	1.2011	0.533895 8	8630	344324	0.41563636		pain and other conditions associated with female genital organs and menstrual cycle		Medical history
X132152	0.9466	0.8328	1.0759	0.400758 6	19758	344324	0.360096345		menopausal and other perimenopausal disorders		Medical history
X132276	1.2111	0.9974	1.4708	0.053196 9	11919	344324	0.428252824		single spontaneous delivery		Medical history
X6146	2.3513	2.1602	2.5593	5.79E-87	16440	344324	3.26E-17		Disability		Medical history
X6149_1	1.0562	0.9597	1.1624	0.263099 4	35449	344324	0.648188546		Mouth/teeth dental problems	Mouth ulcers	Medical history
X6149_2	1.4716	1.2742	1.6995	1.45E-07	10232	344324	0.012709417		Mouth/teeth dental problems	Painful gums	Medical history
X6149_3	0.9943	0.9026	1.0953	0.907612 8	47589	344324	0.317454792		Mouth/teeth dental problems	Bleeding gums	Medical history

X6149_4	1.2308	1.0842	1.3973	0.001329 3	13955	344324	0.198211146		Mouth/teeth dental problems	Loose teeth	Medical history
X6149_5	1.4403	1.2664	1.6381	2.76E-08	15912	344324	0.592116018		Mouth/teeth dental problems	Toothache	Medical history
X6149_6	1.2847	1.2037	1.3713	4.95E-14	47694	344324	0.230699703		Mouth/teeth dental problems	Dentures	Medical history
X6149_7	0.8214	0.7752	0.8703	2.65E-11	214685	344324	0.157728248		Mouth/teeth dental problems	non of the above	Medical history
X1677	0.9801	0.9046	1.062	0.623922 6	261942	344324	0.074867184		Breastfed as a baby		Early life
X1687	1.0308	0.9671	1.0987	0.351497 9	112516	344324	0.067161182		Comparative body size at age 10	Thinner	Early life
X1687	1.082	0.9936	1.1782	0.069848	54807	344324	0.067161182		Comparative body size at age 10	Plumper	Early life
X1697	1.0253	0.9508	1.1055	0.516463 6	68896	344324	0.016278614		Comparative height size at age 10	Shorter	Early life
X1697	0.914	0.8531	0.9792	0.010586 1	91603	344324	0.016278614		Comparative height size at age 10	Taller	Early life
X1787	0.9902	0.9247	1.0605	0.779260 5	92071	344324	0.064928248		Maternal smoking around birth		Early life
NOx	0.9982	0.9281	1.0735	0.960557 2	114780	344324	0.053643286	(27.5,34.3]	Nitrogen oxides		Local environment
NOx	1.1399	1.051	1.2363	0.001580 4	115441	344324	0.053643286	(34.3,153]	Nitrogen oxides		Local environment
PM10	0.9232	0.8583	0.993	0.031604 3	113707	344324	0.373488221	(18.4,19.8]	Particulate matter air pollution (pm10)		Local environment
PM10	1.0592	0.9765	1.1489	0.165624 2	113525	344324	0.373488221	(19.8,57]	Particulate matter air pollution (pm10)		Local environment
X24006	0.997	0.9275	1.0717	0.934243 9	113686	344324	0.057714317	(9.54,10.2]	Particulate matter air pollution (pm2.5)		Local environment
X24006	1.1685	1.0859	1.2574	3.16E-05	116542	344324	0.057714317	(10.2,21.3]	Particulate matter air pollution (pm2.5)		Local environment
X24008	0.9494	0.8845	1.0191	0.151033 7	113521	344324	0.610440154	(5.94,6.39]	Particulate matter air pollution 2.5-10um		Local environment
X24008	0.9385	0.8728	1.0092	0.086595 4	115260	344324	0.610440154	(6.39,12.8]	Particulate matter air pollution 2.5-10um		Local environment
X24020	0.9504	0.8861	1.0194	0.154677 5	113675	344324	0.41414967	(53.5,55.4]	Average daytime sound level of noise pollution		Local environment
X24020	0.95	0.8853	1.0194	0.153925 5	116926	344324	0.41414967	(55.4,88.5]	Average daytime sound level of noise pollution		Local environment
X24021	0.9968	0.9283	1.0704	0.930042 9	102012	308144	0.457385957	(49.6,51.6]	Average evening sound level of noise pollution		Local environment

X24021	0.9823	0.914	1.0557	0.627559 5	104623	308144	0.457385957	(51.6,84.8]	Average evening sound level of noise pollution	Local environment
X24022	0.9931	0.9249	1.0664	0.849136 8	101552	308143	0.43748232	(44.5,46.5]	Average night-time sound level of noise pollution	Local environment
X24022	0.9799	0.9119	1.053	0.580924 7	104765	308143	0.43748232	(46.5,79.7]	Average night-time sound level of noise pollution	Local environment
X24023	0.9996	0.9309	1.0734	0.991368 3	101617	308139	0.473148154	(52.4,54.4]	Average 16-hour sound level of noise pollution	Local environment
X24023	0.9824	0.9142	1.0558	0.629152 6	104914	308139	0.473148154	(54.4,87.6]	Average 16-hour sound level of noise pollution	Local environment
X24024	0.9991	0.9304	1.0729	0.980356 2	101633	308137	0.472128733	(54,56]	Average 24-hour sound level of noise pollution	Local environment
X24024	0.9818	0.9136	1.0551	0.616664 4	104930	308137	0.472128733	(56,89.2]	Average 24-hour sound level of noise pollution	Local environment
X24500	0.8702	0.8055	0.9401	0.000418 2	118186	344324	0.093978484	(51.6,99.2]	Greenspace percentage, buffer 1000m	Local environment
X24500	0.8552	0.7932	0.922	4.58E-05	114155	344324	0.093978484	(32.5,51.6]	Greenspace percentage, buffer 1000m	Local environment
X24501	0.8913	0.8284	0.959	0.002057 3	114278	344324	0.325660421	(20.3,28.9]	Domestic garden percentage, buffer 1000m	Local environment
X24501	1.0253	0.9524	1.1037	0.506587 5	115873	344324	0.325660421	(28.9,66.1]	Domestic garden percentage, buffer 1000m	Local environment
X24502	0.9082	0.8448	0.9763	0.009073 6	114020	344324	0.09486875	(0.341,0.978]	Water percentage, buffer 1000m	Local environment
X24502	0.9452	0.8791	1.0163	0.127722 3	116825	344324	0.09486875	(0.978,59.4]	Water percentage, buffer 1000m	Local environment
X24503	0.9173	0.853	0.9864	0.019848 8	114175	344324	0.055198067	(21.6,39.4]	Greenspace percentage, buffer 300m	Local environment
X24503	0.8764	0.8137	0.9441	0.000504 2	117725	344324	0.055198067	(39.4,99.2]	Greenspace percentage, buffer 300m	Local environment
X24504	0.9248	0.8595	0.995	0.036357 4	113996	344324	0.879809443	(26.4,37.8]	Domestic garden percentage, buffer 300m	Local environment
X24504	0.98	0.9124	1.0527	0.580684 6	116731	344324	0.879809443	(37.8,77.4]	Domestic garden percentage, buffer 300m	Local environment
X24505	0.9227	0.8585	0.9917	0.028737 6	114591	344324	0.084529615	(0.057,0.448]	Water percentage, buffer 300m	Local environment
X24505	0.8625	0.8017	0.9281	7.52E-05	116702	344324	0.084529615	(0.448,91.8]	Water percentage, buffer 300m	Local environment
X24506	0.8692	0.8045	0.939	0.000378 5	117788	344324	0.068233285	(49.3,100]	Natural environment percentage, buffer 1000m	Local environment
X24506	0.8602	0.7978	0.9276	8.99E-05	114288	344324	0.068233285	(25.7,49.3]	Natural environment percentage, buffer 1000m	Local environment

X24507	0.9139	0.8499	0.9827	0.0150905	113190	344324	0.390108344	(10.9,30.4]	Natural environment percentage, buffer 300m	Local environment
X24507	0.8586	0.7972	0.9247	5.64E-05	117342	344324	0.390108344	(30.4,100]	Natural environment percentage, buffer 300m	Local environment
Alcohol_level	1.1497	1.0834	1.2201	4.16E-06	193924	344324	0.712644285		Improper consumption of alcohol	Lifestyles
Ever_smoke	1.05	0.9905	1.1131	0.1010316	152384	344324	0.855389318		Ever_smoke	Lifestyles
fish	0.9736	0.9019	1.051	0.4925879	266779	344324	0.93014719		Fish eaten above recommendation	Lifestyles
fruit	1.0604	0.9993	1.1253	0.052895	176398	344324	0.449722315		Fruit eaten above recommendation	Lifestyles
Healthy_diet_level	1.0734	1.003	1.1488	0.0408132	84899	344324	0.222776213	(3,4]	Healthy diet score	Lifestyles
Healthy_diet_level	1.0679	0.9802	1.1635	0.1326964	45002	344324	0.222776213	(4,8]	Healthy diet score	Lifestyles
processed_meat	0.9914	0.933	1.0535	0.7811285	139960	344324	0.307044337		Processed meat intake reduced	Lifestyles
red_meat	1.0385	0.9795	1.101	0.2053274	156193	344324	0.754016035		Red meat intake reduced	Lifestyles
Sleep_level	1.2512	1.1789	1.328	1.66E-13	107402	344324	0.066003631		Improper sleep duration	<7/>9 hours
vegetable	0.9545	0.8884	1.0255	0.2036388	120642	344324	0.034678991	(3,5]	vegetable	Lifestyles
vegetable	1.0767	1.0016	1.1574	0.0450201	106812	344324	0.034678991	(5,100]	vegetable	Lifestyles
white_bread_refined_cereals	1.0069	0.9084	1.116	0.8964223	314599	344324	0.76068848		White bread/refined cereals intake reduced	Lifestyles
wholegrain	1.0935	0.8772	1.3631	0.4267152	4312	344324	0.00964492		Wholegrain eaten above recommendation	Lifestyles
X1050	1.0438	0.9692	1.1242	0.2568691	119951	344324	0.184045979	(2,4]	Time spend outdoors in summer	Lifestyles
X1050	1.0984	1.0195	1.1834	0.0136109	99217	344324	0.184045979	(4,24]	Time spend outdoors in summer	Lifestyles
X1060	1.0818	1.009	1.1598	0.0269495	86400	344324	0.151632396	(1,2]	Time spent outdoors in winter	Lifestyles
X1060	1.1651	1.0855	1.2504	2.28E-05	72395	344324	0.151632396	(2,24]	Time spent outdoors in winter	Lifestyles
X1070	1.0992	1.0205	1.184	0.0125553	82365	344324	0.303193775	(2,3]	Time spent watching television (TV)	Lifestyles
X1070	1.3439	1.2566	1.4373	6.38E-18	87116	344324	0.303193775	(3,24]	Time spent watching television (TV)	Lifestyles
X1080	1.3458	1.2532	1.4453	3.23E-16	153257	344324	0.769228924	[0,0.5]	Time spent using computer	Lifestyles
X1080	1.2197	1.1246	1.3228	1.61E-06	84747	344324	0.769228924	(1,24]	Time spent using computer	Lifestyles

X1120	1.0138	0.9475	1.0848	0.691663 8	141982	344324	0.171343523		Weekly usage of mobile phone in last 3 months	5-29 mins	Lifestyles
X1120	0.9301	0.8437	1.0253	0.144856 1	56494	344324	0.171343523		Weekly usage of mobile phone in last 4 months	30-59 mins	Lifestyles
X1120	0.9757	0.8704	1.0937	0.673021 7	45926	344324	0.171343523		Weekly usage of mobile phone in last 5 months	1-3 hours	Lifestyles
X1120	0.9375	0.7449	1.1798	0.581974 9	12781	344324	0.171343523		Weekly usage of mobile phone in last 6 months	4-6 hours	Lifestyles
X1120	0.8632	0.6663	1.1183	0.265586 7	13330	344324	0.171343523		Weekly usage of mobile phone in last 7 months	More than 6 hours	Lifestyles
X1130	0.8793	0.7678	1.007	0.063056 6	29825	344324	0.674783657		Hands-free device/speakerphone use with mobile phone in last 3 month	Less than half the time	Lifestyles
X1130	0.846	0.6734	1.0628	0.150757 3	11221	344324	0.674783657		Hands-free device/speakerphone use with mobile phone in last 4 month	About half the time	Lifestyles
X1130	0.9343	0.691	1.2631	0.658525 1	6279	344324	0.674783657		Hands-free device/speakerphone use with mobile phone in last 5 month	More than half the time	Lifestyles
X1130	1.0603	0.8455	1.3296	0.612393 5	7908	344324	0.674783657		Hands-free device/speakerphone use with mobile phone in last 6 month	Always or almost always	Lifestyles
X1190	1.0719	1.0086	1.1392	0.025477 2	129354	344324	0.001793452		Nap during day	Sometimes	Lifestyles
X1190	1.2465	1.1211	1.3859	4.64E-05	17427	344324	0.001793452		Nap during day	Usually	Lifestyles
X1200	0.8252	0.7679	0.8869	1.73E-07	164167	344324	0.00505386		Sleeplessness / insomnia	Sometimes	Lifestyles
X1200	0.8764	0.8105	0.9478	0.000954 8	95487	344324	0.00505386		Sleeplessness / insomnia	Usually	Lifestyles
X1210	0.9366	0.8809	0.9958	0.036178 7	126667	344324	0.841848234		Snoring		Lifestyles
X1220	1.4891	1.2939	1.7138	2.78E-08	8907	344324	0.001957812		Daytime dozing / sleeping (narcolepsy)	Often	Lifestyles
X1220	1.1517	1.079	1.2293	2.19E-05	69457	344324	0.001957812		Daytime dozing / sleeping (narcolepsy)	Sometimes	Lifestyles
X1220	1.8054	0.2535	12.858	0.555339	22	344324	0.001957812		Daytime dozing / sleeping (narcolepsy)	All of the time	Lifestyles
X1269	1.3159	1.1646	1.4869	1.06E-05	16352	344324	0.067485459		Exposure to tobacco smoke at home	NA	Lifestyles
X1279	1.1294	1.0485	1.2164	0.001320 2	62082	344324	0.018714028		Exposure to tobacco smoke outside home	NA	Lifestyles
X1488	0.8951	0.8346	0.96	0.001914 7	100260	344324	0.388670383	(4,99]	Tea intake		Lifestyles

X1488	0.8527	0.7958	0.9138	6.30E-06	101032	344324	0.388670383	(2,4]	Tea intake		Lifestyles
X1498	1.0096	0.9358	1.0893	0.804716 4	66214	344324	0.592177768	(1,2]	Coffee intake		Lifestyles
X1498	1.0482	0.9816	1.1194	0.159474 1	111327	344324	0.592177768	(2,80]	Coffee intake		Lifestyles
X1528	1.0385	0.9557	1.1285	0.372833 9	53140	344324	0.063135462	(2,3]	Water intake		Lifestyles
X1528	1.2526	1.1714	1.3395	4.50E-11	101510	344324	0.063135462	(3,80]	Water intake		Lifestyles
X2139	0.9731	0.9005	1.0517	0.491823 3	125468	344324	0.883103093	(17.4,20]	Age first had sexual intercourse		Lifestyles
X2139	0.9379	0.8685	1.0129	0.102218 4	104502	344324	0.883103093	(20,69]	Age first had sexual intercourse		Lifestyles
X2149	1.0901	1.02	1.165	0.010955 3	122557	344324	0.070373676	(2,6]	Lifetime number of sexual partners		Lifestyles
X2149	1.0577	0.9784	1.1435	0.158332 7	99905	344324	0.070373676	(6,1.7e+04]	Lifetime number of sexual partners		Lifestyles
X2159	1.0007	0.8269	1.2111	0.994235	11958	344324	0.056624961		Ever had same-sex intercourse		Lifestyles
X22036	1.1026	1.0206	1.1911	0.013265 1	60154	344324	0.054447727		Below moderate/vigorous/walking recommendation		Lifestyles
X22037	0.9916	0.9228	1.0655	0.818514	104271	344324	0.179498102	(462,1.16e+03]	MET minutes per week for walking		Lifestyles
X22037	1.0593	0.9886	1.135	0.102077 3	116420	344324	0.179498102	(1.16e+03,4.16e+03]	MET minutes per week for walking		Lifestyles
X22038	0.9232	0.8591	0.9921	0.029607	111475	344324	0.344505381	(240,840]	MET minutes per week for moderate activity		Lifestyles
X22038	0.9237	0.861	0.991	0.026863 4	110864	344324	0.344505381	(840,5.04e+03]	MET minutes per week for moderate activity		Lifestyles
X22039	0.8353	0.7784	0.8963	5.72E-07	115911	344324	0.378686926	(560,1.01e+04]	MET minutes per week for vigorous activity		Lifestyles
X22039	0.8624	0.8052	0.9236	2.36E-05	113344	344324	0.378686926	(41.6,560]	MET minutes per week for vigorous activity		Lifestyles
X22040	0.9755	0.9082	1.0477	0.495676 2	113250	344324	0.477233903	(1.23e+03,2.72e+03]	Summed MET minutes per week for all activity		Lifestyles
X22040	0.9496	0.8848	1.0192	0.151653 3	117592	344324	0.477233903	(2.72e+03,1.93e+04]	Summed MET minutes per week for all activity		Lifestyles
X6155_0	0.9288	0.8745	0.9866	0.016449 3	231524	344324	0.248381314		No vitamin and mineral supplements		Lifestyles
X6155_1	1.0129	0.8442	1.2153	0.890615	6862	344324	0.247453395		Vitamin A supplements		Lifestyles
X6155_2	1.2645	1.1188	1.4292	0.000172 3	15246	344324	0.595432667		Vitamin B supplements		Lifestyles
X6155_3	1.0031	0.9117	1.1036	0.949572 1	31120	344324	0.109010757		Vitamin C supplements		Lifestyles
X6155_4	0.9724	0.8536	1.1076	0.673122 9	14464	344324	0.718436748		Vitamin D supplements		Lifestyles

X6155_5	1.108	0.9583	1.2811	0.1659907	10787	344324	0.194841042		Vitamin E supplements		Lifestyles
X6155_6	1.3411	1.1503	1.5636	0.0001784	8027	344324	0.62465868		Folic acid or Folate (Vit B9) supplements		Lifestyles
X6155_7	1.0302	0.962	1.1033	0.3946209	78837	344324	0.250681522		Multivitamins +/- minerals supplements		Lifestyles
X6160_1	0.746	0.6974	0.798	1.56E-17	109745	344324	0.239231202		Leisure/social activities	Sports club or gym	Lifestyles
X6160_2	0.9789	0.9139	1.0485	0.5429456	88199	344324	0.80333756		Leisure/social activities	Pub or social club	Lifestyles
X6160_3	1.0306	0.9602	1.1061	0.4034085	57555	344324	0.04943537		Leisure/social activities	Religious group	Lifestyles
X6160_4	0.7788	0.7001	0.8663	4.16E-06	28321	344324	0.080055063		Leisure/social activities	Adult education class	Lifestyles
X6160_5	0.9176	0.8592	0.98	0.010429	79714	344324	0.000307513		Leisure/social activities	Other group activity	Lifestyles
X6160_7	1.1875	1.1139	1.266	1.42E-07	98944	344324	0.02186813		Leisure/social activities	non of the above	Lifestyles
X6179_1	0.9545	0.8994	1.0131	0.1253222	108169	344324	0.142980861		Fish oil (including cod liver oil) supplements		Lifestyles
X6179_2	0.8652	0.8079	0.9265	3.41E-05	69158	344324	0.073929614		Glucosamine supplements		Lifestyles
X6179_3	1.1325	1.0259	1.2501	0.0136244	24958	344324	0.06100856		Calcium supplements		Lifestyles
X6179_4	0.9406	0.8167	1.0833	0.3953895	15129	344324	0.75815336		Zinc supplements		Lifestyles
X6179_5	1.2849	1.0933	1.5102	0.0023494	11877	344324	0.081558502		Iron supplements		Lifestyles
X6179_6	0.9969	0.8488	1.1709	0.9700411	9009	344324	0.591996945		Selenium supplements		Lifestyles
X6179_7	1.0453	0.9861	1.1081	0.1367105	194487	344324	0.725311976		No mineral or other dietary supplements		Lifestyles
X1920	1.2583	1.1871	1.3338	1.07E-14	152718	344324	0.468937877		Mood swings		Psychosocial factors
X1930	1.143	1.0769	1.2132	1.10E-05	146640	344324	0.932767451		Miserableness		Psychosocial factors
X1940	0.9749	0.9118	1.0424	0.4565939	97803	344324	0.254575282		Irritability		Psychosocial factors
X1950	1.0319	0.9733	1.0941	0.292095	189957	344324	0.457014509		Sensitivity / hurt feelings		Psychosocial factors
X1960	1.2235	1.1522	1.2993	4.63E-11	135825	344324	0.156923764		Fed-up feelings		Psychosocial factors
X1970	1.1359	1.0609	1.2161	0.0002529	77925	344324	0.78717427		Nervous feelings		Psychosocial factors
X1980	1.1509	1.0849	1.2209	3.13E-06	193614	344324	0.72342634		Worrier / anxious feelings		Psychosocial factors

X1990	1.1799	1.095	1.2715	1.43E-05	58843	344324	0.174918046		Tense / 'highly strung'		Psychosocial factors
X2000	0.9179	0.8657	0.9733	0.0041479	165968	344324	0.984870045		Worry too long after embarrassment		Psychosocial factors
X2010	1.2735	1.1879	1.3653	9.94E-12	71626	344324	0.021589869		Suffer from 'nerves'		Psychosocial factors
X2020	1.4224	1.3228	1.5294	1.81E-21	61496	344324	0.011732253		Loneliness, isolation		Psychosocial factors
X2030	1.0219	0.9568	1.0914	0.5183582	100077	344324	0.443300212		Guilty feelings		Psychosocial factors
X2110	1.2103	1.1365	1.2888	2.71E-09	83791	344324	0.007308975		Able to confide	Rarely	Psychosocial factors
arm_fat_percentage	1.0412	0.9667	1.1216	0.2865564	112874	344324	0.010035705	(23.4,33.2]	Arm fat percentage		Physical measures
arm_fat_percentage	1.0111	0.9169	1.1148	0.8254171	118950	344324	0.010035705	(33.2,74]	Arm fat percentage		Physical measures
hand_grip_strength	0.487	0.4395	0.5396	5.59E-43	115201	344324	0.34818752	(35,85]	Hand grip strength		Physical measures
hand_grip_strength	0.6989	0.6429	0.7598	4.17E-17	107971	344324	0.34818752	(25,35]	Hand grip strength		Physical measures
Leg_fat_percentage	0.9681	0.8901	1.0529	0.4491695	113234	344324	0.002392657	(24.4,38.1]	Leg fat percentage		Physical measures
Leg_fat_percentage	0.8677	0.7681	0.9803	0.0226603	118615	344324	0.002392657	(38.1,64.1]	Leg fat percentage		Physical measures
Lung_function	0.5587	0.5085	0.6139	9.42E-34	114852	344324	0.412550359	(0.383,22.4]	Lung_function		Physical measures
Lung_function	0.7387	0.6843	0.7974	8.51E-15	113086	344324	0.412550359	(-0.396,0.383]	Lung_function		Physical measures
X23099	0.9464	0.8787	1.0194	0.1459981	113971	344324	0.000380626	(26.7,34.6]	Body fat percentage		Physical measures
X23099	0.9173	0.8331	1.01	0.078827	119031	344324	0.000380626	(34.6,69.3]	Body fat percentage		Physical measures
X23105	0.8493	0.7777	0.9274	0.0002746	114265	344324	0.133766681	(5.72e+03,7.21e+03]	Basal metabolic rate		Physical measures
X23105	0.7042	0.6282	0.7894	1.79E-09	117316	344324	0.133766681	(7.21e+03,1.55e+04]	Basal metabolic rate		Physical measures
X23127	0.8804	0.8186	0.9467	0.000591	113285	344324	0.000790574	(27.4,33.9]	Trunk fat percentage		Physical measures
X23127	0.937	0.8679	1.0115	0.0955266	119952	344324	0.000790574	(33.9,77.6]	Trunk fat percentage		Physical measures
X4079	0.9297	0.8665	0.9975	0.0424333	115094	344324	0.011378715	(77.5,86]	Diastolic blood pressure		Physical measures

X4079	0.8827	0.8218	0.948	0.000611 5	113663	344324	0.011378715	(86,148]	Diastolic blood pressure	Physical measures
X4080	0.9457	0.8717	1.026	0.179093 5	116010	344324	0.418313691	(128,143]	Systolic blood pressure	Physical measures
X4080	0.9519	0.881	1.0285	0.212061 4	117194	344324	0.418313691	(143,268]	Systolic blood pressure	Physical measures
X189	1.0137	0.9439	1.0887	0.708367	114062	344324	0.09363826	(-3.29,0.98]	Townsend deprivation index at recruitment	SES
X189	1.242	1.1536	1.3371	8.64E-09	117055	344324	0.09363826	(-0.98,11]	Townsend deprivation index at recruitment	SES
X6138	1.1677	1.102	1.2374	1.54E-07	148635	344324	0.132427421		Lower education	SES
X6142	1.9643	1.7216	2.2412	1.08E-23	20965	344324	0.000134085		Unemployed	SES
X738	0.5041	0.4508	0.5636	2.60E-33	74294	344324	0.000844076		Average total household income before tax	52,000 to 100,000 SES
X738	0.6173	0.5685	0.6704	2.04E-30	97400	344324	0.000844076		Average total household income before tax	31,000 to 51,999 SES
X738	0.7819	0.7295	0.8381	3.66E-12	90543	344324	0.000844076		Average total household income before tax	18,000 to 30,999 SES
X738	0.4721	0.3783	0.5891	3.09E-11	18039	344324	0.000844076		Average total household income before tax	Greater than 100,000 SES

Z-tests were used to assess statistical significance and derive Z statistics and corresponding two-sided P values.

Supplementary Table 8 Associations between 210 risk factors and dementia in EWAS (by age and length of follow up).

	Whole sample		Age groups				Follow-up time							
			<65 years		≥65 years		≥ 5 years		≥ 10 years					
FieldID	HR	P.adjust	HR	P	HR	P	HR	P	HR	P	Section	Exposures	Choices	Category
X1488	0.8527	0.0016889	0.8697	0.0083391	0.8439	0.0003385	0.8668	0.0001338	0.8174	0.0001567	(2,4]	Tea intake		Lifestyles
X1488	0.8951	0.5131397	0.9102	0.0731241	0.8871	0.0139195	0.9132	0.0164897	0.9054	0.0624397	(4,99]	Tea intake		Lifestyles
X1498	1.0096	215.66398	1.042	0.4806163	0.9813	0.7152089	0.9951	0.9058844	1.0097	0.8690845	(1,2]	Coffee intake		Lifestyles
X1498	1.0482	42.739056	1.0852	0.0975499	1.014	0.7598823	1.025	0.48715	1.0492	0.3395585	(2,80]	Coffee intake		Lifestyles
X1528	1.0385	99.919481	1.0122	0.8512004	1.0571	0.3245066	1.0109	0.8100993	0.8678	0.0339005	(2,3]	Water intake		Lifestyles
X1528	1.2526	1.21E-08	1.2903	2.82E-07	1.2143	4.03E-05	1.236	5.26E-09	1.1827	0.000998	(3,80]	Water intake		Lifestyles
vegetable	0.9545	54.575211	0.9219	0.1377902	0.9826	0.7216854	0.9482	0.1696143	0.9711	0.5891521	(3,5]	vegetable		Lifestyles
vegetable	1.0767	12.065397	1.1386	0.0157206	1.0342	0.5078392	1.0473	0.2378628	0.9764	0.6710943	(5,100]	vegetable		Lifestyles
Healthy_diet_level	1.0734	10.937943	1.0631	0.2377073	1.0852	0.0790637	1.0788	0.0389024	1.0143	0.7867351	(3,4]	Healthy diet score		Lifestyles
Healthy_diet_level	1.0679	35.562637	1.0919	0.1748013	1.052	0.3925037	1.0894	0.0636909	1.0439	0.5132057	(4,8]	Healthy diet score		Lifestyles
X24006	0.997	250.37738	0.986	0.8008288	1.0031	0.9497253	0.9909	0.8150499	0.8997	0.055732	(9.54,10.2]	Particulate matter air pollution (pm2.5)		Local environment
X24006	1.1685	0.0084742	1.2419	8.49E-05	1.0996	0.0633004	1.1599	0.0001856	1.0403	0.481876	(10.2,21.3]	Particulate matter air pollution (pm2.5)		Local environment
X24008	0.9494	40.477023	0.9661	0.5234121	0.9402	0.2057224	0.9421	0.1192428	0.9719	0.5939029	(5.94,6.39]	Particulate matter air pollution 2.5-10um		Local environment
X24008	0.9385	23.207554	0.9419	0.2758836	0.9384	0.2052336	0.9183	0.0300206	0.8646	0.0094294	(6.39,12.8]	Particulate matter air pollution 2.5-10um		Local environment
X24020	0.9504	41.453583	0.9491	0.3259766	0.9494	0.2822946	0.9756	0.5144472	0.9749	0.6358916	(53.5,55.4]	Average daytime sound level of noise pollution		Local environment
X24020	0.95	41.252028	0.936	0.2167238	0.9637	0.4470223	0.9666	0.3741537	0.9699	0.5719013	(55.4,88.5]	Average daytime sound level of noise pollution		Local environment
X24021	0.9968	249.2515	1.0242	0.6598015	0.9733	0.580723	1.0261	0.5045137	1.015	0.7853938	(49.6,51.6]	Average evening sound level of noise pollution		Local environment
X24021	0.9823	168.18594	0.9881	0.8276791	0.9797	0.6784363	0.9997	0.9943561	1.0028	0.9597068	(51.6,84.8]	Average evening sound level of noise pollution		Local environment
X24022	0.9931	227.56865	1.0197	0.7191231	0.9701	0.5349753	1.0222	0.5690286	1.0181	0.7436643	(44.5,46.5]	Average night-time sound level of noise pollution		Local environment
X24022	0.9799	155.68781	0.986	0.7969787	0.977	0.6379547	0.9973	0.9445701	1.0025	0.9643603	(46.5,79.7]	Average night-time sound level of		Local environment

												noise pollution		
X24023	0.9996	265.6867	1.0281	0.6097121	0.975	0.6056959	1.0287	0.4638539	1.0172	0.7563493	(52.4,54.4]	Average 16-hour sound level of noise pollution		Local environment
X24023	0.9824	168.61291	0.9897	0.8497671	0.9785	0.660358	1	0.9999649	1.0015	0.9780626	(54.4,87.6]	Average 16-hour sound level of noise pollution		Local environment
X24024	0.9991	262.73547	1.0268	0.6260276	0.9752	0.608759	1.0281	0.471957	1.0169	0.7599777	(54,56]	Average 24-hour sound level of noise pollution		Local environment
X24024	0.9818	165.26607	0.9886	0.8344161	0.9782	0.6564697	0.9993	0.986208	1.0008	0.9881026	(56,89.2]	Average 24-hour sound level of noise pollution		Local environment
X24500	0.8552	0.0122638	0.8593	0.0072821	0.8527	0.0023369	0.8555	0.000134	0.9415	0.3027528	(32.5,51.6]	Greenspace percentage, buffer 1000m		Local environment
X24500	0.8702	0.1120735	0.8664	0.0140443	0.8771	0.0142307	0.8828	0.0028573	0.9993	0.991088	(51.6,99.2]	Greenspace percentage, buffer 1000m		Local environment
X24501	0.8913	0.5513454	0.9155	0.1079526	0.8649	0.004369	0.8996	0.007439	0.9168	0.1131416	(20.3,28.9]	Domestic garden percentage, buffer 1000m		Local environment
X24501	1.0253	135.76545	1.0032	0.9550114	1.0405	0.4321164	1.0301	0.4569997	0.9543	0.4117662	(28.9,66.1]	Domestic garden percentage, buffer 1000m		Local environment
X24502	0.9082	2.4317354	0.9108	0.0914806	0.909	0.054084	0.909	0.014581	0.8713	0.0132049	(0.341,0.978]	Water percentage, buffer 1000m		Local environment
X24502	0.9452	34.229569	0.9952	0.9303979	0.9044	0.0463045	0.9321	0.073602	0.9365	0.2354651	(0.978,59.4]	Water percentage, buffer 1000m		Local environment
X24503	0.9173	5.319465	0.9173	0.1137393	0.9193	0.0958538	0.9197	0.0335035	0.9311	0.2070127	(21.6,39.4]	Greenspace percentage, buffer 300m		Local environment
X24503	0.8764	0.1351285	0.8615	0.0083747	0.8925	0.0261188	0.8768	0.0010886	0.9705	0.5980249	(39.4,99.2]	Greenspace percentage, buffer 300m		Local environment
X24504	0.9248	9.7437964	0.9443	0.2979694	0.9042	0.0475952	0.9463	0.1632578	0.9652	0.522916	(26.4,37.8]	Domestic garden percentage, buffer 300m		Local environment
X24504	0.98	155.62349	0.9996	0.9945953	0.9638	0.4532568	0.9936	0.8692408	0.9639	0.5056123	(37.8,77.4]	Domestic garden percentage, buffer 300m		Local environment
X24505	0.9227	7.7016633	1.0063	0.9097239	0.8641	0.0030967	0.9228	0.0393335	0.9737	0.6271529	(0.057,0.448]	Water percentage, buffer 300m		Local environment
X24505	0.8625	0.0201428	0.9797	0.7112386	0.7769	6.29E-07	0.863	0.0001988	0.8625	0.0089658	(0.448,91.8]	Water percentage, buffer 300m		Local environment
X24506	0.8602	0.0241016	0.8288	0.0008505	0.8911	0.0287541	0.8509	8.05E-05	0.9302	0.2159287	(25.7,49.3]	Natural environment percentage, buffer 1000m		Local environment
X24506	0.8692	0.1014277	0.8389	0.0025026	0.8992	0.0484048	0.8763	0.0015779	0.9786	0.7167095	(49.3,100]	Natural environment percentage, buffer 1000m		Local environment
X24507	0.9139	4.0442617	0.9532	0.3788811	0.8834	0.0142788	0.9282	0.0586133	0.9448	0.31258	(10.9,30.4]	Natural environment percentage, buffer 300m		Local environment
X24507	0.8586	0.0151194	0.8797	0.0234967	0.8428	0.0007965	0.8686	0.0004621	0.9328	0.2212985	(30.4,100]	Natural environment percentage, buffer 300m		Local environment
NOx	0.9982	257.42933	1.0074	0.8945826	0.9904	0.8453427	0.9904	0.8068673	0.9262	0.1617646	(27.5,34.3]	Nitrogen oxides		Local environment

NOx	1.1399	0.4235594	1.2068	0.0019633	1.0819	0.166166	1.1263	0.0068303	0.9361	0.292635	(34.3,153]	Nitrogen oxides		Local environment
PM10	0.9232	8.469946	0.9389	0.2574483	0.9129	0.0685381	0.9156	0.0250666	0.8898	0.0328156	(18.4,19.8]	Particulate matter air pollution (pm10)		Local environment
PM10	1.0592	44.387291	1.0739	0.2425178	1.0462	0.426474	1.043	0.3387219	0.9228	0.1985878	(19.8,57]	Particulate matter air pollution (pm10)		Local environment
X1050	1.0438	68.840925	1.1565	0.0076362	0.9394	0.2329654	1.0417	0.3099079	1.0469	0.4246184	(2,4]	Time spend outdoors in summer		Lifestyles
X1050	1.0984	3.6477138	1.263	2.48E-05	0.9664	0.5113928	1.1126	0.0081864	1.1493	0.0149921	(4,24]	Time spend outdoors in summer		Lifestyles
X1060	1.0818	7.2224717	1.1209	0.0316755	1.042	0.3891037	1.0908	0.0211471	1.1088	0.0527615	(1,2]	Time spent outdoors in winter		Lifestyles
X1060	1.1651	0.0061007	1.3468	2.20E-08	1.0366	0.4611721	1.1691	4.49E-05	1.1818	0.0020528	(2,24]	Time spent outdoors in winter		Lifestyles
X2139	0.9731	131.80864	0.9246	0.1450091	1.0348	0.5634893	1.0041	0.9232498	0.9317	0.2312718	(17.4,20]	Age first had sexual intercourse		Lifestyles
X2139	0.9379	27.394538	0.9227	0.149992	0.9625	0.4987528	0.944	0.1682589	0.906	0.0906097	(20,69]	Age first had sexual intercourse		Lifestyles
X2149	1.0901	2.9360323	0.9925	0.8833338	1.1752	0.0003385	1.0651	0.0793309	1.0895	0.0898388	(2,6]	Lifetime number of sexual partners		Lifestyles
X2149	1.0577	42.43315	0.9975	0.9645436	1.0981	0.0923593	1.034	0.4288662	1.0112	0.8539975	(6,1.7e+04]	Lifetime number of sexual partners		Lifestyles
X22037	0.9916	219.36176	1.0026	0.9616163	0.9826	0.7228298	0.9975	0.9497075	1.0637	0.2706719	(462,1.16e+03]	MET minutes per week for walking		Lifestyles
X22037	1.0593	27.35671	1.0993	0.068269	1.0226	0.6409747	1.0804	0.0384692	1.2116	0.0002979	(1.16e+03,4.16e+03]	MET minutes per week for walking		Lifestyles
X22038	0.9232	7.9346675	0.8841	0.0211056	0.9629	0.4568518	0.9158	0.024467	0.9874	0.822077	(240,840]	MET minutes per week for moderate activity		Lifestyles
X22038	0.9237	7.1993945	0.9034	0.0535416	0.9472	0.2704867	0.9358	0.0808974	1.0486	0.3820089	(840,5.04e+03]	MET minutes per week for moderate activity		Lifestyles
X22039	0.8624	0.0063184	0.8616	0.0041867	0.8659	0.0023874	0.8604	5.11E-05	0.8673	0.0068594	(41.6,560]	MET minutes per week for vigorous activity		Lifestyles
X22039	0.8353	0.0001532	0.8164	0.0001477	0.8498	0.0008339	0.8264	6.16E-07	0.8522	0.0030877	(560,1.01e+04]	MET minutes per week for vigorous activity		Lifestyles
X22040	0.9755	132.84122	0.9841	0.7632263	0.9682	0.5196827	0.9656	0.3660168	0.9955	0.9354317	(1.23e+03,2.72e+03]	Summed MET minutes per week for all activity		Lifestyles
X22040	0.9496	40.643094	0.9283	0.1633304	0.9654	0.4733114	0.9573	0.2532565	1.058	0.2980801	(2.72e+03,1.93e+04]	Summed MET minutes per week for all activity		Lifestyles
X1070	1.0992	3.3648259	1.1454	0.0158758	1.0567	0.2814391	1.0848	0.0430523	1.0885	0.132565	(2,3]	Time spent watching television (TV)		Lifestyles
X1070	1.3439	1.71E-15	1.5072	6.93E-16	1.2269	1.01E-05	1.327	7.01E-15	1.2747	2.68E-06	(3,24]	Time spent watching television (TV)		Lifestyles
X1080	1.3458	8.65E-14	1.325	1.50E-07	1.3551	9.32E-10	1.2973	1.12E-11	1.273	7.78E-06	[0,0.5]	Time spent using computer		Lifestyles
X1080	1.2197	0.0004304	1.2857	3.03E-05	1.1626	0.0081781	1.1795	0.0001612	1.1415	0.034012	(1,24]	Time spent using computer		Lifestyles

X23099	0.9464	39.127477	1.0846	0.1442014	0.8446	0.0011167	0.9638	0.3594012	1.1282	0.0372552	(26.7,34.6]	Body fat percentage		Physical measures
X23099	0.9173	21.125637	0.9364	0.3651681	0.9037	0.1302059	0.9346	0.1949691	1.1505	0.0587761	(34.6,69.3]	Body fat percentage		Physical measures
X23105	0.8493	0.0735818	0.832	0.0051386	0.8718	0.0262553	0.8621	0.001736	0.8667	0.0302079	(5.72e+03,7.21e+03]	Basal metabolic rate		Physical measures
X23105	0.7042	4.78E-07	0.6928	2.02E-05	0.7154	2.50E-05	0.7196	9.94E-08	0.7598	0.0016346	(7.21e+03,1.55e+04]	Basal metabolic rate		Physical measures
X23127	0.8804	0.158383	0.946	0.3118856	0.8262	0.000149	0.8852	0.0020092	0.9938	0.9142808	(27.4,33.9]	Trunk fat percentage		Physical measures
X23127	0.937	25.601116	1.0088	0.8796814	0.879	0.0147404	0.9486	0.2027849	1.1309	0.038787	(33.9,77.6]	Trunk fat percentage		Physical measures
X4079	0.9297	11.372129	0.94	0.2514854	0.9187	0.0789764	0.9607	0.2919981	1.0426	0.4377854	(77.5,86]	Diastolic blood pressure		Physical measures
X4079	0.8827	0.1638767	0.8856	0.0256721	0.8756	0.0069504	0.8877	0.0021486	0.9022	0.063855	(86,148]	Diastolic blood pressure		Physical measures
X4080	0.9457	47.997066	0.9139	0.1192787	0.9695	0.6075245	0.9651	0.421491	1.0511	0.4273844	(128,143]	Systolic blood pressure		Physical measures
X4080	0.9519	56.832444	0.9763	0.6688478	0.9267	0.1738265	0.9635	0.3767502	0.9843	0.7928847	(143,268]	Systolic blood pressure		Physical measures
Leg_fat_percentage	0.9681	120.37744	1.0027	0.9665675	0.9447	0.3260677	0.9781	0.6270164	1.1596	0.019216	(24.4,38.1]	Leg fat percentage		Physical measures
Leg_fat_percentage	0.8677	6.0729489	0.9371	0.4804658	0.8133	0.0144704	0.8844	0.062729	1.0795	0.4090074	(38.1,64.1]	Leg fat percentage		Physical measures
arm_fat_percentage	1.0412	76.797118	1.1235	0.037617	0.9768	0.6494738	1.0402	0.3283369	1.1942	0.0020602	(23.4,33.2]	Arm fat percentage		Physical measures
arm_fat_percentage	1.0111	221.21179	1.0406	0.5878634	0.9861	0.8362521	1.0042	0.9374148	1.1536	0.0563955	(33.2,74]	Arm fat percentage		Physical measures
hand_grip_strength	0.6989	1.12E-14	0.666	5.09E-11	0.7386	3.37E-07	0.7349	9.68E-12	0.8135	0.0012269	(25,35]	Hand grip strength		Physical measures
hand_grip_strength	0.487	1.50E-40	0.4317	1.30E-27	0.5354	2.80E-18	0.5303	7.04E-30	0.6513	8.23E-08	(35,85]	Hand grip strength		Physical measures
Lung_function	0.7387	2.28E-12	0.6819	1.18E-11	0.808	0.0001007	0.7583	2.68E-11	0.7911	6.98E-05	(-0.396,0.383]	Lung_function		Physical measures
Lung_function	0.5587	2.52E-31	0.458	2.06E-29	0.6633	7.15E-10	0.5901	5.85E-25	0.6438	1.59E-09	(0.383,22.4]	Lung_function		Physical measures
X189	1.0137	189.84237	0.9723	0.6135255	1.0453	0.357085	1.0115	0.767712	0.9859	0.7934555	(-3.29,-0.98]	Townsend deprivation index at recruitment		Socioeconomic status
X189	1.242	2.31E-06	1.3092	9.61E-07	1.1712	0.0023247	1.2325	1.69E-07	1.1318	0.0290758	(-0.98,11]	Townsend deprivation index at recruitment		Socioeconomic status
X6179_1	0.9545	33.586355	0.9758	0.5935362	0.936	0.1021246	0.9762	0.4540944	0.977	0.6093718		Fish oil (including cod liver oil) supplements		Lifestyles
X6179_2	0.8652	0.0091402	0.8345	0.000875	0.8869	0.0087011	0.8776	0.0004007	0.8012	3.13E-05		Glucosamine supplements		Lifestyles
X6179_3	1.1325	3.6513421	1.1456	0.0836888	1.1177	0.0908722	1.1298	0.0222398	1.1869	0.0191195		Calcium supplements		Lifestyles
X6179_4	0.9406	105.96437	1.0555	0.60053	0.8465	0.0982545	0.9079	0.213211	1.0074	0.9438416		Zinc supplements		Lifestyles

X6179_5	1.2849	0.6296524	1.309	0.0224686	1.2547	0.0490859	1.2678	0.0070418	1.1662	0.2344645		Iron supplements		Lifestyles
X6179_6	0.9969	259.97101	1.0205	0.8732074	0.9841	0.881593	0.9982	0.9831053	0.9348	0.5959959		Selenium supplements		Lifestyles
X6179_7	1.0453	36.638405	1.0418	0.3549877	1.0501	0.2250835	1.0308	0.3376058	1.0588	0.2014262		No mineral or other dietary supplements		Lifestyles
X1677	0.9801	167.21126	0.9673	0.5593231	0.9923	0.8959633	0.9658	0.4209173	0.9649	0.5582699		Breastfed as a baby		Early life
X1687	1.0308	94.201446	1.0367	0.4594639	1.0204	0.6441159	1.0057	0.8689252	0.9637	0.4490779		Comparative body size at age 10	Thinner	Early life
X1687	1.082	18.719252	1.0673	0.2929091	1.0957	0.1343728	1.0832	0.0818276	0.9475	0.4226561		Comparative body size at age 10	Plumper	Early life
X1697	1.0253	138.41225	1.068	0.2386073	0.9859	0.7892177	0.9764	0.563065	0.9669	0.56668		Comparative height size at age 10	Shorter	Early life
X1697	0.914	2.8370717	0.8491	0.0021401	0.9652	0.450315	0.8997	0.0045297	0.8735	0.0103192		Comparative height size at age 10	Taller	Early life
X1787	0.9902	208.84182	0.942	0.2279884	1.0497	0.3249593	0.9859	0.7021475	0.969	0.5526738		Maternal smoking around birth		Early life
X1920	1.2583	2.86E-12	1.2913	5.43E-09	1.2298	3.36E-07	1.2446	4.01E-12	1.2332	2.71E-06		Mood swings		Psychosocial factors
X1930	1.143	0.0029598	1.147	0.0020423	1.1373	0.0020342	1.1334	0.0001045	1.1217	0.0119549		Miserableness		Psychosocial factors
X1940	0.9749	122.36717	0.9767	0.6331767	0.9781	0.6389863	0.9793	0.5628522	0.9708	0.563921		Irritability		Psychosocial factors
X1950	1.0319	78.281472	1.0144	0.7476973	1.0522	0.2063489	1.0289	0.3687133	1.0339	0.4577305		Sensitivity / hurt feelings		Psychosocial factors
X1960	1.2235	1.24E-08	1.2416	1.12E-06	1.2095	7.30E-06	1.2154	2.05E-09	1.1611	0.0012861		Fed-up feelings		Psychosocial factors
X1970	1.1359	0.0677701	1.2118	0.0001237	1.0762	0.1306638	1.1511	0.0001316	1.0626	0.2518947		Nervous feelings		Psychosocial factors
X1980	1.1509	0.000838	1.1739	0.0003638	1.1356	0.0017471	1.1557	6.14E-06	1.1372	0.0045701		Worrier / anxious feelings		Psychosocial factors
X1990	1.1799	0.0038256	1.211	0.0005361	1.157	0.0056346	1.1786	4.79E-05	1.1048	0.0857664		Tense / 'highly strung'		Psychosocial factors
X2000	0.9179	1.1116473	0.8707	0.0017696	0.965	0.3783672	0.922	0.0104111	0.8825	0.0055066		Worry too long after embarrassment		Psychosocial factors
X2010	1.2735	2.66E-09	1.3127	6.04E-08	1.2375	2.30E-05	1.2584	1.21E-09	1.1522	0.0098399		Suffer from 'nerves'		Psychosocial factors
X2020	1.4224	4.85E-19	1.5264	2.85E-16	1.321	1.74E-07	1.3654	4.40E-15	1.2595	5.65E-05		Loneliness, isolation		Psychosocial factors
X2030	1.0219	138.91999	1.0115	0.8151065	1.0337	0.4722652	1.0243	0.4997843	1.0116	0.8195318		Guilty feelings		Psychosocial factors
X2110	1.2103	7.27E-07	1.1971	0.0001823	1.2116	8.56E-06	1.1772	1.88E-06	1.1597	0.0023164		Able to confide	Rarely	Psychosocial factors
X6155_0	0.9288	4.4084024	0.9259	0.0942621	0.9332	0.0950728	0.9326	0.0327714	0.9674	0.4755234		No vitamin and mineral supplements		Lifestyles
X6155_1	1.0129	238.68482	1.0791	0.5987662	0.9601	0.7371711	0.9462	0.5872327	0.9818	0.8970191		Vitamin A supplements		Lifestyles
X6155_2	1.2645	0.0461708	1.3837	0.0003557	1.1702	0.067705	1.1848	0.0126976	1.1172	0.2594206		Vitamin B supplements		Lifestyles
X6155_3	1.0031	254.48534	0.9665	0.6504464	1.0291	0.653818	0.9664	0.5145167	0.9395	0.4027291		Vitamin C supplements		Lifestyles

X6155_4	0.9724	180.39693	1.0042	0.9680946	0.9456	0.5144286	0.9455	0.4317647	0.928	0.4627387		Vitamin D supplements		Lifestyles
X6155_5	1.108	44.485502	1.0795	0.5082578	1.1186	0.2454186	1.0521	0.5272939	1.0303	0.7929246		Vitamin E supplements		Lifestyles
X6155_6	1.3411	0.0478198	1.7593	1.49E-07	1.0492	0.6748715	1.2623	0.0061305	1.1456	0.2714388		Folic acid or Folate (Vit B9) supplements		Lifestyles
X6155_7	1.0302	105.75841	1.0122	0.8160354	1.0427	0.3756095	1.04	0.2889544	0.9995	0.9920672		Multivitamins +/- minerals supplements		Lifestyles
fruit	1.0604	14.17585	1.0553	0.2273904	1.062	0.1454023	1.0661	0.0466956	1.0029	0.9498702		Fruit eaten above recommendation		Lifestyles
fish	0.9736	132.01355	0.9612	0.4655448	0.987	0.8166382	0.9775	0.5829153	1.0064	0.9148646		Fish eaten above recommendation		Lifestyles
white bread/ refined_cereals	1.0069	240.24117	0.8983	0.1613352	1.1043	0.1694122	0.9975	0.9640746	1.0464	0.5729211		White bread/refined cereals intake reduced		Lifestyles
wholegrain	1.0935	114.35966	0.823	0.3247494	1.2974	0.0571191	1.1594	0.2030281	1.3872	0.0305662		Wholegrain eaten above recommendation		Lifestyles
red_meat	1.0385	55.027735	1.0209	0.6396861	1.0573	0.1690337	1.0439	0.1745616	1.0065	0.8856004		Red meat intake reduced		Lifestyles
processed_meat	0.9914	209.34244	0.9942	0.8994167	0.9915	0.8383288	0.9969	0.9246781	1.0038	0.9342205		Processed meat intake reduced		Lifestyles
X130176	0.4992	0.0033794	0.5534	0.0027895	0.4242	0.0013937	0.5569	0.0002833	0.5085	0.0055952		varicella [chickenpox]		Medical history
X130178	1.1535	29.775187	1.1134	0.4514251	1.1882	0.1347814	1.1527	0.1375806	1.1681	0.2562277		zoster [herpes zoster]		Medical history
X130188	0.9906	246.19262	1.0017	0.9895891	0.987	0.9184519	1.0045	0.9631977	1.0997	0.498661		viral warts		Medical history
X130226	0.9806	205.18085	1.1255	0.1984141	0.8661	0.1274112	0.9961	0.9558426	0.9553	0.6605836		dermatophytosis		Medical history
X130622	1.3832	0.0325822	1.2722	0.0674291	1.5	0.0002322	1.2676	0.0110896	1.3788	0.0112021		iron deficiency anaemia		Medical history
X130648	1.6175	7.26E-08	1.7923	2.40E-07	1.5023	7.97E-05	1.5094	7.75E-07	1.4255	0.0031912		other anaemias		Medical history
X130696	0.992	239.95245	0.9853	0.8735949	0.9949	0.9497967	0.9079	0.1488934	0.941	0.5097363		other hypothyroidism		Medical history
X130792	1.3284	0.0112217	1.3302	0.0051033	1.3241	0.0029926	1.3407	5.93E-05	1.1859	0.1136248		obesity		Medical history
X130814	1.1394	0.0227333	1.297	3.15E-07	1.0406	0.3618515	1.1379	0.000246	1.0898	0.0866187		disorders of lipoprotein metabolism and other lipidaemias		Medical history
X130868	1.1405	17.860572	1.3198	0.0038515	0.972	0.7933161	1.1722	0.0345347	1.0648	0.5811517		mental and behavioural disorders due to use of tobacco		Medical history
X130906	1.6008	5.27E-07	1.5947	2.09E-05	1.6209	1.67E-05	1.5096	1.35E-06	1.2536	0.0876734		other anxiety disorders		Medical history
X130910	1.4325	0.0247262	1.6379	1.49E-05	1.189	0.2702893	1.3655	0.0017807	1.4224	0.0143364		reaction to severe stress, and adjustment disorders		Medical history
X131052	0.8953	45.584587	0.8277	0.1031845	0.982	0.8715834	0.8681	0.102994	0.9178	0.468246		migraine)		Medical history

X131060	1.2964	0.2833568	1.4692	0.000398	1.1475	0.2360947	1.2683	0.0050697	1.2236	0.0960253		sleep disorders)		Medical history
X131074	1.1976	4.0703428	1.4485	0.0002115	0.9851	0.8928181	1.1735	0.0433468	1.1326	0.2694739		mononeuropathies of upper limb		Medical history
X131128	1.1512	42.681312	1.2272	0.1434114	1.0817	0.5834617	1.1629	0.1541414	1.2881	0.0813363		hordeolum and chalazion		Medical history
X131142	1.0173	218.91903	1.0001	0.9990532	1.0293	0.7684169	1.0147	0.8534548	1.0195	0.8667181		conjunctivitis		Medical history
X131166	1.2157	0.2302633	1.5042	3.65E-05	1.1011	0.1847453	1.2036	0.0029972	1.1298	0.1799565		other cataract		Medical history
X131222	0.8875	51.057492	0.8229	0.1574875	0.9492	0.6678693	0.8357	0.0733265	0.8312	0.2048469		otitis externa		Medical history
X131224	0.9971	259.05183	1.065	0.5494797	0.9482	0.5687487	0.9923	0.917353	0.9	0.3573078		other disorders of external ear		Medical history
X131262	0.8317	24.829154	0.9099	0.5297827	0.7623	0.0907883	0.8645	0.2034444	0.7506	0.1098628		otalgia and effusion of ear		Medical history
X131264	1.0347	195.49869	1.1601	0.297052	0.9496	0.7058707	1.0426	0.6909487	1.0743	0.6364933		other disorders of ear, not elsewhere classified		Medical history
X131350	1.3909	4.10E-05	1.2932	0.0217045	1.4492	1.09E-06	1.3304	2.46E-05	1.3049	0.0046511		atrial fibrillation and flutter		Medical history
X131396	1.1874	5.5242424	1.2051	0.1107871	1.1659	0.1099799	1.1647	0.0540616	0.9543	0.6988985		phlebitis and thrombophlebitis		Medical history
X131402	0.8734	22.817062	0.8182	0.1122845	0.9151	0.3770099	0.8989	0.1956183	0.8089	0.0863964		varicose veins of lower extremities		Medical history
X131404	0.9946	251.80287	1.022	0.8367069	0.9744	0.7897126	0.9901	0.8950114	0.9276	0.4891315		haemorrhoids		Medical history
X131426	0.9008	51.277576	0.9788	0.8452219	0.826	0.1026261	0.87	0.1067051	1.04	0.7367394		acute sinusitis		Medical history
X131428	1.0094	243.16654	0.9892	0.9250351	1.0347	0.7590225	1.0142	0.8686406	0.9833	0.8938754		acute pharyngitis		Medical history
X131430	0.7356	2.3236151	0.6439	0.0101387	0.8498	0.3096036	0.7306	0.0124653	0.9213	0.6325775		acute tonsillitis		Medical history
X131436	1.0175	204.83848	1.0833	0.3304031	0.9617	0.6333944	1.0121	0.8462919	0.9091	0.3102317		acute upper respiratory infections of multiple and unspecified sites		Medical history
X131442	0.8263	31.513403	0.9704	0.8515177	0.6939	0.051333	0.8704	0.273231	0.863	0.4181615		influenza, virus not identified		Medical history
X131456	1.1942	5.6522521	1.1745	0.1721767	1.2145	0.0560073	1.1822	0.0412283	0.8882	0.3624754		pneumonia, organism unspecified		Medical history
X131462	1.1345	4.1790852	1.1859	0.0266684	1.0978	0.1892336	1.094	0.1098311	1.0338	0.6866545		unspecified acute lower respiratory infection		Medical history
X131464	0.9571	112.8366	0.8231	0.0173743	1.103	0.1787868	0.9486	0.3633151	1.0438	0.5873803		vasomotor and allergic rhinitis		Medical history
X131468	0.8325	24.329381	0.7393	0.0689332	0.917	0.5450618	0.7499	0.0170722	0.7262	0.0646989		chronic sinusitis		Medical history
X131472	0.961	177.9414	1.0112	0.9311552	0.9079	0.457348	0.957	0.6517941	0.86	0.2983531		other disorders of nose and nasal sinuses		Medical history
X131494	1.0582	53.787431	1.0223	0.7370831	1.0877	0.1594543	1.0111	0.8173557	0.936	0.3325998		asthma		Medical history
X131584	1.0274	155.77613	1.0324	0.6676818	1.0293	0.6589497	1.0024	0.9635511	0.9508	0.5016489		gastro-oesophageal reflux disease		Medical history

X131598	1.2441	0.0110107	1.3766	5.56E-05	1.1561	0.0436057	1.213	0.0006838	1.0169	0.8452708		gastritis and duodenitis		Medical history
X131600	1.1432	32.16345	1.0883	0.5331815	1.1703	0.1581967	1.1872	0.0540699	1.1554	0.2534294		dyspepsia		Medical history
X131612	0.986	221.40349	0.9302	0.4926465	1.0305	0.7100013	0.9871	0.8482065	0.9124	0.3622762		inguinal hernia		Medical history
X131620	1.1237	7.3779701	1.19	0.034059	1.0817	0.2565213	1.1043	0.0775004	0.8775	0.1283176		diaphragmatic hernia		Medical history
X131630	1.2609	0.1053617	1.3034	0.0060165	1.2293	0.0204473	1.199	0.0100648	1.0846	0.4251369		other non-infective gastro-enteritis and colitis		Medical history
X131636	1.0776	47.168431	0.8976	0.2756526	1.1834	0.0116076	1.046	0.4469563	1.0137	0.870792		diverticular disease of intestine		Medical history
X131638	0.9563	133.12235	0.9641	0.7004148	0.953	0.5977237	0.9217	0.2483068	0.7873	0.0232563		irritable bowel syndrome		Medical history
X131640	1.534	1.18E-05	1.471	0.0014469	1.5931	5.41E-06	1.399	9.91E-05	1.357	0.0129657		other functional intestinal disorders		Medical history
X131646	1.0279	171.60345	1.1864	0.0415498	0.9145	0.2792819	1.0179	0.7763888	0.9889	0.9011295		other diseases of anus and rectum		Medical history
X131650	0.9526	125.62917	0.9729	0.7818376	0.9391	0.4896821	0.98	0.7741878	1.0283	0.7829768		haemorrhoids and perianal venous thrombosis		Medical history
X131674	1.1044	37.19933	1.2749	0.0146332	0.9946	0.9525414	1.0531	0.4739277	1.0437	0.6685041		cholelithiasis		Medical history
X131702	1.1361	21.325874	1.0983	0.3948198	1.1659	0.1133447	1.0696	0.3964503	0.9746	0.8278628		cellulitis		Medical history
X131708	1.1593	35.432837	1.1846	0.2363359	1.1483	0.3058353	1.186	0.0996463	1.2627	0.1133663		other local infections of skin and subcutaneous tissue		Medical history
X131720	1.0632	141.76045	0.842	0.2621126	1.2931	0.041527	1.0715	0.5055128	0.8774	0.4338332		atopic dermatitis		Medical history
X131740	1.163	1.1650858	1.1584	0.0598672	1.1703	0.0290319	1.1428	0.0184145	1.0874	0.3122515		other dermatitis		Medical history
X131742	1.1642	22.325965	1.2594	0.0636791	1.0891	0.491394	1.153	0.1279308	1.054	0.7010668		psoriasis		Medical history
X131768	0.9656	179.88236	1.0183	0.8934507	0.946	0.5935962	0.9963	0.9656989	0.9457	0.6635239		skin changes due to chronic exposure to nonionising radiation		Medical history
X131794	1.0774	115.46351	1.0965	0.5128773	1.0668	0.6129179	1.1018	0.3301456	1.1058	0.4746482		follicular cysts of skin and subcutaneous tissue		Medical history
X131806	0.8157	1.3514685	0.8187	0.0810335	0.8155	0.0300096	0.8455	0.0272493	0.7853	0.0342943		seborrhoeic keratosis		Medical history
X131836	0.8245	0.5084465	0.8614	0.110338	0.7987	0.006928	0.8278	0.0041899	0.8364	0.0626318		other disorders of skin and subcutaneous tissue, not elsewhere classified		Medical history
X131858	1.0295	190.61634	1.3874	0.0023815	0.7868	0.0367411	1.0038	0.9645077	1.0512	0.6696284		gout		Medical history
X131864	1.1508	11.272833	1.2988	0.0120491	1.055	0.5620833	1.1137	0.1467527	0.8819	0.2849401		other arthritis		Medical history
X131872	1.2112	0.2444896	1.2689	0.0089689	1.1678	0.0377983	1.2022	0.0026119	1.1209	0.1952404		gonarthrosis [arthrosis of knee]		Medical history
X131876	1.0785	14.074573	1.1669	0.0106812	1.0202	0.6937335	1.0687	0.1072139	0.9844	0.7887404		other arthrosis		Medical history

X131878	1.1104	47.178781	1.0669	0.5948415	1.142	0.1859845	1.1108	0.1996903	0.9284	0.5528921		acquired deformities of fingers and toes		Medical history
X131884	0.9526	152.82185	1.058	0.6338729	0.8564	0.211147	0.9415	0.5073443	0.8279	0.1635566		internal derangement of knee		Medical history
X131888	1.0586	48.223094	1.1433	0.0293635	0.9917	0.8874181	1.0631	0.1746276	0.9902	0.881175		other joint disorders, not elsewhere classified		Medical history
X131916	1.37	1.50E-05	1.5205	1.62E-06	1.276	0.0016634	1.3306	4.29E-06	1.1964	0.0509559		spondylosis		Medical history
X131924	1.1882	3.5503695	1.0759	0.4973664	1.2754	0.0076927	1.1583	0.0481511	0.81	0.0857456		other intervertebral disk disorders		Medical history
X131928	1.1038	8.7366632	1.2294	0.0015754	1.002	0.9751895	1.0724	0.1586774	1.0024	0.9734095		dorsalgia		Medical history
X131938	1.0259	207.18707	1.1079	0.4284524	0.9642	0.7644819	0.9882	0.9016848	0.8801	0.3789116		synovitis and tenosynovitis		Medical history
X131946	1.1146	64.906757	0.9325	0.6339294	1.2709	0.0454254	1.1095	0.2944849	0.921	0.6084138		soft tissue disorders related to use, overuse and pressure		Medical history
X131950	0.9673	180.56802	1.037	0.7634697	0.9274	0.4698678	0.9576	0.6074363	1.0148	0.901969		fibroblastic disorders		Medical history
X131954	1.141	10.964427	1.1217	0.2397825	1.1561	0.0915474	1.1352	0.0647085	1.2269	0.0311332		shoulder lesions		Medical history
X131958	0.952	125.46712	0.8533	0.1101946	1.062	0.5180673	0.9468	0.4508016	0.9097	0.3832391		other enthesopathies		Medical history
X131960	1.0994	8.9934096	1.1327	0.0571072	1.071	0.2603183	1.1103	0.0267711	1.0632	0.3734251		other soft tissue disorders, not elsewhere classified		Medical history
X131964	1.2651	0.3328412	1.3076	0.0294022	1.2284	0.0230007	1.1812	0.0352739	1.0649	0.5813823		osteoporosis without pathological fracture		Medical history
X132054	0.9733	205.55669	1.084	0.5439353	0.8987	0.3968212	0.9484	0.5897493	1.0876	0.5237733		cystitis		Medical history
X132070	1.3158	4.23E-05	1.2391	0.00994	1.3805	1.78E-06	1.2093	0.0008959	1.0914	0.2931566		other disorders of urinary system		Medical history
X132072	1.025	179.48658	1.1594	0.1289365	0.984	0.8226207	0.9965	0.9553412	0.9535	0.5944409		hyperplasia of prostate		Medical history
X132096	0.8696	50.466242	0.7969	0.1706573	0.9237	0.5665948	0.822	0.0905238	0.768	0.125867		benign mammary dysplasia		Medical history
X132102	1.1087	75.543837	1.1454	0.3032787	1.0779	0.5920868	1.0989	0.3562577	1.0004	0.9981413		unspecified lump in breast		Medical history
X132104	1.1535	50.37738	1.0427	0.7937802	1.2962	0.078841	1.1863	0.1331534	1.208	0.2431283		other disorders of breast		Medical history
X132124	1.2356	2.7100521	1.0364	0.7985397	1.3573	0.0027113	1.1817	0.0589734	1.274	0.041521		female genital prolapse		Medical history
X132128	1.0448	188.6861	1.1164	0.4902072	0.9668	0.8397717	0.9871	0.9175062	0.9203	0.6491076		noninflammatory disorders of ovary, fallopian tube and broad ligament		Medical history
X132130	0.8817	55.915513	0.9308	0.6088574	0.8381	0.2176226	0.8366	0.1003651	0.9235	0.5909463		polyp of female genital tract		Medical history
X132140	0.885	85.900891	0.8456	0.3535813	0.924	0.6377404	0.9083	0.455204	1.072	0.6884446		other noninflammatory disorders of vagina		Medical history
X132146	1.0315	197.74025	1.1039	0.3702538	0.935	0.7002123	1.0029	0.9767134	0.9416	0.6873376		excessive, frequent and irregular menstruation		Medical history

X132148	0.9718	226.50678	0.7778	0.2245733	1.3472	0.1489716	1.0301	0.8445619	1.176	0.4337077		other abnormal uterine and vaginal bleeding		Medical history
X132150	0.9183	143.08408	0.8856	0.5003546	0.9814	0.9289585	0.8788	0.3827384	0.7259	0.1571118		pain and other conditions associated with female genital organs and menstrual cycle		Medical history
X132152	0.9466	107.4033	0.8961	0.2635745	0.9972	0.9747902	0.91	0.1794781	0.879	0.2030984		menopausal and other perimenopausal disorders		Medical history
X132276	1.2111	14.256765	1.0814	0.5990003	1.3533	0.0229283	1.1863	0.1100997	1.0686	0.6816524		single spontaneous delivery		Medical history
X6149_1	1.0562	70.510632	1.0669	0.3580762	1.0482	0.4886535	1.0644	0.226729	1.0213	0.7766315		Mouth/teeth dental problems	Mouth ulcers	Medical history
X6149_2	1.4716	3.90E-05	1.6384	1.71E-06	1.339	0.0053172	1.3439	0.0002639	1.1308	0.313261		Mouth/teeth dental problems	Painful gums	Medical history
X6149_3	0.9943	243.24024	1.106	0.1146766	0.861	0.0562931	0.9871	0.8052336	0.9638	0.6230551		Mouth/teeth dental problems	Bleeding gums	Medical history
X6149_4	1.2308	0.356246	1.2938	0.0059916	1.1778	0.0674747	1.2052	0.0071114	1.1818	0.0913923		Mouth/teeth dental problems	Loose teeth	Medical history
X6149_5	1.4403	7.38E-06	1.5282	1.98E-06	1.3374	0.0027904	1.435	2.25E-07	1.3965	0.0008383		Mouth/teeth dental problems	Toothache	Medical history
X6149_6	1.2847	1.33E-11	1.4179	3.90E-11	1.2072	9.84E-06	1.3118	1.08E-14	1.3163	2.51E-08		Mouth/teeth dental problems	Dentures	Medical history
X6149_7	0.8214	7.11E-09	0.757	2.10E-10	0.8795	0.0012923	0.8122	3.14E-11	0.8365	5.70E-05		Mouth/teeth dental problems	non of the above	Medical history
Diabetes	1.8559	2.56E-43	2.1772	1.03E-33	1.6454	3.72E-17	1.8235	1.92E-38	1.7849	5.55E-19		Diabetes		Medical history
CHD	1.5055	1.87E-20	1.7166	4.86E-16	1.4059	1.31E-10	1.4342	5.89E-16	1.1703	0.0172623		CHD		Medical history
stroke	2.5015	4.21E-49	2.9196	3.14E-29	2.2793	1.04E-25	2.3348	1.72E-37	1.8723	6.29E-10		stroke		Medical history
COPD	1.3926	0.0946449	1.3943	0.027787	1.3935	0.00473	1.2646	0.0219414	1.1256	0.423821		COPD		Medical history
Depression	1.7731	5.38E-36	1.7392	3.49E-19	1.8203	2.38E-21	1.6382	1.12E-24	1.4276	6.10E-07		Depression		Medical history
Arthritis	1.1099	0.8119697	1.1752	0.0031424	1.0634	0.180235	1.0964	0.0135522	0.9983	0.9747718		Arthritis		Medical history
Hypertension	1.3827	1.34E-19	1.5625	4.92E-18	1.2718	4.81E-08	1.3164	1.99E-14	1.1496	0.0070575		Hypertension		Medical history
Hearing_loss	1.2888	0.3473316	1.2948	0.0339266	1.2915	0.0134734	1.2471	0.0096526	1.1839	0.1753888		Hearing_loss		Medical history
Vision_impairment	1.2543	0.0082369	1.5407	2.88E-06	1.1398	0.0510563	1.238	0.0002281	1.1463	0.1050388		Vision_impairment		Medical history
X6146	2.3513	1.55E-84	2.7252	6.64E-58	2.0752	4.41E-34	2.0881	7.59E-54	1.4281	2.62E-06		Disability		Medical history
X6160_1	0.746	4.19E-15	0.7265	2.94E-10	0.7644	9.46E-09	0.7484	1.87E-15	0.7759	7.18E-07		Leisure/social activities	Sports club or gym	Lifestyles
X6160_2	0.9789	145.50943	0.9484	0.299442	1.0083	0.8634257	0.9912	0.8117215	1.0127	0.8100504		Leisure/social activities	Pub or social club	Lifestyles
X6160_3	1.0306	108.11348	1.1835	0.002175	0.9308	0.1326085	1.0346	0.3731058	0.9154	0.1131124		Leisure/social activities	Religious group	Lifestyles
X6160_4	0.7788	0.0011136	0.9034	0.2130196	0.6989	8.39E-07	0.7743	9.25E-06	0.8136	0.0097702		Leisure/social activities	Adult education	Lifestyles

X1130	0.8793	16.899163	0.8247	0.0337263	0.9446	0.5937487	0.9136	0.213358	0.9858	0.888571		Hands-free device/speakerphone use with mobile phone in last 3 month	Less than half the time	Lifestyles
X1130	0.846	40.402956	0.893	0.4114135	0.7106	0.1198375	0.8863	0.319982	1.0365	0.8258661		Hands-free device/speakerphone use with mobile phone in last 4 month	About half the time	Lifestyles
X1130	0.9343	176.48473	0.8983	0.5681232	0.9672	0.9011799	1.0115	0.9420873	0.8203	0.4315662		Hands-free device/speakerphone use with mobile phone in last 5 month	More than half the time	Lifestyles
X1130	1.0603	164.12145	0.9756	0.8692042	1.1614	0.4092282	1.0571	0.651087	1.0845	0.6360171		Hands-free device/speakerphone use with mobile phone in last 6 month	Always or almost always	Lifestyles
X6138	1.1677	4.12E-05	1.1144	0.0137372	1.2155	1.04E-06	1.1641	1.25E-06	1.1179	0.0122101		Lower education		Socioeconomic status
X6142	1.9643	2.90E-21	2.0172	5.79E-23	1.615	0.0333363	1.8695	8.65E-18	1.3972	0.0039347		Unemployed		Socioeconomic status
X738	0.7819	9.80E-10	0.7246	5.47E-09	0.8295	5.16E-05	0.785	1.28E-10	0.866	0.0074021		Average total household income before tax	18,000 to 30,999	Socioeconomic status
X738	0.6173	5.47E-28	0.5206	3.34E-26	0.7258	2.33E-08	0.628	2.13E-25	0.7178	1.64E-07		Average total household income before tax	31,000 to 51,999	Socioeconomic status
X738	0.5041	6.96E-31	0.4665	5.70E-25	0.5236	2.30E-12	0.5327	6.08E-26	0.6392	8.59E-08		Average total household income before tax	52,000 to 100,000	Socioeconomic status
X738	0.4721	8.27E-09	0.4176	4.67E-10	0.5354	0.0012315	0.4923	3.08E-09	0.58	0.0014837		Average total household income before tax	Greater than 100,000	Socioeconomic status
TBI	2.9403	2.00E-28	3.4789	4.14E-20	2.6009	1.22E-13	2.5801	2.78E-19	1.178	0.4452235		Traumatic brain injury		Medical history

Blue cells indicate HR<1. HR, hazard ratios; P.adjust, after Bonferroni correction.

Z-tests were used to assess statistical significance and derive Z statistics and corresponding two-sided P values.

Supplementary Table 9 Variables excluded in the scrutiny of multicollinearity by hierarchical clustering.

FieldID	Exposure
X24500	Greenspace percentage, buffer 1000m
X24506	Natural environment percentage, buffer 1000m
X24507	Natural environment percentage, buffer 300m
hand_grip_strength	Hand grip strength
Lung_function	Lung_function
X1930	Miserableness
X1960	Fed-up feelings
X6149_7	Mouth/teeth dental problems_non of the above

Supplementary Table 10 GWAS ID in MRC IEU OpenGWAS database for exposures

Trait	Id.exposure
Qualifications: College or University degree	ukb-b-16489
Basal metabolic rate	ukb-b-16446
Qualifications: None of the above	ukb-b-17729
Qualifications: A levels/AS levels or equivalent	ukb-b-11615
Number of days/week of moderate physical activity 10+ minutes	ukb-b-4710
Leisure/social activities: Adult education class	ukb-b-1553
Qualifications: O levels/GCSEs or equivalent	ukb-b-18099
Time spent using computer	ukb-b-4522
Time spent watching television (TV)	ukb-b-5192
Traumatic subdural haemorrhage	ukb-b-10810
Average total household income before tax	ukb-b-7408
Forced expiratory volume in 1-second (FEV1)	ukb-b-19657
Sleep duration (dichotomized as <7/>9h, 7-9h)	GWAS was conducted based on dichotomization in EWAS
Stroke, including SAH	ukb-d-I9_STR_SAH
Particulate matter air pollution (pm2.5); 2010	ukb-b-10817
Vitamin and mineral supplements: Folic acid or Folate (Vit B9)	ukb-b-3563
Qualifications: Other professional qualifications eg: nursing, teaching	ukb-b-13799
Diagnoses - main ICD10: I48 Atrial fibrillation and flutter	ukb-b-964
Diagnoses - main ICD10: D50.9 Iron deficiency anaemia, unspecified	ukb-b-3355
Mouth/teeth dental problems: None of the above	ukb-b-17601
Hand grip strength (left)	ukb-a-374
Diagnoses - main ICD10: K59 Other functional intestinal disorders	ukb-d-K59
Suffer from 'nerves'	ukb-b-19957
Mental health problems ever diagnosed by a professional: Depression	ukb-d-20544_11
Hypertension	ukb-d-I9_HYPTENS
Number of days/week of vigorous physical activity 10+ minutes	ukb-b-151
Attendance/disability/mobility allowance: Disability living allowance	ukb-b-20214
Diabetes diagnosed by doctor	ukb-a-306
Tense / 'highly strung'	ukb-b-10093

Diagnoses - secondary ICD10: E78.0 Pure hypercholesterolaemia	ukb-b-12651
Attendance/disability/mobility allowance: Blue badge	ukb-b-1867
Mouth/teeth dental problems: Dentures	ukb-b-12930
Diagnoses - secondary ICD10: E66.9 Obesity, unspecified	ukb-b-15541
Water intake	ukb-b-14898
Nap during day	ukb-b-4616
Mouth/teeth dental problems: Toothache	ukb-b-19191
Tea intake	ukb-b-6066
Miserableness	ukb-b-18994
Mouth/teeth dental problems: Painful gums	ukb-b-11161
Major coronary heart Medical history event	ukb-d-I9_CHD
Diagnoses - main ICD10: N39 Other disorders of urinary system	ukb-d-N39
Current employment status: Looking after home and/or family	ukb-b-9056
Current employment status: Unemployed	ukb-b-12885
Anxiety disorders	ukb-d-KRA_PSY_ANXIETY
Qualifications: CSEs or equivalent	ukb-b-10505
Fed-up feelings	ukb-b-19809
Mood swings	ukb-b-14180
Loneliness, isolation	ukb-b-8476
Diagnoses - main ICD10: M47 Spondylosis	ukb-d-M47
Able to confide	ukb-b-4982
Diagnoses - main ICD10: D64 Other anaemias	ukb-d-D64
Exposure to tobacco smoke at home	ukb-b-4462
Attendance/disability/mobility allowance: None of the above	ukb-b-3855
Qualifications: NVQ or HND or HNC or equivalent	ukb-b-7926
Diagnoses - secondary ICD10: Z72.1 Alcohol use (alcohol level)	ukb-b-391
Forced vital capacity (FVC)	ukb-b-7953
Mineral and other dietary supplements: Glucosamine	ukb-a-494
Leisure/social activities: None of the above	ukb-b-5076
Hand grip strength (right)	ukb-a-379
Daytime dozing / sleeping (narcolepsy)	ukb-b-5776
Leisure/social activities: Sports club or gym	ukb-b-4000

Time spent outdoors in winter	ukb-b-6811
Current employment status: Full or part-time student	ukb-b-20437
Townsend deprivation index at recruitment	ukb-b-10011
Peak expiratory flow (PEF)	ukb-b-12019
Diagnoses - main ICD10: H25 Senile cataract	ukb-a-529
Sleeplessness / insomnia	ukb-b-3957
Current employment status: Retired	ukb-b-19852
Worrier / anxious feelings	ukb-b-6519
Attendance/disability/mobility allowance: Attendance allowance	ukb-b-16671
Current employment status: Unable to work because of sickness or disability	ukb-b-14080
Diagnoses - main ICD10: K29 Gastritis and duodenitis	ukb-a-547
Current employment status: Doing unpaid or voluntary work	ukb-b-8810
Current employment status: In paid employment or self-employed	ukb-b-6740
Vitamin and mineral supplements: Vitamin B	ukb-b-10188
Diagnoses - main ICD10: F43 Reaction to severe stress and adjustment disorders	ukb-a-526

Supplementary Table 11 Two-sample mendelian randomisation analyses identified risk factors for dementia (using GWAS for Alzheimer's disease)

Trait	IVW				MREgger				Wmedian				Instrument strength	Heterogeity		Horizontal Pleiotropy		
	OR	LCI	UCI	p	OR	LCI	UCI	p	OR	LCI	UCI	p	Mean.F.statitics	Q	Qpval	p.for.Egger.intercept	p.for.MR.PRESSO.global.test	p.for.MR.PRESSO.distortion.test
Qualifications: College or University degree	0.465 3576	0.331 2397	0.6537 79326	1.03 E-05	0.5633 73251	0.138 2405	2.2959 22863	0.424 325	0.497 3271	0.308 7301	0.8011 33955	0.004 0853	271.5773594	0.0026 794	46.876 327	0.783700 53	0.001	0.84
Basal metabolic rate	0.751 5179	0.660 2014	0.8554 64878	1.55 E-05	0.8288 36459	0.607 0204	1.1317 08107	0.238 0742	0.746 0195	0.617 7054	0.9009 87835	0.002 3443	578.197211	3.98E- 05	80.250 952	0.498285 191	<0.001	0.958
Qualifications: None of the above	4.102 1062	2.022 0271	8.3219 83184	9.20 E-05	51.625 61371	1.019 7912	2613.4 80207	0.052 459	3.355 0952	1.311 1777	8.5851 54934	0.011 564	101.238459	0.0396 582	42.677 898	0.202395 245	0.043	NA
Qualifications: A levels/AS levels or equivalent	0.400 913	0.214 4424	0.7495 31317	0.004 195	0.0672 74475	0.001 9781	2.2880 10688	0.138 1111	0.518 63	0.223 4598	1.2036 93215	0.126 4084	84.09142812	0.1372 623	41.369 204	0.316920 219	0.144	NA
Number of days/week of moderate physical activity 10+ minutes	1.957 4466	1.154 6782	3.3183 24885	0.012 6289	54.635 04948	1.89 E-05	15760 4367	0.650 7242	1.808 4574	1.005 6518	3.2521 37519	0.047 8359	3.497845341	0.3210 417	32.912 357	0.703517 012	0.388	NA
Leisure/social activities: Adult education class	0.067 8705	0.005 8919	0.7818 23921	0.030 9759	0.0010 75387	9.62 E-07	1.2023 1145	0.067 004	0.201 1974	0.009 6606	4.1902 48213	0.300 6225	38.25522445	0.0936 667	24.802 907	0.228170 923	0.096	NA
Qualifications: O levels/GCSEs or equivalent	0.286 0062	0.091 1945	0.8969 78892	0.031 8384	0.0893 06477	1.19 E-05	670.34 80024	0.602 9882	0.095 0627	0.020 4116	0.4427 34061	0.002 7171	19.4875702	0.3012 74	35.546 655	0.799770 134	0.297	NA
Time spent using computer	0.700 5407	0.505 9666	0.9699 40213	0.032 0451	0.4666 49941	0.092 9726	2.3422 19575	0.357 5061	0.597 1179	0.390 4885	0.9130 865	0.017 3313	92.98832379	0.0670 571	38.708 093	0.615740 92	0.064	NA
Time spent watching television (TV)	1.383 1817	1.023 8783	1.8685 73386	0.034 5352	4.6597 47285	1.228 5343	17.674 10498	0.025 845	1.387 0757	0.910 0239	2.1142 07188	0.128 1211	118.5407232	0.0995 273	41.105 766	0.069985 264	0.101	NA

Traumatic subdural haemorrhage	1.062 7306 93	1.003 6197 55	1.1253 23131	0.037 1831	43.054 90	3.597 050E -03	5.1534 57E+0 5	0.467 8162	12.47 7557 2	6.963 545E -02	2235.7 78	0.340 3608	75.80433	2.2877 036	0.8914 295	0.614741 4	0.836	NA
Average total household income before tax	0.746 3034	0.566 6967	0.9828 33985	0.037 2265	1.2530 09929	0.367 4266	4.2730 54205	0.720 4322	0.739 7916	0.497 109	1.1009 48917	0.137 3157	48.77527537	0.4662 911	0.9766 985	0.400431 737	0.246	NA
Forced expiratory volume in 1-second (FEV1)	0.840 1829	0.707 8078	0.9973 15045	0.046 5099	0.6963 70968	0.406 1334	1.1940 2271	0.189 82	0.778 956	0.617 3591	0.9828 51791	0.035 2206	265.6569209	0.0048 534	56.225 233	0.472410 293	0.005	0.611
Sleep duration (dichotomized as <7/>9h, 7-9h)	0.895 6833	0.703 0749	1.1410 57	0.372 4909	0.7784 713	0.423 0200	1.4325 98	0.433 5330	0.963 4480	0.686 2383	1.3526 38	0.829 6846	22.86979	12.811 249	0.6865 053	0.630482 5	0.666	NA
Stroke, including SAH	0.002 121	2.79 E-06	1.6134 69874	0.068 9623	5.3947 81751	1.82 E-08	16014 61024	0.868 6097	0.001 9183	1.98 E-07	18.547 49682	0.181 4649	6.652641388	0.8796 894	23.791 147	0.420016 062	0.916	NA
Particulate matter air pollution (pm2.5); 2010	1.773 5137	0.900 2732	3.4937 73463	0.097 6602	1.5198 48755	0.524 6946	4.4024 47321	0.475 2987	1.665 0355	0.692 7664	4.0018 44285	0.254 4655	4.081538664	0.6656 429	36.575 509	0.727093 78	0.733	NA
Vitamin and mineral supplements: Folic acid or Folate (Vit B9)	0.003 3639	3.52 E-06	3.2156 16942	0.103 8625	2.50E- 08	3.37 E-15	0.1855 58306	0.055 2915	0.002 4453	5.73 E-07	10.444 14821	0.158 5551	16.34435708	0.1288 205	36.596 432	0.140466 966	0.089	NA
Qualifications: Other professional qualifications eg: nursing, teaching	0.288 7545	0.058 0005	1.4375 59879	0.129 3148	41.932 94782	0.001 5997	10991 84.445	0.482 7461	0.311 7924	0.044 0001	2.2094 15044	0.243 4037	27.7411434	0.0339 448	36.346 487	0.346823 09	0.028	NA
Diagnoses - main ICD10: I48 Atrial fibrillation and flutter	0.081 6032	0.003 1077	2.1427 38374	0.132 8566	0.0109 51809	1.11 E-05	10.792 75047	0.220 1256	0.047 5113	0.000 4297	5.2537 52614	0.204 4311	11.11576638	0.7443 455	79.888 968	0.527104 729	0.76	NA
Diagnoses - main ICD10: D50.9 Iron deficiency anaemia, unspecified	1.35 E-10	1.56 E-23	1163.6 10937	0.134 7905	2.07E +94	1.17 E- 294	Inf	0.666 5036	7.32 E-12	2.39 E-24	22.392 74958	0.080 4542	9.407299143	0.0516 873	22.989 701	0.635096 65	0.1	NA

Mouth/teeth dental problems: None of the above	2.957 4059	0.663 0966	13.190 00891	0.155 1907	1.4278 58514	0.020 3755	100.06 05863	0.873 1446	2.198 561	0.452 4974	10.682 20721	0.328 6708	17.2893818	0.0682 01	55.455 597	0.726277 881	0.089	NA
Hand grip strength (left)	1.302 9981	0.902 8039	1.8805 8992	0.157 4201	2.7249 86814	0.577 6128	12.855 59021	0.209 5728	1.169 7451	0.766 5163	1.7850 93893	0.467 2141	119.1627923	0.0002 269	46.991 206	0.340698 547	0.001	0.428
Diagnoses - main ICD10: K59 Other functional intestinal disorders	1903. 1558	0.052 1	69520 174.47	0.158 899	0.1324 21949	1.11 E-11	15789 51141	0.872 6794	1719 9.714	0.024 5915	12029 77244 2	0.155 5033	4.804611406	0.4401 882	23.010 166	0.415703 727	0.483	NA
Suffer from 'nerves'	6.392 5968	0.437 3919	93.429 4669	0.175 1948	26939 2943	2.753 4531	2.64E+ 16	0.057 65	1.028 8294	0.111 6145	9.4834 46652	0.979 9909	57.07841339	7.96E- 07	35.454 033	0.080002 404	<0.001	0.001
Mental health problems ever diagnosed by a professional: Depression	2.218 7386	0.661 8559	7.4378 74363	0.196 6043	21.357 89021	0.563 3425	809.73 73006	0.124 7306	2.329 7486	0.579 6961	9.3630 58025	0.233 3727	20.29010123	0.0882 09	24.933 751	0.221092 828	0.067	NA
Hypertension	9127. 0868	0.004 2999	19373 31102 2	0.219 8713	7.48E +15	4.479 9045	1.25E+ 31	0.071 319	0.470 3261	2.38 E-09	92949 795.8	0.938 3054	10.67422719	0.3834 554	23.428 444	0.128157 805	0.396	NA
Number of days/week of vigorous physical activity 10+ minutes	39.10 8242	0.085 5508	17877. 72603	0.240 7032	1.43E +26	1.70 E+11	1.21E+ 41	0.007 452	1.271 9367	0.740 9709	2.1833 82645	0.382 9177	3572.324728	0	39.647 324	0.009962 927	<0.001	NA
Attendance/disability/mobility allowance: Disability living allowance	0.006 0114	1.11 E-06	32.536 98845	0.243 6079	2.32E- 08	5.19 E-26	10381 80870 6	0.444 3258	0.142 7808	9.35 E-05	218.13 43672	0.602 8143	15.2291084	0.0094 27	32.410 028	0.570576 553	0.018	0.08
Diabetes diagnosed by doctor	0.452 171	0.118 7491	1.7217 69415	0.244 6306	0.5298 95804	0.022 3272	12.576 13246	0.696 3681	0.740 5833	0.128 2154	4.2776 7216	0.737 1419	70.28025281	0.0029 753	66.472 493	0.914072 645	0.009	NA
Tense / 'highly strung'	2.305 1967	0.544 9817	9.7506 606	0.256 3568	0.3088 12051	2.55 E-05	3743.0 94689	0.808 232	1.179 986	0.233 2146	5.9703 24295	0.841 4201	56.67162352	0.0022 841	36.307 862	0.674559 791	0.003	NA
Diagnoses - secondary ICD10: E78.0 Pure	82.66 1892	0.033 2961	20521 8.8893	0.268 3257	61159 8.4264	0.067 6701	5.53E+ 12	0.131 2814	2.799 362	0.206 257	37.993 51576	0.439 1583	222.3301275	7.80E- 41	93.911 256	0.240508 222	<0.001	<0.001

hypercholesterolaemia																		
Attendance/disability/mobility allowance: Blue badge	0.1722255	0.0068535	4.327937364	0.284924	0.001333431	1.31E-09	1360.119312	0.3553562	0.1832654	0.0034642	9.695239949	0.4020022	48.2009652	0.0425311	23.448882	0.483792501	0.049	0.259
Mouth/teeth dental problems: Dentures	1.9394961	0.5740987	6.5522618	0.2861898	9.162676653	0.1435849	584.7036462	0.3042474	4.1371851	0.9958956	17.18684254	0.0506611	53.51760404	0.0099246	47.18934	0.449270793	0.007	NA
Diagnoses - secondary ICD10: E66.9 Obesity, unspecified	0.0077264	7.19E-07	82.97009802	0.3044455	1083819595	1.06E-07	1.11E+25	0.294589	0.0016873	4.66E-08	61.10199934	0.2332156	17.38761977	0.0969213	28.734741	0.189972447	0.101	NA
Water intake	9.5907313	0.1274551	721.6825939	0.3051062	8.90013529	6.86E-05	1154824.839	0.7183808	0.8027248	0.433038	1.488015266	0.4852778	3942.471288	0	48.520413	0.989388321	<0.001	<0.001
Nap during day	0.8107831	0.5423578	1.212058384	0.306547	1.009794307	0.2580093	3.952123562	0.9888646	0.7100538	0.3991793	1.26303236	0.2438967	95.09007582	0.1355924	47.024333	0.742172563	0.119	NA
Mouth/teeth dental problems: Toothache	0.0940465	0.0007712	11.46816678	0.3347579	0.004027012	3.33E-10	48738.23193	0.5211313	0.0138233	3.67E-05	5.203769968	0.1570939	16.39052693	0.1739967	22.771664	0.698488297	0.189	NA
Tea intake	1.1480416	0.8574449	1.537124259	0.3538512	1.406692153	0.7365198	2.68666614	0.3095619	1.4005181	0.9076469	2.161028691	0.1279772	27.50444893	0.646616	61.581928	0.495750982	0.658	NA
Miserableness	1.455853	0.628917	3.37009143	0.3804529	0.032610916	0.0008492	1.252274079	0.0751905	0.9725017	0.3143503	3.008616806	0.9614042	38.73846693	0.226574	39.277361	0.044480174	0.245	NA
Mouth/teeth dental problems: Painful gums	0.0355225	1.92E-05	65.59971426	0.3844264	0.015364189	8.24E-17	2.86E+12	0.8132102	0.4595801	7.19E-05	2937.179982	0.8619474	8.20983318	0.22313	23.181226	0.960813678	0.255	NA
Major coronary heart Medical history event	3.8931436	0.1683763	90.016035	0.3963154	0.334614775	0.0004348	257.5013573	0.7577456	6.9185267	0.0760634	629.2910599	0.4006214	6.391329356	0.4948692	77.073005	0.442911489	0.419	NA
Diagnoses - main ICD10: N39 Other disorders of urinary system	9.9808553	0.0461661	2157.804145	0.4016052	1455.1909	0.000419	5053869632	0.3635853	2.0830859	0.0045722	949.0407414	0.8142378	18.82518829	0.0928354	23.517215	0.500622822	0.106	NA
Current employment status: Looking after	5.0123288	0.114757	218.9272698	0.4028738	0.019490861	4.18E-08	9087.843329	0.566264	3.5102937	0.0301054	409.3013078	0.6050246	6.57319391	0.8844834	22.59463	0.402570688	0.887	NA

home and/or family																		
Current employment status: Unemployed	0.027 0855	5.20 E-06	141.07 24982	0.408 5239	5.39E-21	1.07 E-39	0.0272 32563	0.059 6212	2.917 7663	0.000 3828	22241. 56255	0.814 3683	19.38822291	0.0544 709	22.709 488	0.074500 712	0.07	NA
Anxiety disorders	0.001 9772	7.19 E-10	5434.8 97542	0.410 4799	3.07E-15	7.23 E-36	13075 05.589	0.201 2386	11.40 8139	2.25 E-08	57844 24093	0.811 8508	9.891923554	0.4500 259	22.602 837	0.267888 194	0.476	NA
Qualifications: CSEs or equivalent	4.561 5737	0.120 0856	173.27 60786	0.413 4539	0.0003 25337	7.26 E-10	145.77 97569	0.293 0349	7.982 8569	0.218 7595	291.30 6273	0.257 6803	9.730989164	0.0832 264	39.767 611	0.211547 073	0.113	NA
Fed-up feelings	0.767 804	0.401 2899	1.4690 70076	0.424 7898	0.5387 13379	0.020 1817	14.379 97438	0.713 6826	0.646 596	0.256 5517	1.6296 37829	0.355 2114	50.79937314	0.3638 742	39.211 956	0.830075 817	0.342	NA
Mood swings	1.287 4398	0.681 45	2.4323 15043	0.436 3368	3.8092 19677	0.081 0488	179.02 98026	0.499 1726	0.973 7806	0.414 6295	2.2869 78254	0.951 3649	54.5258158	0.3064 819	39.642 584	0.578007 808	0.3	NA
Loneliness, isolation	3.572 1888	0.141 0353	90.477 60421	0.440 0439	0.1768 20017	8.43 E-08	37095 4.996	0.820 7524	0.395 6507	0.030 1686	5.1888 19452	0.480 1145	32.5938486	0.0003 187	36.886 905	0.687112 471	0.001	0.031
Diagnoses - main ICD10: M47 Spondylosis	0.015 2846	2.77 E-07	843.41 06711	0.452 9345	7.16E-11	6.53 E-22	7.8415 8866	0.094 9004	9.222 7235	2.31 E-05	36774 48.126	0.735 62	21.90679519	0.0805 499	23.226 895	0.129504 489	0.072	NA
Able to confide	1.142 8869	0.803 5611	1.6255 02384	0.457 4065	1.1225 85035	0.214 8028	5.8667 62169	0.894 0047	1.274 245	0.812 9478	1.9972 99538	0.290 5592	4.101824975	0.9426 366	38.790 546	0.983127 755	0.928	NA
Diagnoses - main ICD10: D64 Other anaemias	0.041 985	7.39 E-06	238.52 91368	0.472 2578	187.65 01682	2.29 E-07	1.54E+11	0.628 0028	0.403 3243	1.69 E-05	9638.4 91494	0.859 8764	17.676156	0.0894 055	25.170 12	0.396021 474	0.069	NA
Exposure to tobacco smoke at home	0.554 0945	0.109 8171	2.7957 46664	0.474 6159	0.1893 68574	0.001 8554	19.327 28212	0.489 7642	0.703 9436	0.080 1974	6.1789 65073	0.751 4246	17.16221391	0.5788 791	22.905 909	0.633085 413	0.569	NA
Attendance/disability/mobility allowance: None of the above	3.436 6288	0.111 0685	106.33 4563	0.480 8153	11822. 4681	7.35 E-05	1.90E+12	0.368 2596	6.487 2663	0.077 8639	540.48 9396	0.407 2931	6.226685567	0.5135 44	35.577 939	0.423347 316	0.524	NA
Qualifications: NVQ or HND or HNC or equivalent	1.938 4169	0.243 6246	15.423 1533	0.531 6485	0.0730 74776	1.03 E-05	517.99 71719	0.569 8127	2.074 8539	0.189 4097	22.728 60459	0.550 0816	33.79998161	0.0275 1	23.329 349	0.464888 547	0.026	NA
Diagnoses - secondary ICD10: Z72.1 Alcohol use (alcohol level)	0.050 3720 3	2.599 878e -08	97594. 6141	0.685 7841	9.6484 05e-07	2.285 277e -62	4.0735 42e+4 9	0.845 7210	0.351 3002	4.929 263e -09	25036 564.76 4	0.909 7183	23.59081	0.4662 911	0.9766 985	0.877772 3	0.975	NA

Forced vital capacity (FVC)	0.926 0175	0.711 8294	1.2046 54462	0.566 8524	0.7293 16662	0.345 1173	1.5412 23158	0.409 0632	0.851 7709	0.688 8335	1.0532 49755	0.138 588	1017.740432	4.43E- 88	62.414 8	0.504547 364	<0.001	0.441
Mineral and other dietary supplements: Glucosamine	1.383 0824	0.411 3807	4.6499 92353	0.600 1179	0.0133 60255	0.000 1757	1.0156 78678	0.063 0971	0.837 1016	0.151 9963	4.6102 39616	0.838 1406	22.54349291	0.5468 62	24.148 306	0.039155 034	0.585	NA
Leisure/social activities: None of the above	1.842 6869	0.158 0545	21.483 06371	0.625 7076	0.3428 52879	4.40 E-09	26685 561.34	0.911 3157	2.561 5063	0.219 0132	29.958 53175	0.453 4621	18.28426699	0.0191 934	34.978 086	0.859746 268	0.024	NA
Hand grip strength (right)	0.926 4306	0.660 3414	1.2997 42203	0.658 2276	1.9874 37508	0.526 639	7.5002 18388	0.313 7667	0.887 1953	0.592 553	1.3283 46051	0.561 0956	137.3559505	0.0001 247	46.281 22	0.247540 746	<0.001	0.906
Daytime dozing / sleeping (narcolepsy)	1.185 5577	0.553 2159	2.5406 84164	0.661 6092	0.1206 74768	0.003 4128	4.2669 83748	0.254 8723	1.295 3127	0.441 878	3.7970 5495	0.637 2383	29.82597488	0.4227 291	44.077 038	0.209305 392	0.415	NA
Leisure/social activities: Sports club or gym	1.569 6969	0.185 5223	13.281 1467	0.678 9948	0.0167 96086	4.91 E-11	57425 89.565	0.704 4253	1.297 9922	0.091 7053	18.371 71424	0.847 0312	2.400300314	0.7914 294	32.895 944	0.672518 365	0.787	NA
Time spent outdoors in winter	1.612 5303	0.161 938	16.057 09296	0.683 6661	45930. 02062	0.002 874	7.34E+ 11	0.332 2875	3.560 545	0.463 4236	27.356 14289	0.222 2015	6.970269452	0.0728 514	49.774 046	0.346141 154	0.167	NA
Current employment status: Full or part-time student	0.047 0716	1.62 E-08	13707 0.2898	0.687 3663	2.32E- 27	3.70 E-86	1.45E+ 32	0.415 2626	0.864 5545	7.73 E-09	96734 365.12	0.987 7194	8.558108164	0.1999 969	22.880 098	0.434611 132	0.219	NA
Townsend deprivation index at recruitment	1.107 3894	0.616 5008	1.9891 48088	0.732 8382	7.0697 16572	0.047 6769	1048.3 25416	0.455 0917	0.882 593	0.376 4889	2.0690 39858	0.773 8702	16.4853461	0.4196 305	35.432 282	0.475442 222	0.419	NA
Peak expiratory flow (PEF)	0.961 969	0.761 5147	1.2151 89071	0.745 014	0.8186 75032	0.363 6616	1.8430 01118	0.629 8587	1.027 5223	0.746 4827	1.4143 69286	0.867 7327	149.4916421	0.0143 999	50.273 607	0.684808 323	0.02	0.833
Diagnoses - main ICD10: H25 Senile cataract	0.285 4903	5.76 E-05	1413.8 47829	0.772 7387	13.791 91703	1.08 E-06	17666 0867.2	0.757 3782	45.86 7311	0.001 4057	14965 80.633	0.470 6038	26.29876198	0.0691 867	24.944 764	0.591257 995	0.08	NA
Sleeplessness / insomnia	0.920 1245	0.522 7231	1.6196 51347	0.772 9258	0.7442 82589	0.132 9018	4.1681 63515	0.739 2936	0.809 389	0.396 1648	1.6536 3156	0.561 8093	37.39163721	0.1660 017	45.230 163	0.799807 206	0.168	NA

Current employment status: Retired	1.175 6355	0.305 8074	4.5195 72216	0.813 8097	0.1512 30991	0.004 4257	5.1677 42141	0.306 3669	0.582 4463	0.094 6026	3.5859 88518	0.559 9738	27.30533136	0.1998 585	23.600 38	0.232709 118	0.208	NA
Worrier / anxious feelings	1.075 0115	0.489 4425	2.3611 54837	0.857 011	0.0405 07555	0.000 6555	2.5033 2832	0.134 1023	0.877 2763	0.345 6341	2.2266 71758	0.782 918	82.07001655	0.0021 399	40.817 396	0.119189 719	0.002	NA
Attendance/disability/mobility allowance: Attendance allowance	2.240 179	0.000 174	28844. 54986	0.867 3278	0.2914 02892	3.62 E-27	2.35E+ 25	0.968 408	1.680 7943	8.68 E-06	32548 4.647	0.933 3724	3.739952292	0.9876 849	22.825 934	0.947101 362	0.985	NA
Current employment status: Unable to work because of sickness or disability	0.825 7191	0.065 2955	10.441 93817	0.882 3999	4931.8 32848	0.064 2457	37859 3203.2	0.148 5173	2.989 8113	0.080 0009	111.73 58375	0.553 2916	30.09677327	0.5631 332	25.027 082	0.130064 382	0.563	NA
Diagnoses - main ICD10: K29 Gastritis and duodenitis	0.768 0595	0.000 7644	771.73 17315	0.940 355	0.0789 06319	2.52 E-12	24747 81739	0.846 8928	1.846 744	0.000 6612	5158.1 74195	0.879 5642	0.379540058	0.9958 744	23.374 722	0.856664 864	0.995	NA
Current employment status: Doing unpaid or voluntary work	1.243 831	0.001 1994	1289.9 03318	0.950 8921	0.1824 30382	1.72 E-13	1.94E+ 11	0.906 7923	0.089 8419	5.92 E-05	136.29 34787	0.519 0415	21.11077014	0.0203 343	24.297 344	0.891076 834	0.022	NA
Current employment status: In paid employment or self-employed	0.968 3863	0.227 075	4.1297 89352	0.965 3726	2.4449 21708	0.055 4831	107.73 80574	0.648 2213	1.371 3862	0.230 6694	8.1532 27833	0.728 4009	32.1610741	0.0746 911	24.018 948	0.608062 888	0.088	NA
Vitamin and mineral supplements: Vitamin B	0.969 8838	0.013 0136	72.284 23324	0.988 908	816.09 67755	0.001 6035	41534 7731.3	0.340 8654	4.222 2225	0.012 3926	1438.5 32377	0.628 2767	7.860869307	0.7257 101	24.326 675	0.312565 829	0.703	NA

Supplementary Table 12 Two-sample mendelian randomisation analyses identified risk factors for dementia (using GWAS for all-cause dementia)

Trait	IVW				MREgger				Wmedian				Instrument strength	Heterogeity		Horizontal Pleiotropy		
	OR	LCI	UCI	p	OR	LCI	UCI	p	OR	LCI	UCI	p	Mean.F.s tatistics	Q	Qpval	p.for.Egger. intercept	p.for.MR.PRESS O.global.test	p.for.MR.PRESSO .distortion.test
Qualifications: A levels/AS levels or equivalent	0.55 5094	0.30 8639	0.998 348	0.04 9354	0.049 969	0.00 2066	1.208 314	0.06 9603	0.48 1533	0.20 8458	1.112 329	0.08 7121	41.3893	75.5 2337	0.27 5899	0.136509	0.273	NA
Leisure/social activities: None of the above	5.38 3868	0.85 4824	33.90 877	0.07 2983	0.929 747	1.39 E-06	6205 82.6	0.99 1803	5.35 9288	0.61 4535	46.73 775	0.12 8672	34.97809	11.6 8913	0.16 5622	0.802754	0.184	NA
Exposure to tobacco smoke at home	3.63 8289	0.81 1281	16.31 635	0.09 1634	2.984 076	0.03 3109	268.9 527	0.63 9761	3.73 9349	0.49 5114	28.24 146	0.20 1057	22.93684	13.0 1436	0.83 7835	0.928066	0.821	NA
Hypertension	5.23 E-05	5.44 E-10	5.015 227	0.09 2084	2352 498	2.46 E-09	2.25 E+21	0.42 6061	1.96 E-05	4.86 E-12	79.20 75	0.16 2522	23.42844	6.09 7916	0.80 6971	0.173531	0.852	NA
Able to confide	1.39 5687	0.94 2904	2.065 897	0.09 5677	0.222 205	0.03 9972	1.235 227	0.11 643	1.55 7703	0.98 1265	2.472 766	0.06 0137	38.74346	15.5 9954	0.15 6661	0.057822	0.199	NA
Diagnoses - secondary ICD10: E78.0 Pure hypercholesterol aemia	156. 8665	0.39 5768	6217 5.64	0.09 7659	2206 0.09	0.13 696	3.55 E+09	0.13 0314	19.7 4151	1.19 3478	326.5 476	0.03 7201	94.01306	138. 681	1.11 E-23	0.370028	<0.001	<0.001
Qualifications: None of the above	1.69 1286	0.90 1745	3.172 127	0.10 1488	59.60 013	1.77 0005	2006. 873	0.02 56	1.47 3523	0.63 4283	3.423 187	0.36 7375	42.80308	82.6 0531	0.25 6181	0.047369	0.262	NA
Stroke, including SAH	0.00 6371	1.03 E-05	3.937 416	0.12 3072	0.000 11	4.12 E-14	2924 68	0.42 9498	0.00 8828	7.19 E-07	108.4 492	0.32 4852	23.85206	9.76 0943	0.55 201	0.709024	0.547	NA
Mouth/teeth dental problems: None of the above	2.40 2168	0.77 7448	7.422 251	0.12 7853	1.411 242	0.06 4592	30.83 352	0.83 2198	1.79 3065	0.41 0383	7.834 338	0.43 7663	55.4556	4.59 6952	0.86 7934	0.725866	0.869	NA

Leisure/social activities: Sports club or gym	0.15 0087	0.01 2673	1.777 432	0.13 2606	2.315 685	8.98 E-12	5.97 E+11	0.95 3062	0.52 095	0.03 4294	7.913 525	0.63 8512	32.89594	7.02 3472	0.21 8901	0.847414	0.262	NA
Nap during day	1.39 0773	0.90 0532	2.147 895	0.13 6874	8.715 641	1.95 4514	38.86 51	0.00 5809	1.64 0483	0.91 7479	2.933 237	0.09 5015	46.51523	106. 687	0.01 4198	0.014299	0.01	0.212
Diagnoses - main ICD10: D50.9 Iron deficiency anaemia, unspecified	7.25 E-08	2.14 E-17	245.2 078	0.14 1966	#### ###	8.85 E-69	Inf	0.26 5287	3.48 E-08	6.42 E-19	1886. 246	0.17 3233	22.9897	6.23 6543	0.18 2166	0.248213	0.257	NA
Basal metabolic rate	0.91 7169	0.81 3845	1.033 612	0.15 6227	0.910 304	0.68 9193	1.202 354	0.50 8345	0.88 1409	0.73 3793	1.058 722	0.17 7072	80.51271	540. 503	0.00 1138	0.953282	<0.001	0.874
Mouth/teeth dental problems: Painful gums	0.02 9722	0.00 0176	5.017 114	0.17 9068	132.7 693	2.99 E-07	5.89 E+10	0.64 503	0.03 768	5.01 E-05	28.33 911	0.33 1901	23.51062	4.24 151	0.83 4703	0.420215	0.834	NA
Miserableness	0.61 8721	0.29 6153	1.292 625	0.20 1537	0.940 126	0.02 4495	36.08 286	0.97 374	0.66 945	0.22 5244	1.989 675	0.47 0242	39.16619	34.4 0639	0.40 0286	0.819803	0.353	NA
Time spent using computer	0.79 5725	0.55 8841	1.133 022	0.20 5037	0.456 8	0.06 5943	3.164 322	0.43 0317	0.81 373	0.51 7709	1.279 012	0.37 1643	38.84503	101. 4219	0.00 5347	0.569399	0.005	0.556
Qualifications: Other professional qualifications eg: nursing, teaching	0.42 1118	0.10 672	1.661 742	0.21 6886	0.086 687	7.74 E-06	970.4 019	0.61 4669	0.21 107	0.04 3178	1.031 8	0.05 469	36.43899	22.6 5486	0.12 3277	0.741399	0.125	NA
Number of days/week of moderate physical activity 10+ minutes	1.42 9101	0.80 5677	2.534 925	0.22 2065	5.634 912	0.32 5028	97.69 062	0.32 0357	1.31 8959	0.71 4813	2.433 717	0.37 5733	37.46254	8.34 2086	0.07 9821	0.406772	0.133	NA
Daytime dozing / sleeping (narcolepsy)	1.85 5198	0.64 9476	5.299 288	0.24 8482	15.97 088	0.15 2713	1670. 253	0.25 3445	1.65 9726	0.55 3229	4.979 302	0.36 6057	44.64747	52.8 9568	0.00 2072	0.360127	0.001	0.261
sleep duration (dichotomized as <7/>9h, 7-9h)	1.13 6982	0.90 5145	1.428 2	0.26 9848	0.944 023	0.55 3938	1.608 808	0.83 4944	0.92 3109	0.65 9557	1.291 974	0.64 0882	22.78208	15.2 0905	0.58 0427	0.460349	0.581	NA
Water intake	8.61 3072	0.18 533	400.2 858	0.27 16	9.862 49	0.00 0256	3805 27.9	0.67 4363	0.92 0766	0.52 5899	1.612 115	0.77 2679	50.98719	2880 .022	0	0.978609	<0.001	<0.001

Attendance/disability/mobility allowance: None of the above	9.50 4716	0.15 9665	565.8 083	0.28 0129	2.32 E+10	26.3 4982	2.05 E+19	0.07 232	3.19 1801	0.02 998	339.8 179	0.62 6028	36.1645	8.17 7696	0.22 5372	0.091201	0.266	NA
Diagnoses - main ICD10: M47 Spondylosis	0.01 6191	8.47 E-06	30.93 816	0.28 4769	1.65 E-07	3.74 E-17	725.0 963	0.18 9629	0.00 0651	1.68 E-08	25.23 073	0.17 3456	23.306	10.9 858	0.75 36	0.298779	0.749	NA
Current employment status: In paid employment or self-employed	0.55 9978	0.19 2985	1.624 87	0.28 6031	0.802 795	0.07 5895	8.491 726	0.85 6844	0.78 4103	0.17 0061	3.615 268	0.75 5117	24.04746	23.3 2473	0.44 1948	0.739157	0.425	NA
Current employment status: Looking after home and/or family	0.16 1807	0.00 5179	5.055 419	0.29 9642	0.330 344	2.64 E-07	4136 69.6	0.87 9921	0.31 586	0.00 2205	45.24 238	0.64 9113	22.29723	9.45 3362	0.66 3797	0.919995	0.661	NA
Diagnoses - main ICD10: D64 Other anaemias	0.00 8012	8.69 E-07	73.85 991	0.30 0053	0.000 46	6.88 E-16	3.08 E+08	0.59 5317	0.00 0139	5.85 E-09	3.299 772	0.08 4031	25.43963	16.1 0061	0.06 481	0.831431	0.063	NA
Time spent watching television (TV)	1.16 0227	0.87 3661	1.540 789	0.30 4506	4.560 193	1.12 5474	18.47 698	0.03 6104	1.42 5339	0.95 1309	2.135 574	0.08 5792	41.23208	110. 6978	0.16 1582	0.053267	0.147	NA
Sleeplessness / insomnia	1.36 8666	0.74 5043	2.514 277	0.31 1796	1.448 458	0.26 6412	7.875 127	0.67 1077	1.44 4566	0.65 5001	3.185 905	0.36 2048	45.00876	53.6 5541	0.00 6999	0.944299	0.007	NA
Current employment status: Full or part-time student	4650 .033	0.00 0186	1.16 E+11	0.33 1194	8.26 E-15	4.13 E-83	1.65 E+54	0.70 2826	3354 55.1	0.00 8479	1.33 E+13	0.15 4001	22.8801	11.9 796	0.06 2425	0.629929	0.094	NA
Worrier / anxious feelings	0.69 7592	0.32 8799	1.480 038	0.34 8051	0.465 562	0.01 3883	15.61 239	0.67 171	0.64 0384	0.26 1336	1.569 212	0.32 973	41.50107	75.6 4315	0.00 3825	0.818267	0.004	0.458
Anxiety disorders	0.00 428	3.57 E-08	512.4 073	0.36 0617	4043 164	6.46 E-11	2.53 E+23	0.45 6988	5.73 4279	3.63 E-07	9050 5756	0.83 638	22.65681	10.9 7525	0.53 1041	0.29535	0.486	NA
Mineral and other dietary supplements: Glucosamine	1.69 708	0.53 2804	5.405 516	0.37 0882	4.867 673	0.07 1212	332.7 296	0.47 0226	1.51 2484	0.31 465	7.270 316	0.60 5493	24.15666	25.6 422	0.37 1566	0.615601	0.389	NA

Diagnoses - main ICD10: I48 Atrial fibrillation and flutter	5.38 4358	0.12 8705	225.2 536	0.37 6835	22.68 479	0.00 8856	5810 5.4	0.44 8608	28.8 9815	0.54 9845	1518. 797	0.09 6092	79.88897	24.1 6146	0.06 2402	0.686663	0.086	NA
Qualifications: NVQ or HND or HNC or equivalent	0.41 5377	0.05 5742	3.095 3	0.39 1239	64.38 546	0.03 0783	1346 68.5	0.30 0591	0.08 1872	0.00 779	0.860 414	0.03 7045	23.21586	31.7 3626	0.02 3621	0.198557	0.03	NA
Tea intake	1.11 3169	0.84 8222	1.460 873	0.43 9493	1.105 52	0.61 8626	1.975 628	0.73 7145	1.16 2262	0.78 67	1.717 114	0.45 0151	60.81215	31.4 0463	0.49 6531	0.979115	0.503	NA
Hand grip strength (right)	0.90 3152	0.69 4515	1.174 465	0.44 721	0.594 351	0.20 2485	1.744 587	0.34 6342	0.99 1202	0.67 5575	1.454 288	0.96 3963	46.50326	97.9 8689	0.15 8667	0.4344	0.176	NA
Average total household income before tax	0.88 0409	0.61 7036	1.256 199	0.48 2486	0.329 146	0.05 3379	2.029 568	0.23 8589	0.78 8243	0.53 9405	1.151 876	0.21 8905	40.84422	74.2 4319	0.00 0568	0.286575	<0.001	0.698
Qualifications: O levels/GCSEs or equivalent	0.67 7539	0.22 5243	2.038 058	0.48 8415	0.212 938	7.52 E-06	6029. 047	0.77 1485	1.15 1299	0.27 408	4.836 141	0.84 7425	35.72095	17.7 6522	0.33 7823	0.826791	0.337	NA
Qualifications: CSEs or equivalent	0.41 539	0.03 2649	5.284 995	0.49 8395	0.171 377	1.04 E-05	2831. 564	0.74 5432	0.35 3292	0.01 6362	7.628 279	0.50 684	41.63135	2.76 4568	0.59 7964	0.864938	0.634	NA
Current employment status: Unemployed	7.04 1024	0.01 8083	2741. 572	0.52 1286	7.93 E+08	1.12 E-07	5.63 E+24	0.29 4654	122. 0666	0.03 2329	4609 01.6	0.25 2899	22.71449	9.50 2788	0.65 9491	0.334582	0.673	NA
Loneliness, isolation	0.60 4099	0.12 7731	2.857 069	0.52 4924	0.156 109	0.00 0228	106.6 996	0.58 8231	0.88 2789	0.10 9413	7.122 73	0.90 6837	36.18789	10.9 9792	0.52 9097	0.683734	0.534	NA
Current employment status: Doing unpaid or voluntary work	0.20 3996	0.00 1319	31.55 242	0.53 655	68.24 846	1.36 E-07	3.42 E+10	0.68 8163	0.37 5775	0.00 1175	120.1 616	0.73 9426	24.15345	20.1 9621	0.04 2721	0.568963	0.053	NA
Mouth/teeth dental problems: Toothache	0.26 4844	0.00 2637	26.59 761	0.57 2111	0.016 755	2.35 E-08	1196 0.41	0.56 4126	0.76 4991	0.00 328	178.4 023	0.92 3275	22.77166	18.7 0405	0.09 5925	0.676024	0.107	NA
Mood swings	1.22 9502	0.59 3416	2.547 412	0.57 828	53.71 642	0.44 3711	6503. 001	0.10 9952	1.68 1962	0.68 8287	4.110 194	0.25 4043	39.62921	76.5 8164	0.00 9169	0.124934	0.012	0.736

Diabetes diagnosed by doctor	1.43 4949	0.40 1487	5.128 63	0.57 8409	1.185 493	0.05 7273	24.53 844	0.91 2921	4.16 1159	0.82 6465	20.95 097	0.08 383	68.56895	71.9 3001	0.00 1447	0.892121	<0.001	0.62
Tense / 'highly strung'	0.67 837	0.16 0427	2.868 506	0.59 7835	14.31 988	0.00 1202	1706 52.1	0.58 2452	1.62 6275	0.39 427	6.708 024	0.50 118	36.29216	65.8 3456	0.00 0262	0.524044	<0.001	0.186
Hand grip strength (left)	0.91 3323	0.64 8017	1.287 249	0.60 4579	1.058 667	0.24 4831	4.577 763	0.93 9392	0.99 7989	0.66 7918	1.491 172	0.99 2159	46.24887	107. 2176	0.00 2197	0.839407	0.003	0.767
Major coronary heart Medical history event	0.46 3851	0.02 383	9.028 867	0.61 2021	4.392 708	0.00 5198	3712. 169	0.67 9841	0.05 679	0.00 1055	3.056 306	0.15 8365	73.02848	8.49 6877	0.38 6497	0.48697	0.355	NA
Townsend deprivation index at recruitment	1.21 7465	0.55 764	2.658 026	0.62 1352	0.000 783	3.26 E-06	0.188 375	0.02 3885	0.91 1619	0.37 8161	2.197 603	0.83 6697	35.81294	24.4 3492	0.04 0573	0.020166	0.033	0.088
Diagnoses - main ICD10: H25 Senile cataract	4.18 8823	0.01 2376	1417. 758	0.62 9784	0.007 451	3.90 E-07	142.5 385	0.34 658	0.04 1843	1.30 E-05	134.7 445	0.44 1208	25.18258	11.6 2728	0.70 6994	0.141041	0.698	NA
Diagnoses - main ICD10: K59 Other functional intestinal disorders	4.32 6068	0.00 7894	2370. 73	0.64 8953	17.50 865	0.00 0402	7631 83.6	0.62 1868	40.7 8565	0.00 8245	2017 50	0.39 2857	23.12255	5.37 6691	0.49 6484	0.760172	0.511	NA
Diagnoses - secondary ICD10: Z72.1 Alcohol use (alcohol level)	23.3 4369	2.19 E-05	2484 1273	0.65 6368	4.63 E-36	3.98 E-85	5.38 E+13	0.25 2938	3.38 6555	1.11 E-07	1.03 E+08	0.88 9651	23.60601	4.64 1583	0.32 6088	0.236482	0.408	NA
Mental health problems ever diagnosed by a professional: Depression	0.74 6636	0.17 9652	3.103 032	0.68 7686	0.639 146	0.01 1521	35.45 869	0.83 074	0.79 6206	0.21 96	2.886 81	0.72 8752	24.93375	38.0 8753	0.00 0279	0.936288	0.001	0.17
Suffer from 'nerves'	1.37 5453	0.25 9108	7.301 476	0.70 8181	685.3 496	0.00 2998	1.57 E+08	0.31 507	1.39 7606	0.22 6389	8.628 095	0.71 8504	36.78387	32.8 8416	0.01 1659	0.334212	0.007	0.122
Vitamin and mineral supplements: Vitamin B	2.40 6143	0.01 781	325.0 726	0.72 5754	0.082 626	1.06 E-08	6416 63.5	0.76 3807	6.87 3675	0.05 5028	858.5 99	0.43 3837	24.27236	24.2 9704	0.01 8529	0.668699	0.023	NA

Mouth/teeth dental problems: Dentures	0.83 9914	0.29 5835	2.384 619	0.74 3152	0.585 904	0.01 1629	29.52 049	0.79 1181	0.78 2567	0.18 5073	3.309 018	0.73 8917	45.66444	35.6 6952	0.18 3392	0.852951	0.192	NA
Diagnoses - main ICD10: K29 Gastritis and duodenitis	2.59 3134	0.00 6078	1106. 258	0.75 7779	747.6 382	1.54 E-07	3.64 E+12	0.59 2137	0.78 6288	0.00 0149	4135. 783	0.95 6137	22.80496	4.21 4721	0.51 8934	0.632345	0.58	NA
Attendance/disability/mobility allowance: Attendance allowance	3.93 0945	0.00 0433	3572 3.2	0.76 8482	6.28 E-11	5.73 E-35	6.88 E+13	0.42 4932	2.32 1062	2.56 E-05	2107 49.1	0.88 5057	22.94434	2.71 7654	0.99 3956	0.393112	0.995	NA
Forced expiratory volume in 1-second (FEV1)	1.03 3588	0.81 3462	1.313 282	0.78 6876	0.785 991	0.37 6799	1.639 55	0.52 1594	1.05 3246	0.83 9968	1.320 679	0.65 3163	56.72911	593. 6914	1.19 E-37	0.44084	<0.001	0.055
Reaction to severe stress and adjustment disorders	2.86 5723	0.00 1144	7180. 928	0.79 2039	43.29 979	7.15 E-15	2.62 E+17	0.85 7751	3.49 5419	0.00 0579	2111 6.92	0.77 815	41.3202	0.31 673	0.95 6852	0.894545	0.953	NA
Forced vital capacity (FVC)	1.02 6876	0.84 1414	1.253 217	0.79 412	0.762 055	0.43 9974	1.319 913	0.33 3133	0.98 1372	0.80 2091	1.200 727	0.85 504	62.96477	640. 2519	4.58 E-33	0.254545	<0.001	0.114
Diagnoses - secondary ICD10: E66.9 Obesity, unspecified	3.12 0388	0.00 0442	2200 4.35	0.80 1267	7404 134	2.77 E-10	1.98 E+23	0.43 3593	6.75 5372	0.00 0201	2265 95	0.71 936	29.2168	15.7 4577	0.10 7145	0.453538	0.111	NA
Qualifications: College or University degree	0.95 8383	0.67 1367	1.368 101	0.81 4924	1.885 516	0.44 1331	8.055 56	0.39 2998	0.76 0046	0.48 8193	1.183 28	0.22 4419	46.85865	326. 1075	2.40 E-07	0.347236	<0.001	0.697
Traumatic subdural haemorrhage	0.58 2014	0.00 4084	82.93 529	0.83 0614	0.000 339	5.34 E-08	2.148 693	0.14 8162	0.56 1094	0.00 1549	203.2 697	0.84 7572	80.96207	6.90 9403	0.22 7466	0.129771	0.251	NA
Current employment status: Retired	1.08 2659	0.41 5496	2.821 086	0.87 0882	1.047 923	0.15 6499	7.016 939	0.96 1901	1.85 0387	0.42 7505	8.009 11	0.41 0381	23.36334	26.2 3356	0.45 0325	0.969062	0.443	NA

Particulate matter air pollution (pm2.5); 2010	0.96 3714	0.61 5173	1.509 728	0.87 1792	0.952 808	0.50 8544	1.785 183	0.88 5947	0.95 4398	0.58 8335	1.548 226	0.85 0016	36.57551	6.11 5239	0.41 0406	0.956671	0.678	NA
Attendance/disability/mobility allowance: Disability living allowance	2.41 945	1.46 E-05	4001 90.3	0.88 5408	6.53 E-18	4.59 E-39	9290. 039	0.25 2313	0.27 3346	0.00 0136	550.6 124	0.73 8274	31.25047	13.3 0155	0.00 4028	0.238776	0.026	0.121
Peak expiratory flow (PEF)	1.01 5743	0.79 2732	1.301 493	0.90 1704	0.939 782	0.40 8125	2.164 023	0.88 422	0.82 6826	0.59 1346	1.156 078	0.26 6165	49.56563	192. 9124	9.61 E-06	0.848569	<0.001	0.094
Fed-up feelings	0.94 8419	0.39 6006	2.271 427	0.90 5394	0.090 396	0.00 1172	6.972 021	0.28 3964	1.61 3738	0.63 3491	4.110 796	0.31 5811	39.00644	98.8 142	1.50 E-05	0.284792	<0.001	0.336
Time spent outdoors in winter	1.05 8379	0.24 6978	4.535 49	0.93 9085	2.271 468	2.43 E-05	2123 00.7	0.90 114	1.12 568	0.19 5795	6.471 863	0.89 4459	49.77405	1.01 2119	0.79 832	0.90717	0.805	NA
Attendance/disability/mobility allowance: Blue badge	1.04 7044	0.04 1	26.73 905	0.97 7815	83.67 783	3.46 E-05	2.02 E+08	0.55 9099	1.13 0007	0.02 0585	62.03 143	0.95 2308	23.45615	52.6 6651	0.01 6265	0.553216	0.014	NA
Diagnoses - main ICD10: N39 Other disorders of urinary system	1.02 3057	0.01 8268	57.29 284	0.99 1144	0.011 255	2.61 E-07	485.5 581	0.42 7418	0.39 8556	0.00 2046	77.65 111	0.73 2358	23.51721	13.4 9141	0.33 4357	0.389678	0.349	NA

Supplementary Table 13 Risk of incident dementia according to profiles of 6 risk domains based on unweighted and weighted scores

	Unweighted score (Model1)				Unweighted score (Model2)				Weighted score (Model1)			
Subgroup	Hazard Ratio	5% CI	95% CI	P-value	Hazard Ratio	5% CI	95% CI	P-value	Hazard Ratio	5% CI	95% CI	P-value
Lifestyles												
Favorable	1.00			Reference	1.00			Reference	1.00			Reference
Intermediate	1.36	1.25	1.48	1.53×10-12	1.28	1.18	1.40	9.26×10-9	1.37	1.25	1.50	2.55×10-12
Unfavorable	1.77	1.63	1.93	6.36×10-42	1.55	1.43	1.69	< 2×10-16	1.88	1.73	2.04	1.58×10-51
P for trend				5.17×10-43				< 2×10-16				5.07×10-54
Medical history												
Favorable	1.00			Reference	1.00			Reference	1.00			Reference
Intermediate	1.15	1.05	1.26	2.05×10-3	1.09	0.99	1.19	0.07	1.15	1.05	1.27	0.002561891
Unfavorable	1.74	1.59	1.90	6.70×10-35	1.50	1.37	1.65	< 2×10-16	1.75	1.63	1.89	4.35×10-51
P for trend				4.38×10-41				< 2×10-16				1.59×10-53
Local environment												
Favorable	1.00			Reference	1.00			Reference	1.00			Reference
Intermediate	1.03	0.95	1.13	0.47	1.01	0.92	1.10	0.89	1.17	1.08	1.26	0.000107914
Unfavorable	1.19	1.10	1.28	6.83×10-6	1.10	1.02	1.19	0.01	1.27	1.17	1.38	9.32×10-9
P for trend				1.11×10-5				0.0136				8.06×10-9
	Unweighted score (Model1)				Unweighted score (Model2)				Weighted score (Model1)			
Psychosocial factors												
Favorable	1.00			Reference	1.00			Reference	1.00			Reference
Intermediate	1.33	1.23	1.43	8.22×10-14	1.16	1.08	1.25	0.00	1.21	1.12	1.31	1.18×10-6
Unfavorable	1.08	1.00	1.17	0.05	1.03	0.95	1.12	0.43	1.40	1.30	1.51	5.55×10-18
P for trend				0.0084				0.29				4.76×10-18
Physical measures												
Favorable	1.00			Reference	1.00			Reference	1.00			Reference

Intermediate	1.50	1.37	1.65	4.17×10-18	1.32	1.21	1.45	3.39×10-9	1.49	1.37	1.63	3.10×10-19
Unfavorable	1.84	1.62	2.09	5.11×10-21	1.63	1.43	1.85	5.77×10-14	2.27	1.99	2.59	2.23×10-34
P for trend				8.53×10-21				2×10-14				1.84×10-35
SES												
Favorable	1.00			Reference	1.00			Reference	1.00			Reference
Intermediate	1.17	1.09	1.26	2.77×10-5	1.05	0.98	1.13	0.16	1.25	1.14	1.37	8.50×10-7
Unfavorable	1.58	1.45	1.72	4.41×10-25	1.27	1.16	1.39	1.74×10-7	1.65	1.51	1.80	4.14×10-30
P for trend				8.21×10-24				1.15×10-7				2.07×10-32

Model 1 adjusted for age, sex, APOE ε4 status and assessment center. Model 2 further added all six risk domains which were mutually adjusted. The results of model 2 based on weighted score are shown in the main text Fig. 3. SES, socioeconomic status. Z-tests were used to assess statistical significance and derive Z statistics and corresponding two-sided P values.

Supplementary Table 14 Weighted and unweighted PAF and communality of six domains when shifting the unfavorable variables to favorable and intermediate ones.

Categories	PAF	CI_L	CI_U	OR	CI5	CI95	P	communality	Weighted PAF
Lifestyles	0.185	0.161	0.208	1.68	1.58	1.79	4.15E-56	0.180	0.100
Medical history	0.226	0.202	0.251	1.83	1.72	1.96	9.46E-76	0.399	0.123
Local environment	0.059	0.038	0.080	1.21	1.13	1.30	1.85E-08	0.483	0.032
Psychosocial factors	0.089	0.069	0.110	1.35	1.26	1.45	1.79E-18	0.416	0.049
Physical measures	0.151	0.117	0.185	1.60	1.43	1.79	3.36E-16	0.166	0.082
SES	0.154	0.130	0.178	1.54	1.44	1.65	2.36E-38	0.356	0.084

Weighted PAF was calculated after considering overlap between risk factors. Overall weighted PAF=47.0%. Z-tests were used to assess statistical significance and derive Z statistics and corresponding two-sided P values. PAF, population attributable fraction; OR, odds ratio; CI, confidence interval; SES, socioeconomic status.

Supplementary Table 15 Weighted and unweighted PAF and communality of six domains when reducing the unfavorable and intermediate variables to favorable ones.

Categories	PAF	CI_L	CI_U	OR	CI5	CI95	P	communality	Weighted PAF
Lifestyles	0.323	0.284	0.363	1.74	1.61	1.88	2.16E-44	0.175	0.166
Medical history	0.273	0.238	0.308	1.68	1.56	1.80	1.32E-46	0.224	0.140
Local environment	0.130	0.089	0.171	1.24	1.15	1.33	2.80E-09	0.040	0.067
Psychosocial factors	0.176	0.139	0.212	1.37	1.28	1.46	5.25E-19	0.104	0.090
Physical measures	0.249	0.205	0.294	1.56	1.42	1.70	3.17E-22	0.128	0.128
SES	0.263	0.218	0.309	1.52	1.40	1.65	1.42E-24	0.329	0.135

Weighted PAF was calculated after considering overlap between risk factors. Overall weighted PAF=72.6%. Z-tests were used to assess statistical significance and derive Z statistics and corresponding two-sided P values. PAF, population attributable fraction; OR, odds ratio; CI, confidence interval; SES, socioeconomic status.

Supplementary Table 16 Weighted and unweighted PAF and communality of six domains when shifting the unfavorable variables to favorable and intermediate ones (variables selection was based on machine learning).

predictors	PAF	CI_L	CI_U	OR	CI5	CI95	P	communality	weighted_PAF
Medical history	0.222	0.198	0.247	1.827	1.712	1.948	9.34E-75	0.307	0.130
Lifestyles	0.185	0.161	0.208	1.681	1.576	1.794	4.15E-56	0.195	0.107
SES	0.154	0.130	0.178	1.542	1.444	1.646	2.36E-38	0.379	0.090
Psychosocial factors	0.087	0.066	0.107	1.358	1.269	1.454	1.30E-18	0.222	0.050
Local environment	0.062	0.041	0.083	1.227	1.147	1.312	2.98E-09	0.219	0.036
Physical measures	0.022	-0.007	0.052	1.077	0.977	1.187	0.137	0.679	0.013

Weighted PAF was calculated after considering overlap between risk factors. Overall weighted PAF=42.6%. Z-tests were used to assess statistical significance and derive Z statistics and corresponding two-sided P values. PAF, population attributable fraction; OR, odds ratio; CI, confidence interval; SES, socioeconomic status.

Supplementary Table 17 Weighted and unweighted PAF and communality of six domains when reducing the unfavorable and intermediate variables to favorable ones (variables selection was based on machine learning).

predictors	PAF	CI_L	CI_U	OR	CI5	CI95	P	communality	Weighted PAF
Lifestyles	0.323	0.284	0.363	1.741	1.611	1.882	2.16E-44	0.245	0.162
Medical history	0.272	0.237	0.307	1.666	1.552	1.788	1.76E-45	0.278	0.136
SES	0.263	0.218	0.309	1.522	1.404	1.650	1.42E-24	0.398	0.132
Psychosocial factors	0.174	0.139	0.209	1.379	1.288	1.475	1.65E-20	0.239	0.087
Local environment	0.106	0.067	0.144	1.199	1.120	1.283	1.78E-07	0.062	0.053
Physical measures	0.074	0.022	0.126	1.130	1.034	1.235	0.006800105	0.778	0.037

Weighted PAF was calculated after considering overlap between risk factors. Overall weighted PAF=60.7%. Z-tests were used to assess statistical significance and derive Z statistics and corresponding two-sided P values. PAF, population attributable fraction; OR, odds ratio; CI, confidence interval; SES, socioeconomic status.

Supplementary Table 18 Weighted and unweighted PAF and communality of six domains and PRS when shifting the unfavorable variables to favorable and intermediate ones.

predictors	AF	CI_L	CI_U	OR	CI5	CI95	P	communality	weighted_PAF
Medical history	0.224	0.199	0.249	1.801	1.689	1.920	2.03E-72	0.399	0.111
Lifestyle	0.187	0.163	0.210	1.680	1.576	1.791	8.99E-57	0.180	0.092
SES	0.154	0.130	0.178	1.531	1.435	1.634	9.32E-38	0.356	0.076
Physical measures	0.150	0.116	0.184	1.585	1.417	1.773	7.23E-16	0.182	0.074
Psychosocial factors	0.091	0.070	0.112	1.356	1.268	1.450	4.66E-19	0.417	0.045
Genetic	0.083	0.056	0.111	1.281	1.185	1.385	4.97E-10	0.980	0.041
Local environment	0.056	0.035	0.078	1.200	1.122	1.284	9.57E-08	0.487	0.028

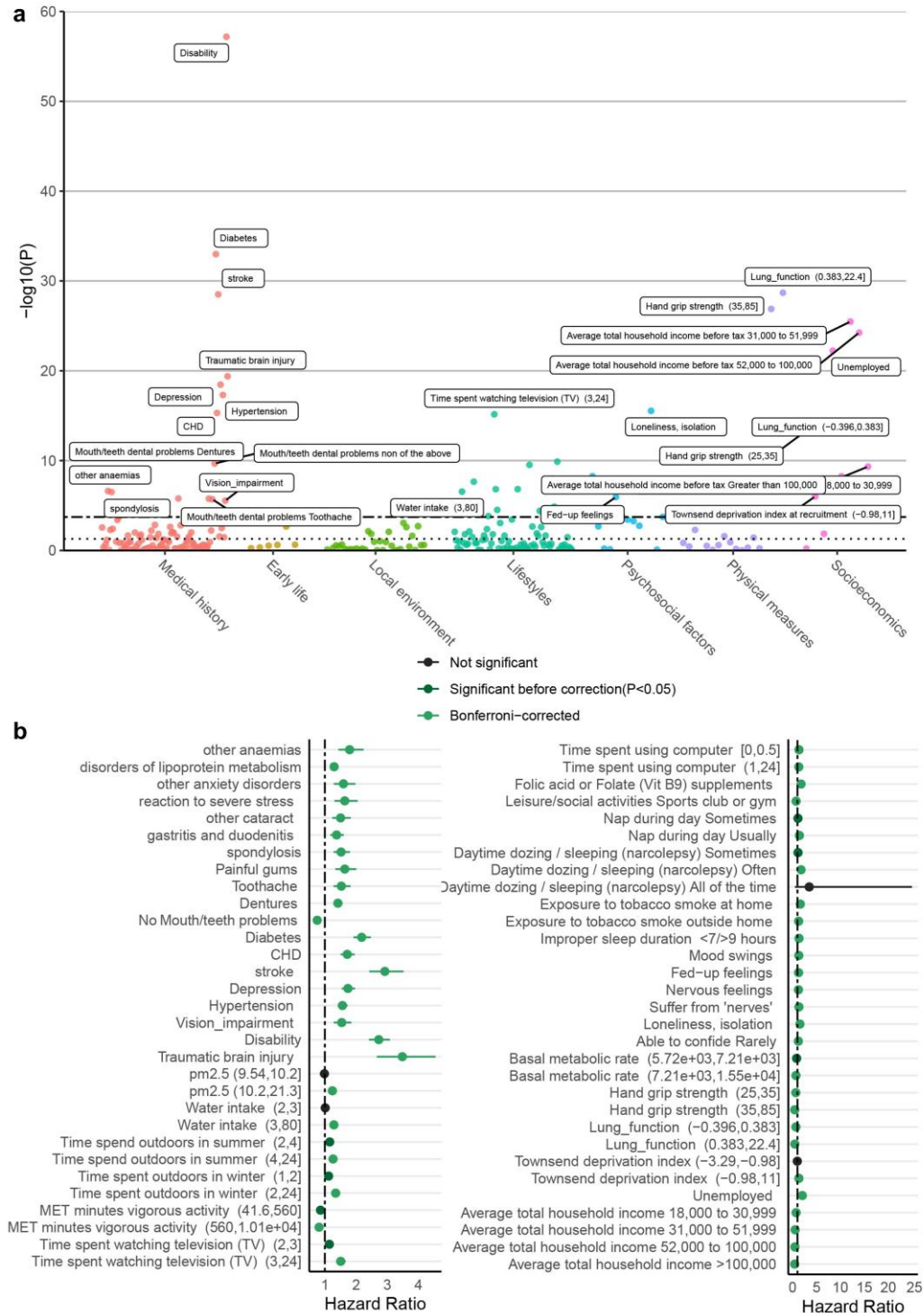
Weighted PAF was calculated after considering overlap between risk factors. Overall weighted PAF=46.8%. Z-tests were used to assess statistical significance and derive Z statistics and corresponding two-sided P values. PAF, population attributable fraction; OR, odds ratio; CI, confidence interval; SES, socioeconomic status.

Supplementary Table 19 Weighted and unweighted PAF and communality of six domains and PRS when shifting the unfavorable and intermediate variables to favorable ones

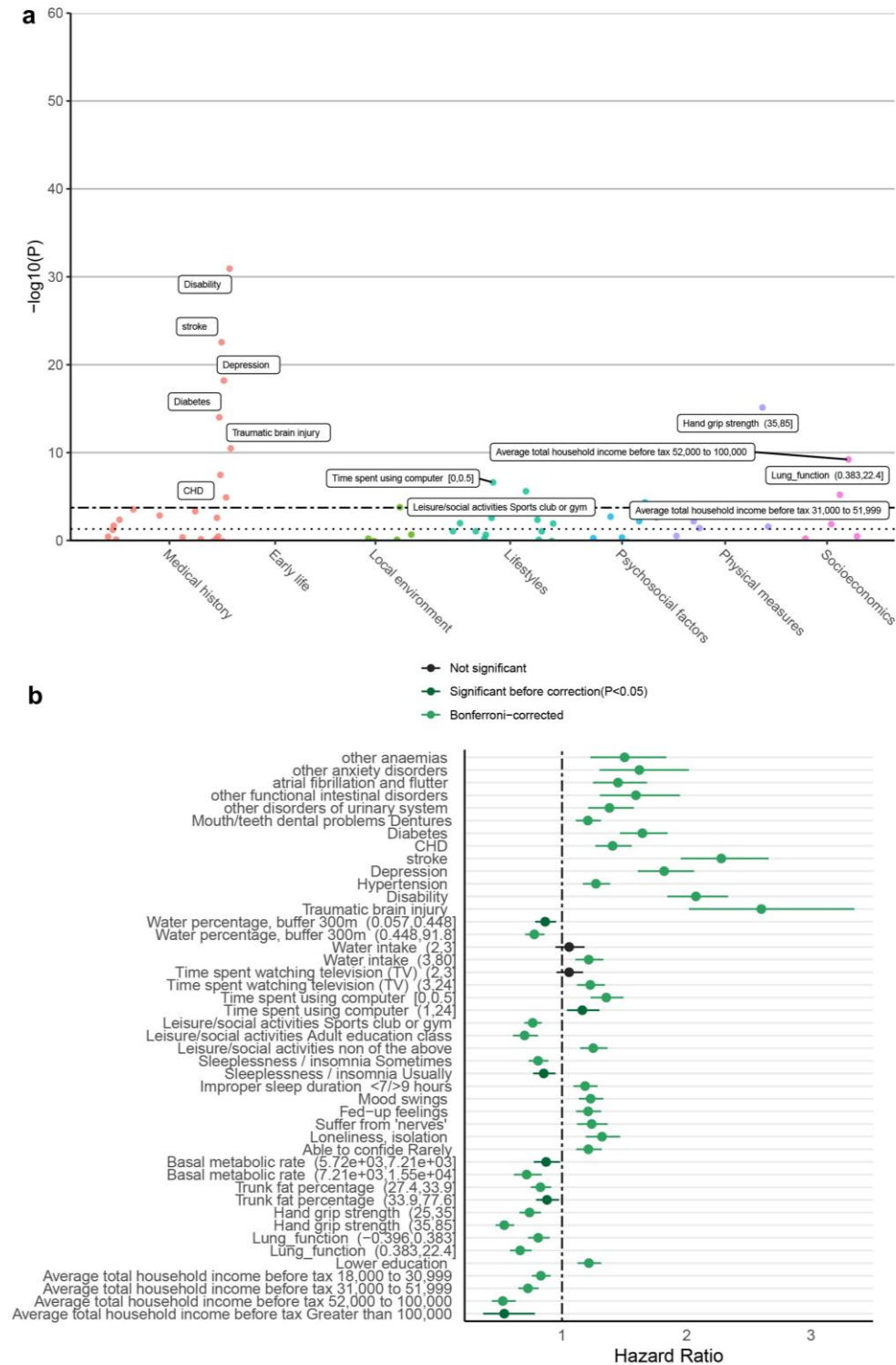
predictors	AF	CI_L	CI_U	OR	CI5	CI95	P	communality	weighted_PAF
Lifestyle	0.325	0.285	0.365	1.732	1.604	1.872	3.91E-44	0.181	0.145
Medical history	0.271	0.236	0.306	1.655	1.543	1.775	5.25E-45	0.239	0.121
SES	0.264	0.218	0.310	1.515	1.398	1.641	2.76E-24	0.337	0.118
Physical measures	0.247	0.202	0.292	1.538	1.407	1.680	1.67E-21	0.256	0.110
Genetic	0.197	0.130	0.264	1.318	1.190	1.461	1.38E-07	0.628	0.088
Psychosocial factors	0.177	0.140	0.214	1.366	1.276	1.463	3.93E-19	0.270	0.079
Local environment	0.132	0.091	0.174	1.239	1.156	1.329	1.75E-09	0.089	0.059

Weighted PAF was calculated after considering overlap between risk factors. Overall weighted PAF=72.2%. Z-tests were used to assess statistical significance and derive Z statistics and corresponding two-sided P values. PAF, population attributable fraction; OR, odds ratio; CI, confidence interval; SES, socioeconomic status.

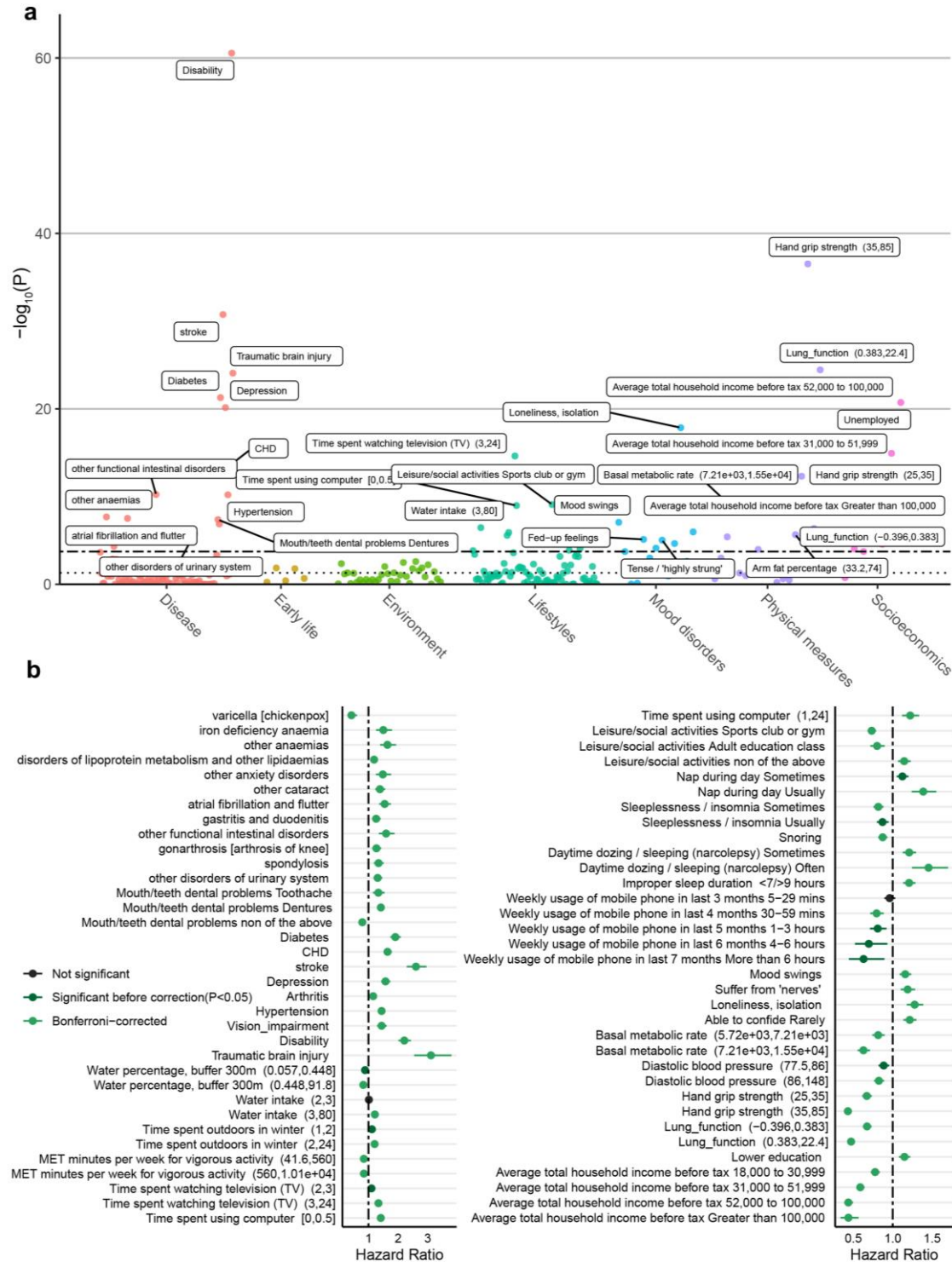
Supplementary Fig. 1. Modifiable factors associated with dementia in people <65 years old. **a**, The x-axis shows the category domains and y-axis represents the statistical significance (i.e., $-\log_{10}$ of the P value). The horizontal dotted line indicates the significance threshold of $P < 0.05$, while the dashed line indicates threshold after correcting for multiple testing (Bonferroni correction, $P < 1.87 \times 10^{-4}$). Z-tests were used to assess statistical significance and derive Z statistics and corresponding two-sided P values ($N=286,345$). **b**, Dots represent hazard ratios, horizontal lines indicate corresponding 95% CIs. Hazard ratios were calculated using Cox proportional hazards regression analysis after adjustments for age at baseline age, sex, *APOE* $\epsilon 4$ status, and assessment center ($N=286,345$). HR, hazard ratio; CI, confidence interval; SES, socioeconomic status; CHD, coronary heart disease.



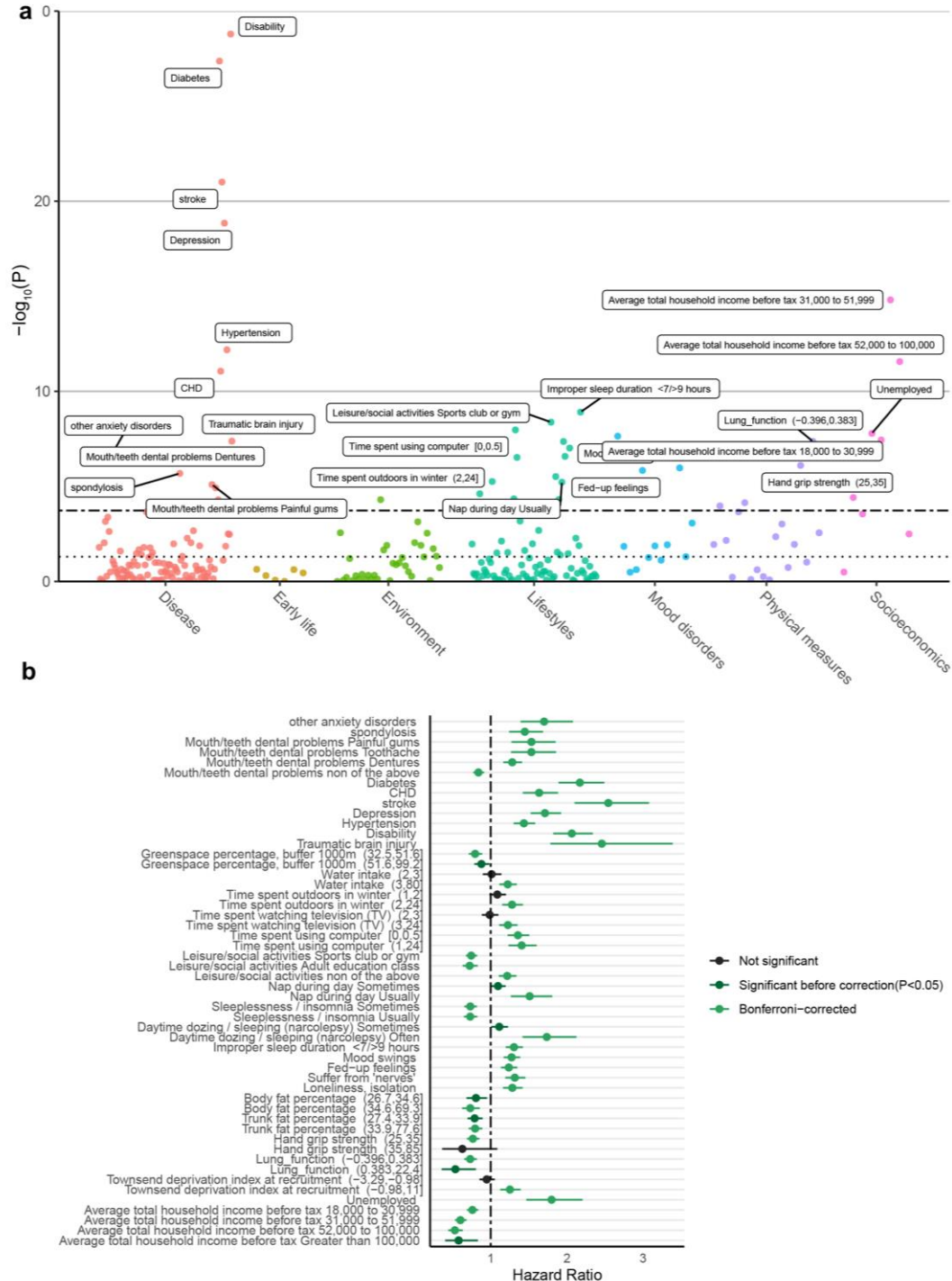
Supplementary Fig. 2. Modifiable factors associated with dementia in people ≥ 65 years old. **a**, The x-axis shows the category domains and y-axis represents the statistical significance (i.e., $-\log_{10}$ of the P value). The horizontal dotted line indicates the significance threshold of $P < 0.05$, while the dashed line indicates threshold after correcting for multiple testing (Bonferroni correction, $P < 1.87 \times 10^{-4}$). Z-tests were used to assess statistical significance and derive Z statistics and corresponding two-sided P values ($N=57,979$). **b**, Dots represent hazard ratios, horizontal lines indicate corresponding 95% CIs. Hazard ratios were calculated using Cox proportional hazards regression analysis after adjustments for age at baseline age, sex, *APOE* $\epsilon 4$ status, and assessment center ($N=57,979$). HR, hazard ratio; CI, confidence interval; SES, socioeconomic status; CHD, coronary heart disease.



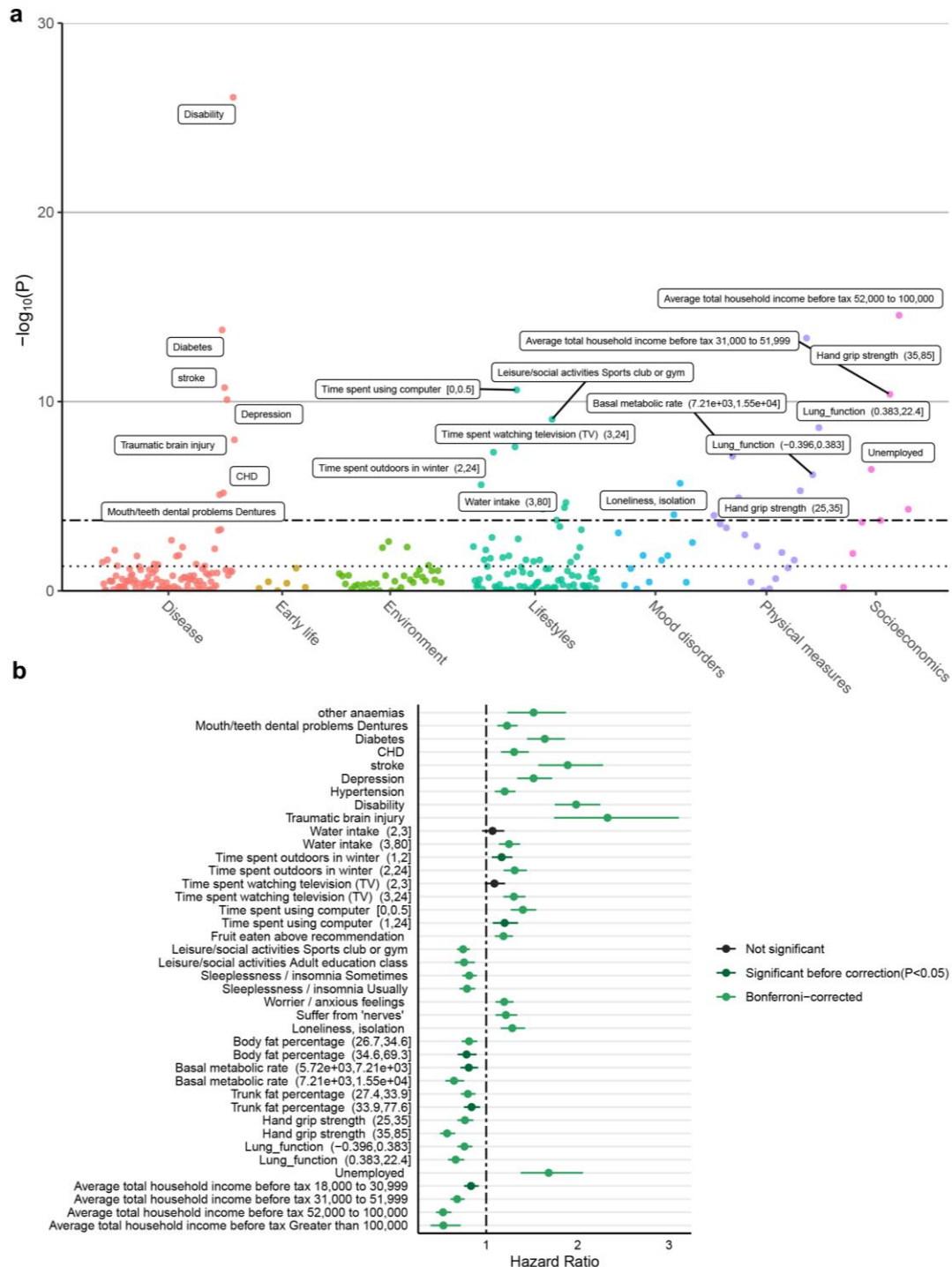
Supplementary Fig. 3. Modifiable factors associated with dementia in males. **a**, The x-axis shows the category domains and y-axis represents the statistical significance (i.e., $-\log_{10}$ of the P value). The horizontal dotted line indicates the significance threshold of $P < 0.05$, while the dashed line indicates threshold after correcting for multiple testing (Bonferroni correction, $P < 1.87 \times 10^{-4}$). Z-tests were used to assess statistical significance and derive Z statistics and corresponding two-sided P values ($N=157,612$). **b**, Dots represent hazard ratios, horizontal lines indicate corresponding 95% CIs. Hazard ratios were calculated using Cox proportional hazards regression analysis after adjustments for age at baseline age, *APOE* $\epsilon 4$ status, and assessment center ($N=157,612$). HR, hazard ratio; CI, confidence interval; SES, socioeconomic status; CHD, coronary heart disease.



Supplementary Fig. 4. Modifiable factors associated with dementia in females. **a**, The x-axis shows the category domains and y-axis represents the statistical significance (i.e., $-\log_{10}$ of the P value). The horizontal dotted line indicates the significance threshold of $P < 0.05$, while the dashed line indicates threshold after correcting for multiple testing (Bonferroni correction, $P < 1.87 \times 10^{-4}$). Z-tests were used to assess statistical significance and derive Z statistics and corresponding two-sided P values ($N=186,712$). **b**, Dots represent hazard ratios, horizontal lines indicate corresponding 95% CIs. Hazard ratios were calculated using Cox proportional hazards regression analysis after adjustments for age at baseline age, *APOE* $\epsilon 4$ status, and assessment center ($N=186,712$). HR, hazard ratio; CI, confidence interval; SES, socioeconomic status; CHD, coronary heart disease.



Supplementary Fig. 5. Modifiable factors associated with dementia in APOE ε4 carriers. **a**, The x-axis shows the category domains and y-axis represents the statistical significance (i.e., $-\log_{10}$ of the P value). The horizontal dotted line indicates the significance threshold of $P < 0.05$, while the dashed line indicates threshold after correcting for multiple testing (Bonferroni correction, $P < 1.87 \times 10^{-4}$). Z-tests were used to assess statistical significance and derive Z statistics and corresponding two-sided P values ($N=98,014$). **b**, Dots represent hazard ratios, horizontal lines indicate corresponding 95% CIs. Hazard ratios were calculated using Cox proportional hazards regression analysis after adjustments for age at baseline age, sex, and assessment center ($N=98,014$). HR, hazard ratio; CI, confidence interval; SES, socioeconomic status; CHD, coronary heart disease.



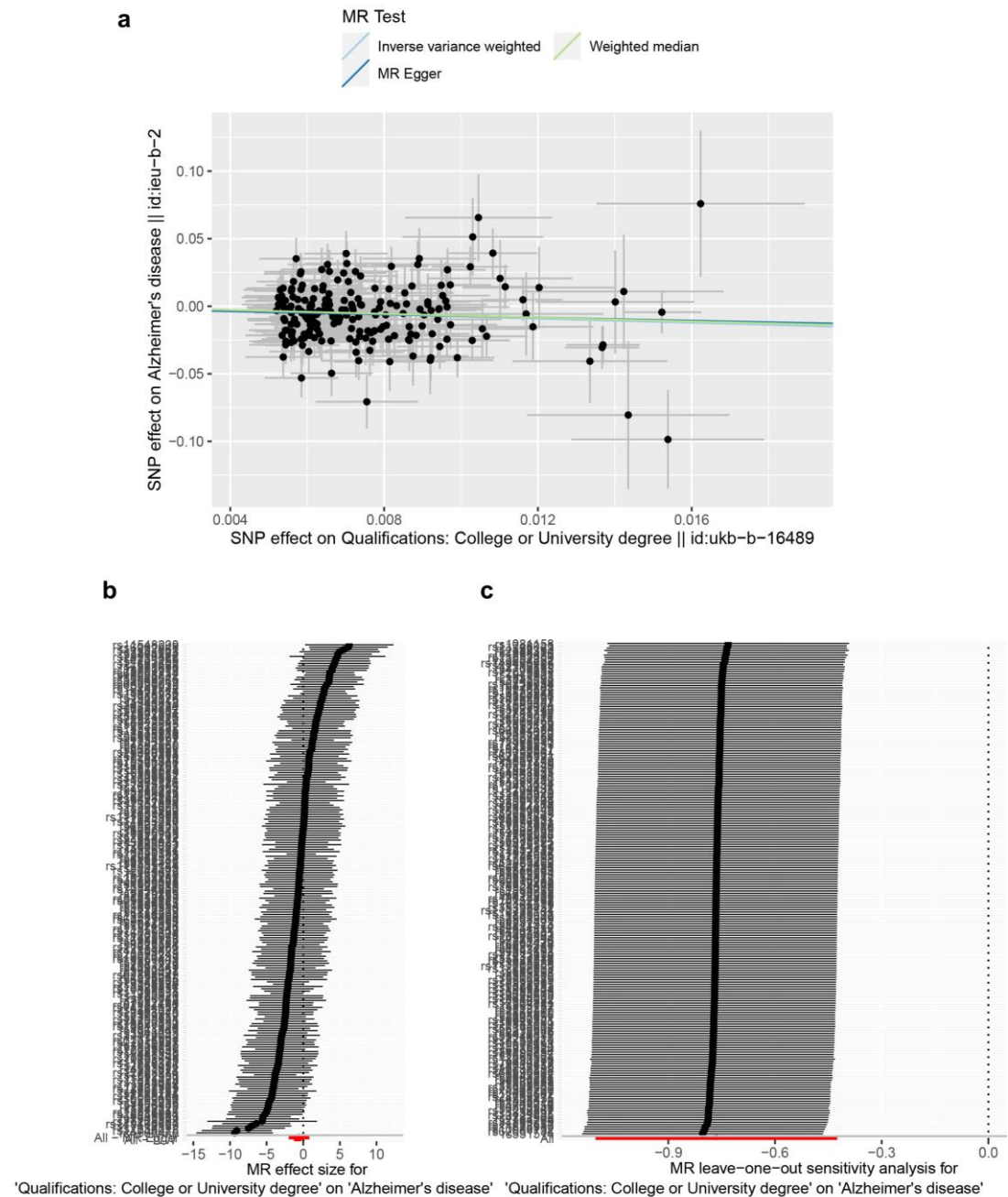
Supplementary Fig. 6. Modifiable factors associated with dementia in APOE ε4 non-carrier. **a**, The x-axis shows the category domains and y-axis represents the statistical significance (i.e., $-\log_{10}$ of the P value). The horizontal dotted line indicates the significance threshold of $P < 0.05$, while the dashed line indicates threshold after correcting for multiple testing (Bonferroni correction, $P < 1.87 \times 10^{-4}$). Z-tests were used to assess statistical significance and derive Z statistics and corresponding two-sided P values ($N=246,310$). **b**, Dots represent hazard ratios, horizontal lines indicate corresponding 95% CIs. Hazard ratios were calculated using Cox proportional hazards regression analysis after adjustments for age at baseline age, sex, and assessment center ($N=246,310$). HR, hazard ratio; CI, confidence interval; SES, socioeconomic status; CHD, coronary heart disease.



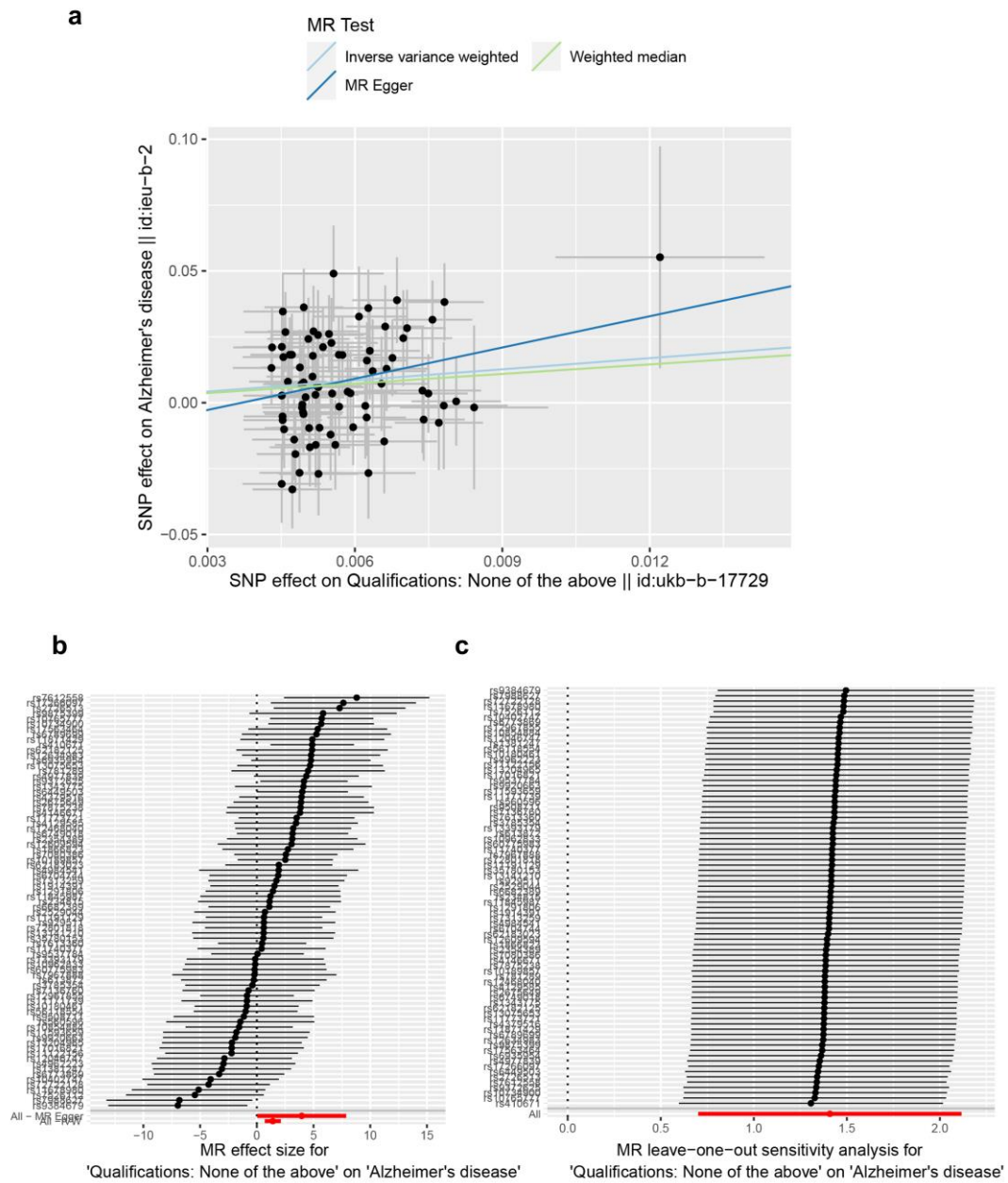
Left panel presents the Spearman rank-order correlations, and right panel shows the results of hierarchical clustering.



Supplementary Fig. 8. Mendelian randomization plots for college or university degree → risk of incident dementia. **a**, Scatterplot of SNP effects on college or university degree versus their effects on dementia, with slope of each line corresponding to estimated MR effect per method (Supplementary Table11). **b**, Forest plot of individual and combined SNP effects. **c**, Forest plot of leave-one-SNP-out sensitivity analyses. Top SNPs $p < 5 \times 10^{-8}$.

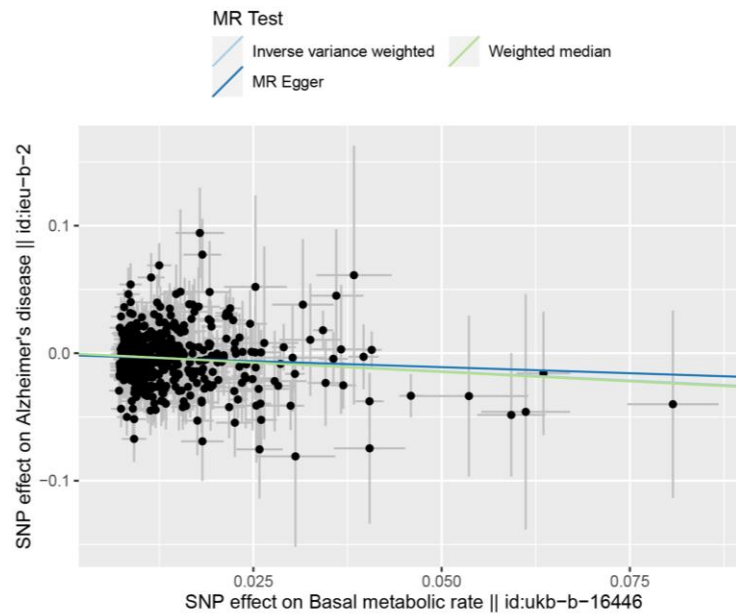


Supplementary Fig. 9. Mendelian randomization plots for no qualification → risk of incident dementia. **a**, Scatterplot of SNP effects on no qualification versus their effects on dementia, with slope of each line corresponding to estimated MR effect per method (Supplementary Table11). **b**, Forest plot of individual and combined SNP effects. **c**, Forest plot of leave-one-SNP-out sensitivity analyses. Top SNPs $p < 5 \times 10^{-8}$.

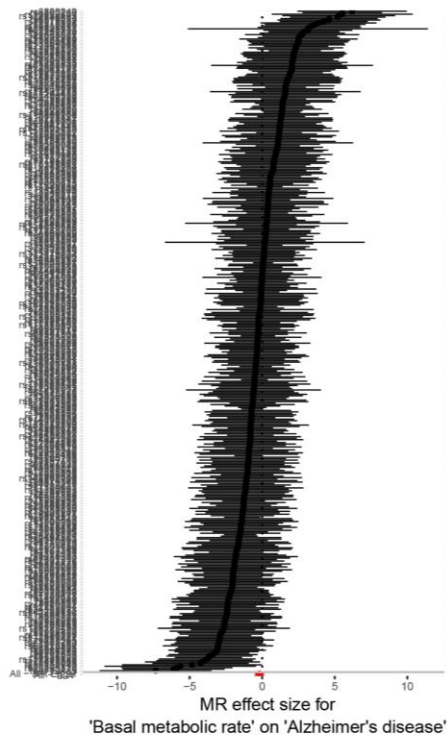


Supplementary Fig. 10. Mendelian randomization plots for basal metabolic rate → risk of incident dementia. **a**, Scatterplot of SNP effects on basal metabolic rate versus their effects on dementia, with slope of each line corresponding to estimated MR effect per method (Supplementary Table11). **b**, Forest plot of individual and combined SNP effects. **c**, Forest plot of leave-one-SNP-out sensitivity analyses. Top SNPs $p < 5 \times 10^{-8}$.

a



b



c

